

# BOOK 2

# KINEMATICS

## 2 Kinematics

### 2.1 Equations of motion

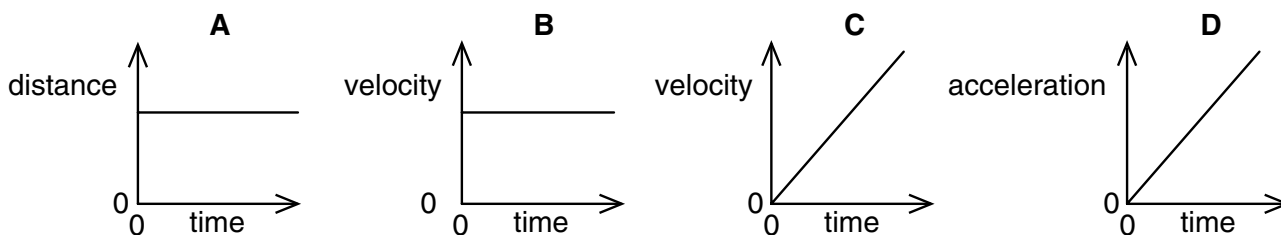
Candidates should be able to:

- 1 define and use distance, displacement, speed, velocity and acceleration
- 2 use graphical methods to represent distance, displacement, speed, velocity and acceleration
- 3 determine displacement from the area under a velocity–time graph
- 4 determine velocity using the gradient of a displacement–time graph
- 5 determine acceleration using the gradient of a velocity–time graph
- 6 derive, from the definitions of velocity and acceleration, equations that represent uniformly accelerated motion in a straight line
- 7 solve problems using equations that represent uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance
- 8 describe an experiment to determine the acceleration of free fall using a falling object
- 9 describe and explain motion due to a uniform velocity in one direction and a uniform acceleration in a perpendicular direction

1 9702/1/O/N/02/Q2\*

A particle is moving in a straight line with uniform acceleration.

Which graph represents the motion of the particle?



2 9702/01/M/J/03/Q4

Which experimental technique reduces the systematic error of the quantity being investigated?

- A adjusting an ammeter to remove its zero error before measuring a current
- B measuring several internodal distances on a standing wave to find the mean internodal distance
- C measuring the diameter of a wire repeatedly and calculating the average
- D timing a large number of oscillations to find a period

3 9702/01/M/J/03/Q5

A student makes measurements from which she calculates the speed of sound as  $327.66 \text{ m s}^{-1}$ .

She estimates that her result is accurate to  $\pm 3\%$ .

Which of the following gives her result expressed to the appropriate number of significant figures?

- A  $327.7 \text{ m s}^{-1}$
- B  $328 \text{ m s}^{-1}$
- C  $330 \text{ m s}^{-1}$
- D  $300 \text{ m s}^{-1}$

4 9702/01/M/J/03/Q6

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

|             |                              |
|-------------|------------------------------|
| length / mm | 892, 891, 892, 891, 891, 892 |
|-------------|------------------------------|

Are the readings accurate and precise to within 1 mm?

|   | results are accurate to within 1 mm | results are precise to within 1 mm |
|---|-------------------------------------|------------------------------------|
| A | no                                  | no                                 |
| B | no                                  | yes                                |
| C | yes                                 | no                                 |
| D | yes                                 | yes                                |

5 9702/13/O/N/12/Q3

A 1.5 V cell supplies 0.20 A to a lamp for seven hours before the lamp goes out.

What is a sensible estimate for the initial chemical energy content of the cell?

- A  $1 \times 10^2 \text{ J}$
- B  $1 \times 10^4 \text{ J}$
- C  $1 \times 10^6 \text{ J}$
- D  $1 \times 10^8 \text{ J}$

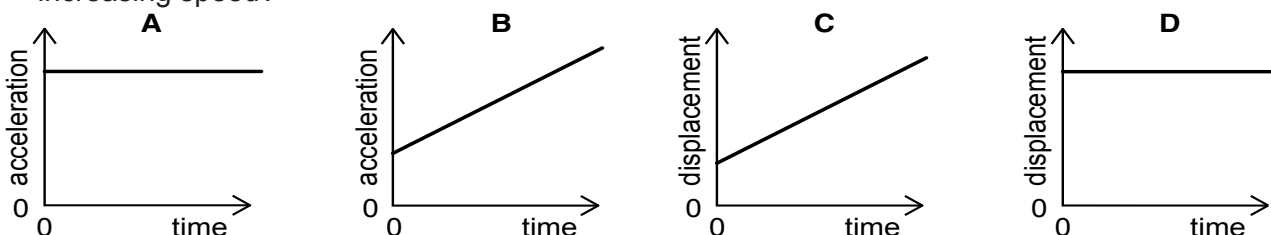
6 9702/1/M/J/02/Q3

Which formula could be correct for the speed  $v$  of ocean waves in terms of the density  $\rho$  of sea-water, the acceleration of free fall  $g$ , the depth  $h$  of the ocean and the wavelength  $\lambda$ ?

- A  $v = \sqrt{g\lambda}$
- B  $v = \sqrt{\frac{g}{h}}$
- C  $v = \sqrt{\rho gh}$
- D  $v = \sqrt{\frac{g}{\rho}}$

7 9702/1/M/J/02/Q7

Which graph represents the motion of a car that is travelling along a straight road with a uniformly increasing speed?



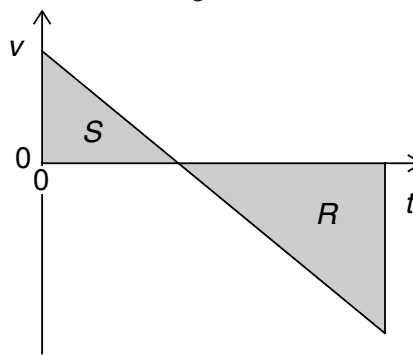
8 9702/1/M/J/02/Q8

A stone is thrown upwards from the top of a cliff. After reaching its maximum height, it falls past the cliff-top and into the sea.

The graph shows how the vertical velocity  $v$  of the stone varies with time  $t$  after being thrown upwards.  $R$  and  $S$  are the magnitudes of the areas of the two triangles.

What is the height of the cliff-top above the sea?

- A  $R$                       B  $S$   
C  $R + S$                 D  $R - S$



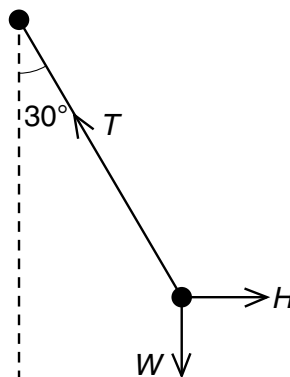
9 9702/1/O/N/02/Q3

A pendulum bob is held stationary by a horizontal force  $H$ . The three forces acting on the bob are shown in the diagram.

The tension in the string of the pendulum is  $T$ .  
The weight of the pendulum bob is  $W$ .

Which statement is correct?

- A  $H = T \cos 30^\circ$             B  $T = H \sin 30^\circ$   
C  $W = T \cos 30^\circ$             D  $W = T \sin 30^\circ$



10 9702/1/O/N/02/Q4

What is meant by the weight of an object?

- A the gravitational field acting on the object  
B the gravitational force acting on the object  
C the mass of the object multiplied by gravity  
D the object's mass multiplied by its acceleration

11 9702/12/O/N/13/Q8

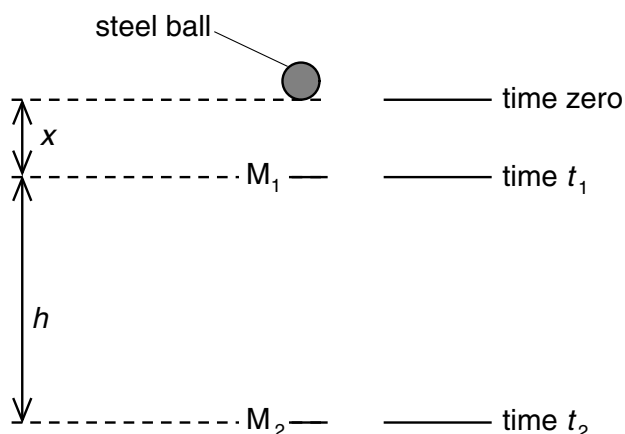
An aeroplane travels at an average speed of  $600 \text{ km h}^{-1}$  on an outward flight and at  $400 \text{ km h}^{-1}$  on the return flight over the same distance.

What is the average speed of the whole flight?

- A  $111 \text{ ms}^{-1}$             B  $167 \text{ ms}^{-1}$             C  $480 \text{ km h}^{-1}$             D  $500 \text{ km h}^{-1}$

12 9702/12/M/J/12/Q8\*\*

Two markers  $M_1$  and  $M_2$  are set up a vertical distance  $h$  apart.



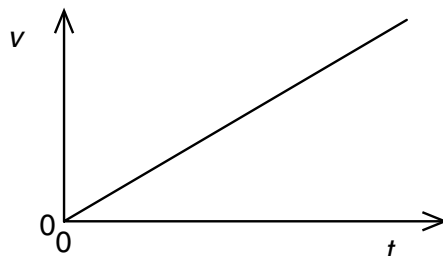
When a steel ball is released from rest from a point a distance  $x$  above  $M_1$ , it is found that the ball takes time  $t_1$  to reach  $M_1$  and time  $t_2$  to reach  $M_2$ .

Which expression gives the acceleration of the ball?

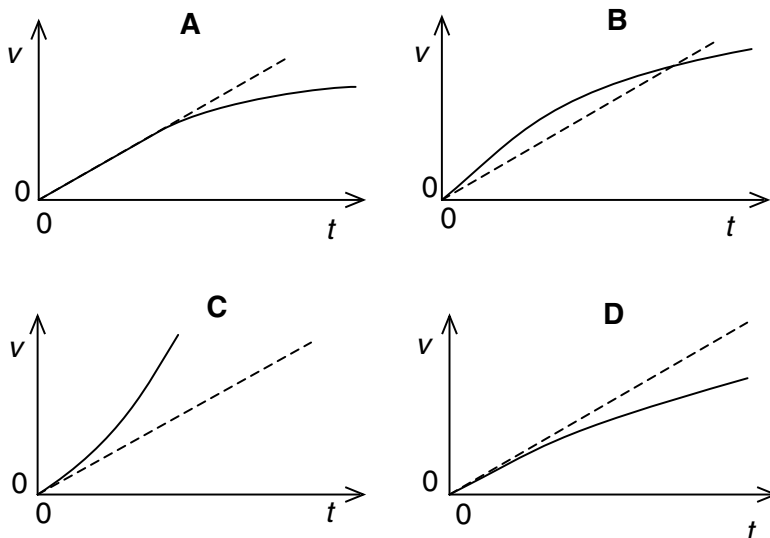
- A  $\frac{2h}{t_2^2}$       B  $\frac{2h}{(t_2 + t_1)}$       C  $\frac{2h}{(t_2 - t_1)^2}$       D  $\frac{2h}{(t_2^2 - t_1^2)}$

13 9702/1/O/N/02/Q10

A body falls from rest in a vacuum near the Earth's surface. The variation with time  $t$  of its speed  $v$  is shown below.



Which graph shows the variation with time  $t$  of the speed  $v$  of the same ball falling in air at the same place on Earth?



14 9702/01/M/J/03/Q7

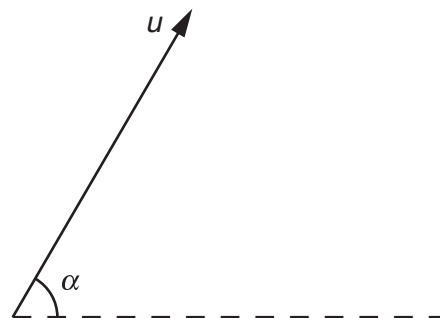
A projectile is fired at an angle  $\alpha$  to the horizontal at a speed  $u$ , as shown.

What will be the vertical and horizontal components of its velocity after a time  $t$ ?

Assume that air resistance is negligible.

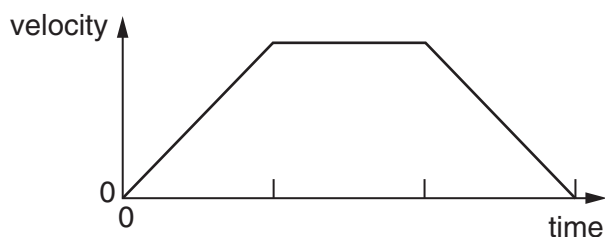
The acceleration of free fall is  $g$ .

|          | vertical component   | horizontal component |
|----------|----------------------|----------------------|
| <b>A</b> | $u \sin \alpha$      | $u \cos \alpha$      |
| <b>B</b> | $u \sin \alpha - gt$ | $u \cos \alpha - gt$ |
| <b>C</b> | $u \sin \alpha - gt$ | $u \cos \alpha$      |
| <b>D</b> | $u \cos \alpha$      | $u \sin \alpha - gt$ |

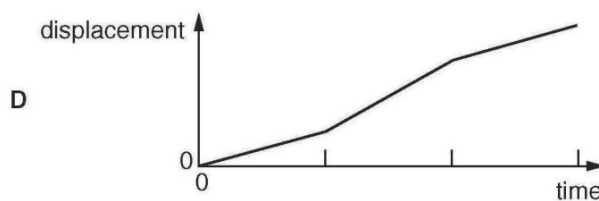
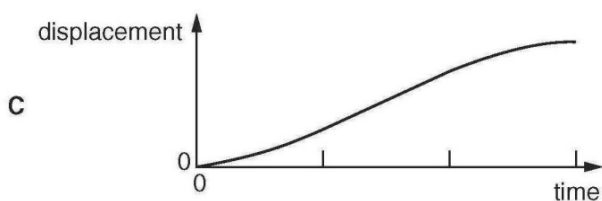
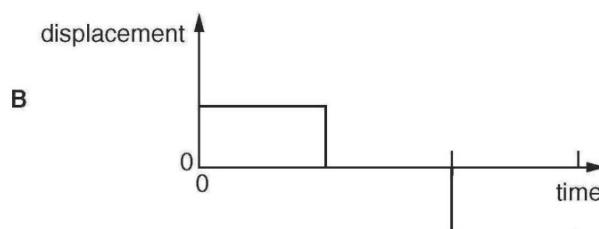
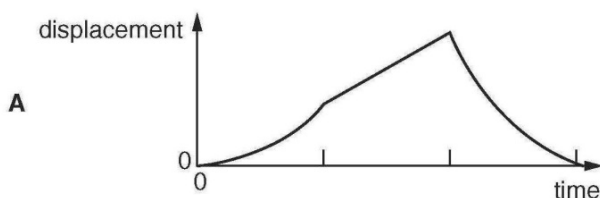


15 9702/12/O/N/12/Q9\*

The graph of velocity against time for an object moving in a straight line is shown.



Which of the following is the corresponding graph of displacement against time?



16 9702/01/M/J/04/Q7

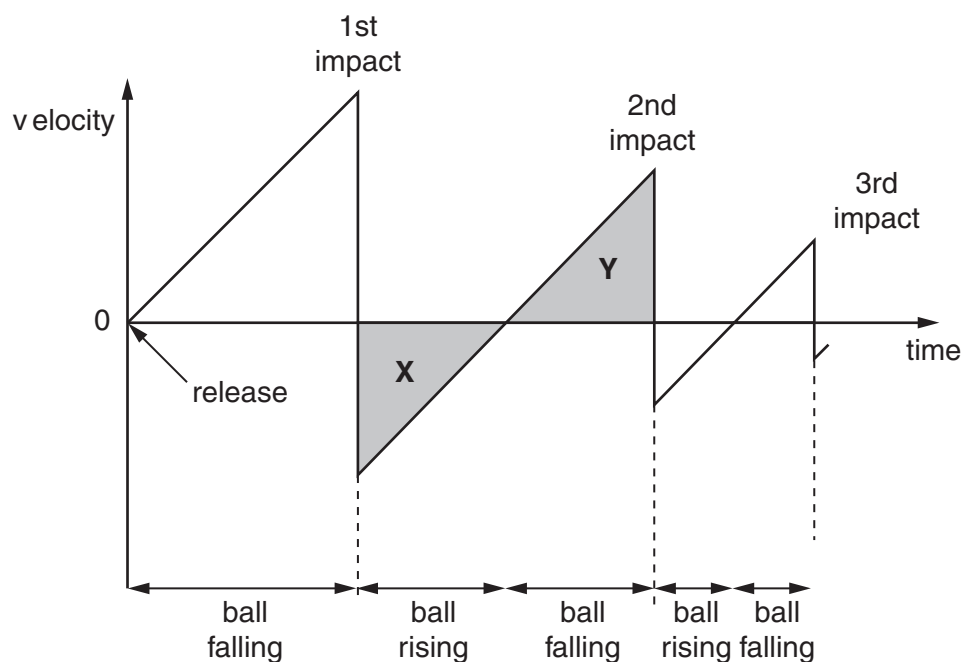
A car is travelling with uniform acceleration along a straight road. The road has marker posts every 100 m. When the car passes one post, it has a speed of  $10 \text{ m s}^{-1}$  and, when it passes the next one, its speed is  $20 \text{ m s}^{-1}$ .

What is the car's acceleration?

- A**  $0.67 \text{ m s}^{-2}$       **B**  $1.5 \text{ m s}^{-2}$       **C**  $2.5 \text{ m s}^{-2}$       **D**  $6.0 \text{ m s}^{-2}$

17 9702/01/M/J/03/Q9

A ball is released from rest above a horizontal surface. The graph shows the variation with time of its velocity.



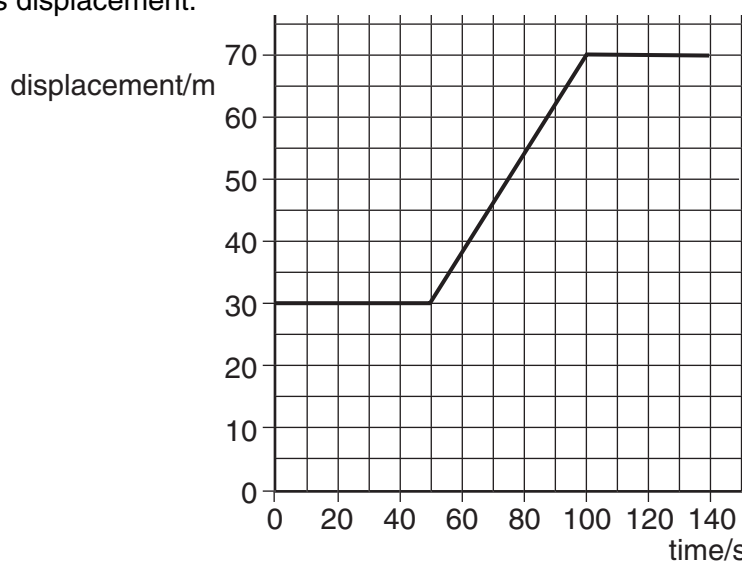
Areas **X** and **Y** are equal.

This is because

- A** the ball's acceleration is the same during its upward and downward motion.
- B** the speed at which the ball leaves the surface after an impact is equal to the speed at which it returns to the surface for the next impact.
- C** for one impact, the speed at which the ball hits the surface equals the speed at which it leaves the surface.
- D** the ball rises and falls through the same distance between impacts.

18 9702/01/O/N/03/Q7

A car at rest in a traffic queue moves forward in a straight line and then comes to rest again. The graph shows the variation with time of its displacement.



What is its speed while it is moving?

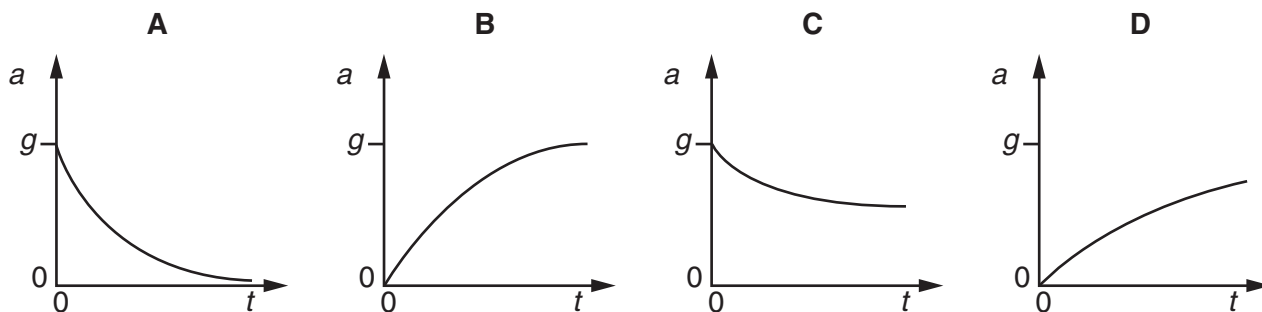
- A**  $0.70 \text{ m s}^{-1}$       **B**  $0.80 \text{ m s}^{-1}$
- C**  $1.25 \text{ m s}^{-1}$       **D**  $1.40 \text{ m s}^{-1}$

19 9702/01/O/N/03/Q8

An object is dropped from a great height and falls through air of uniform density.

The acceleration of free fall is  $g$ .

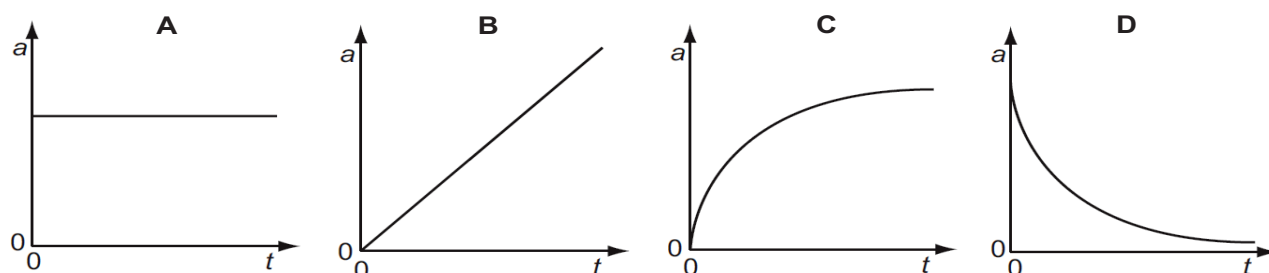
Which graph could show the variation with time  $t$  of the acceleration  $a$  of the object?



20 9702/01/M/J/04/Q8\*

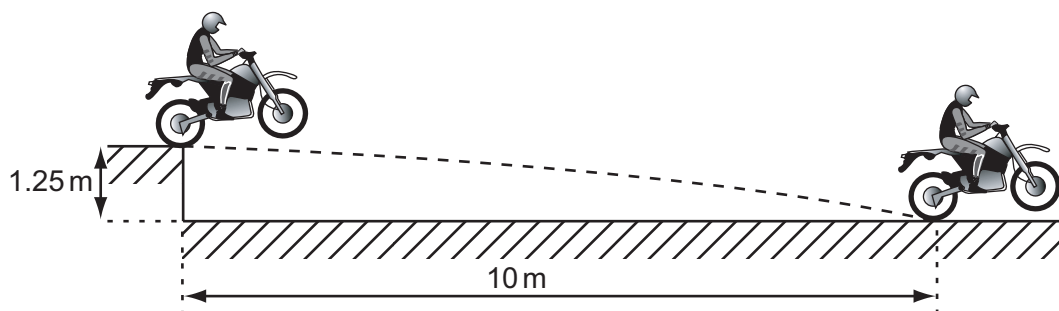
A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time  $t$  of the acceleration  $a$  of the ball as it falls, assuming that the effects of air resistance are appreciable?



21 9702/01/M/J/04/Q9

A motorcycle stunt-rider moving horizontally takes off from a point 1.25 m above the ground, landing 10 m away as shown.



What was the speed at take-off?

- A  $5 \text{ ms}^{-1}$       B  $10 \text{ ms}^{-1}$       C  $15 \text{ ms}^{-1}$       D  $20 \text{ ms}^{-1}$

22 9702/01/O/N/07/Q7

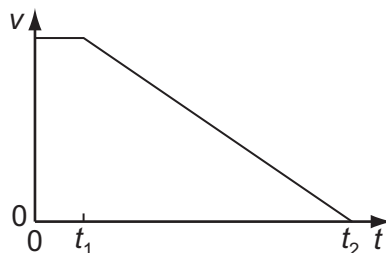
The symbol  $g$  represents the acceleration of free fall.

Which of these statements is correct?

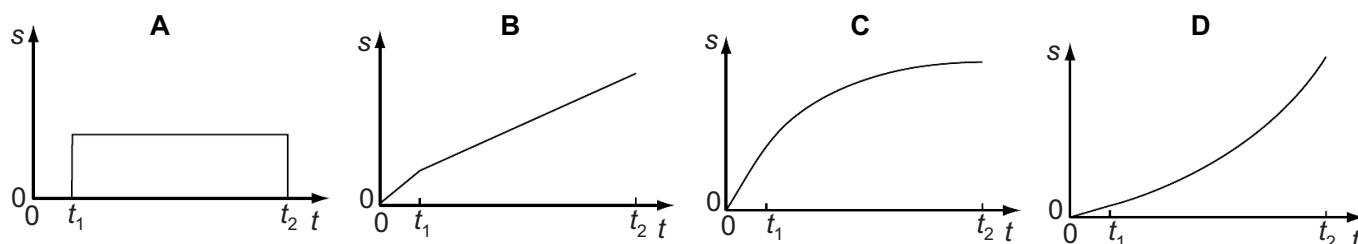
- A  $g$  is gravity.      C  $g$  is the ratio weight / mass.  
 B  $g$  is reduced by air resistance.      D  $g$  is the weight of an object.

23 9702/01/O/N/04/Q8\*

When a car driver sees a hazard ahead, she applies the brakes as soon as she can and brings the car to rest. The graph shows how the speed  $v$  of the car varies with time  $t$  after the hazard is seen.



Which graph represents the variation with time  $t$  of the distance  $s$  travelled by the car after the hazard has been seen?



24 9702/01/O/N/04/Q9

An object falls 10.0 m from rest before entering some water.

Assuming negligible air resistance, what is the time taken to reach the water and the speed with which the object reaches the water?

|          | time / ms | speed / $\text{m s}^{-1}$ |
|----------|-----------|---------------------------|
| <b>A</b> | 1.02      | 10.0                      |
| <b>B</b> | 1.02      | 14.0                      |
| <b>C</b> | 1.43      | 10.0                      |
| <b>D</b> | 1.43      | 14.0                      |

25 9702/01/O/N/04/Q10

A constant mass undergoes uniform acceleration.

Which of the following is a correct statement about the resultant force acting on the mass?

- A** It increases uniformly with respect to time.
- B** It is constant but not zero.
- C** It is proportional to the displacement from a fixed point.
- D** It is proportional to the velocity.

26 9702/01/M/J/07/Q6

What gives the value of a body's acceleration?

- A** the area under its displacement-time graph
- B** the area under its velocity-time graph
- C** the gradient of its displacement-time graph
- D** the gradient of its velocity-time graph

27 9702/01/M/J/07/Q8

A stone is dropped from the top of a tower of height 40 m. The stone falls from rest and air resistance is negligible.

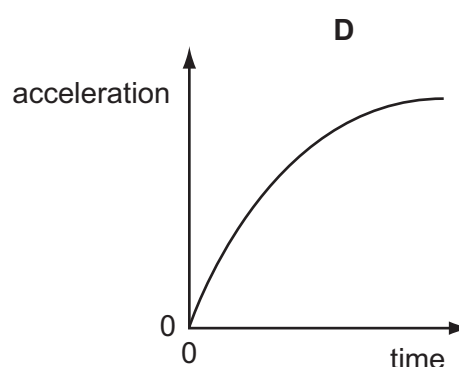
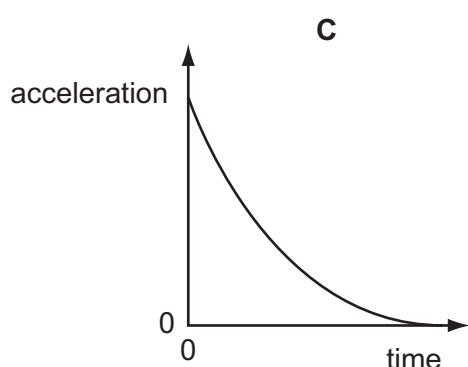
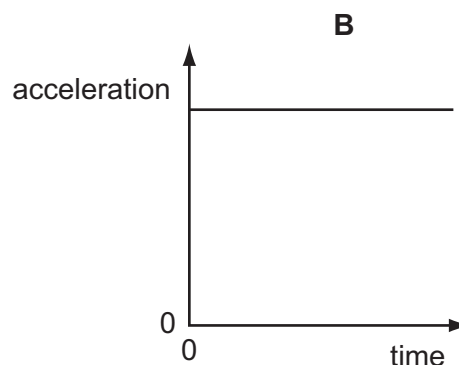
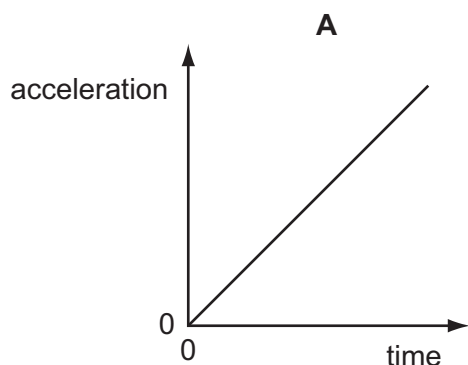
What time is taken for the stone to fall the last 10 m to the ground?

- A** 0.38 s
- B** 1.4 s
- C** 2.5 s
- D** 2.9 s

28 9702/01/M/J/08/Q8\*

A football is dropped from the top of a tall building.

Which acceleration-time graph best represents the motion of the football through the air?



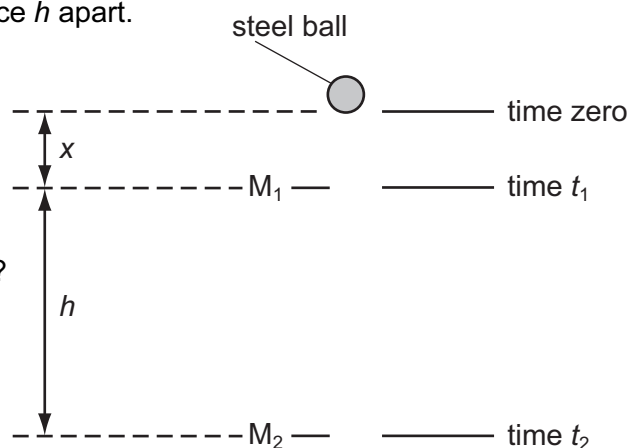
29 9702/12/M/J/12/Q8

Two markers  $M_1$  and  $M_2$  are set up a vertical distance  $h$  apart.

A steel ball is released at time zero from a point a distance  $x$  above  $M_1$ . The ball reaches  $M_1$  at time  $t_1$  and reaches  $M_2$  at time  $t_2$ . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

- A**  $\frac{2h}{t_2^2}$       **B**  $\frac{2h}{(t_2 + t_1)}$
- C**  $\frac{2h}{(t_2 - t_1)^2}$       **D**  $\frac{2h}{(t_2^2 - t_1^2)}$



30 9702/01/M/J/06/Q7

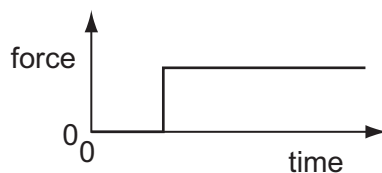
An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

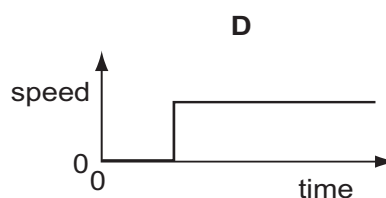
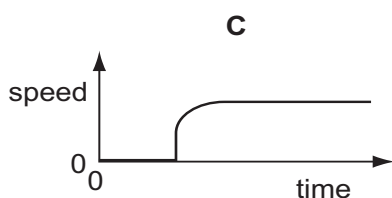
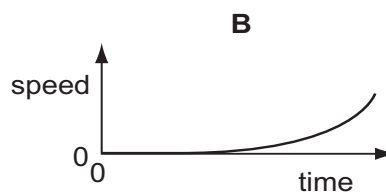
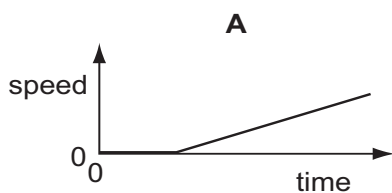
- A** the height of fall and the time of fall      **C** the mass of the body and the height of fall
- B** the height of fall and the weight of the body      **D** the mass of the body and the time of fall

31 9702/01/O/N/05/Q8

A car driver sharply presses down the accelerator when the traffic lights go green. The resultant horizontal force acting on the car varies with time as shown.



Which graph shows the variation with time of the speed of the car?

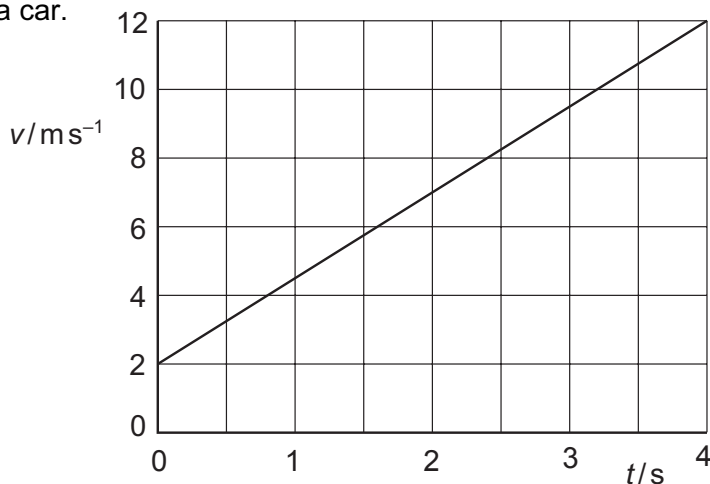


32 9702/01/M/J/05/Q8\*

The diagram shows a velocity-time graph for a car.

What is the distance travelled between time  $t = 0$  and  $t = 4$  s?

- A** 2.5 m      **B** 3.0 m  
**C** 20 m      **D** 28 m



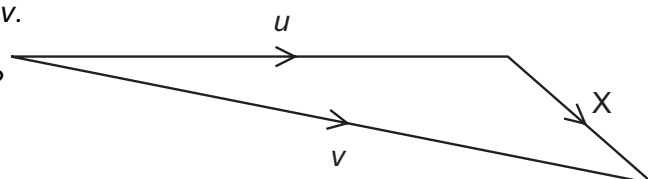
33 9702/01/M/J/07/Q7

An object has an initial velocity  $u$ . It is subjected to a constant force  $F$  for  $t$  seconds, causing a constant acceleration  $a$ . The force is **not** in the same direction as the initial velocity.

A vector diagram is drawn to find the final velocity  $v$ .

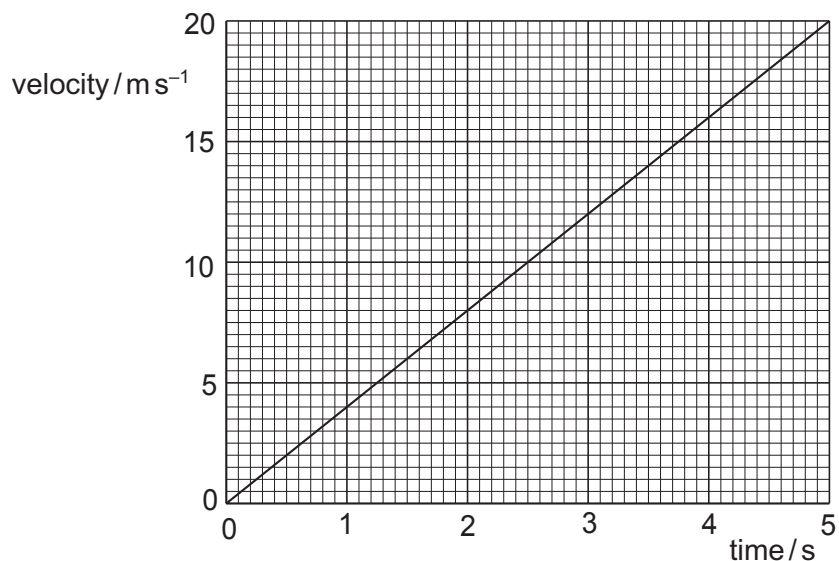
What is the length of side X of the vector diagram?

- A**  $F$       **B**  $Ft$   
**C**  $at$       **D**  $u + at$



34 9702/12/O/N/11/Q6\*

The velocity of an object during the first five seconds of its motion is shown on the graph.

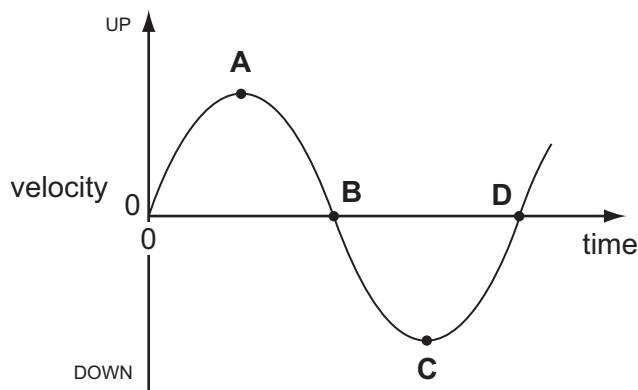


What is the distance travelled by the object in this time?

- A** 4 m                      **B** 20 m  
**C** 50 m                    **D** 100 m

**35 9702/13/O/N/14/Q7\***

The diagram shows a velocity-time graph for a mass moving up and down on the end of a spring. Which point represents the velocity of the mass when at the lowest point of its motion?



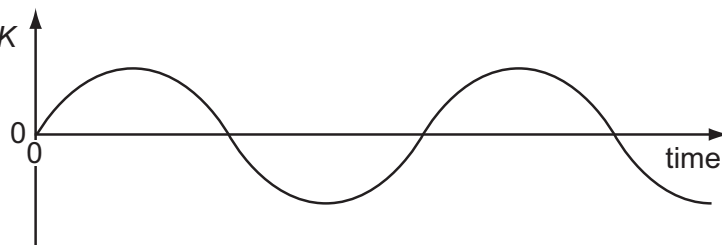
36 9702/01/O/N/07/Q8

A particle moves along a straight line. A particular property  $K$  of the particle's motion is plotted against time.

At any time, the slope of the graph is the  $K$  acceleration of the particle.

What is the property  $K$ ?

- A** the displacement of the particle  
**B** the distance travelled by the particle  
**C** the speed of the particle  
**D** the velocity of the particle



37 9702/01/M/J/08/Q7

The acceleration of free fall on a planet P is  $\frac{1}{6}$  of the acceleration of free fall on Earth.

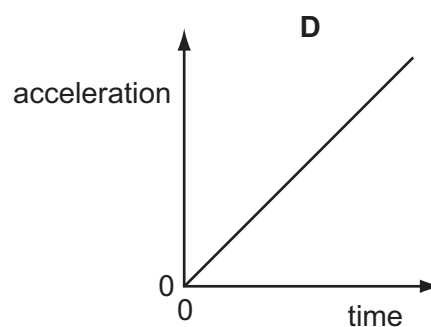
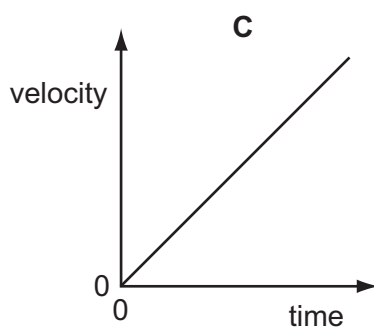
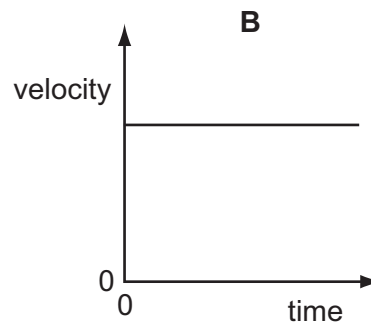
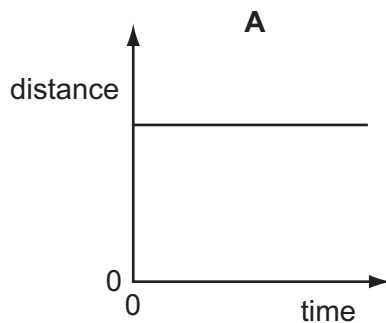
The mass of a body on planet P is 30 kg.

What is its weight on planet P?

- A** 4.9 N                      **B** 49 N                      **C** 180 N                      **D** 290 N

38 9702/1/O/N/02/Q2

A particle is moving in a straight line with uniform acceleration.  
Which graph represents the motion of the particle?

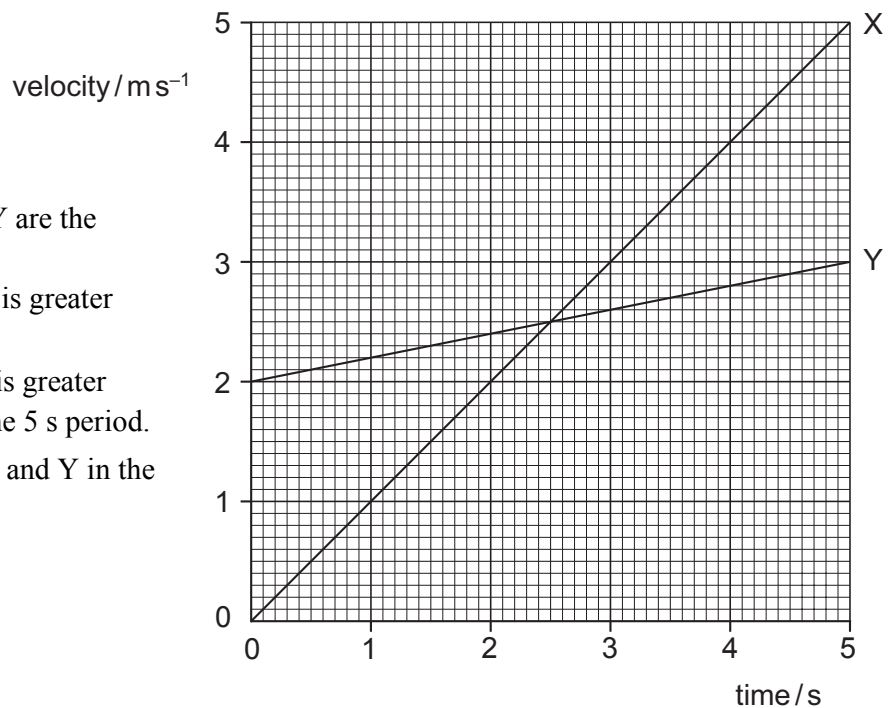


39 9702/01/O/N/06/Q8

The graph shows velocity-time plots for two vehicles X and Y. The accelerations and distances travelled by the two vehicles can be estimated from these plots.

Which statement is correct?

- A** The accelerations of X and Y are the same at 2.5 s.
- B** The initial acceleration of Y is greater than that of X.
- C** The distance travelled by X is greater than that travelled by Y in the 5 s period.
- D** The distances travelled by X and Y in the 5 s period are the same.

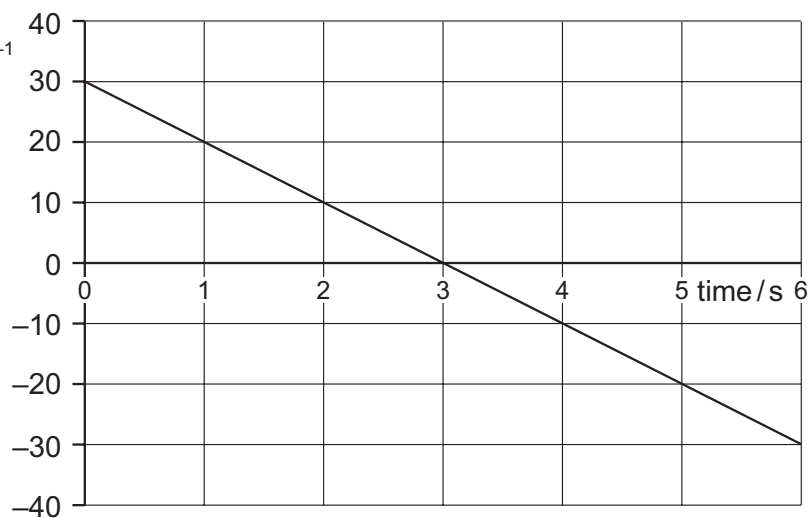


40 9702/01/O/N/07/Q9

A stone is thrown vertically upwards. A student plots the variation with time of its velocity.

What is the vertical displacement of the stone from its starting point after 5 seconds?

- A 20 m                      B 25 m  
C 45 m                      D 65 m



41 9702/01/M/J/08 /Q6

An object accelerates in a direction that is always perpendicular to its motion.

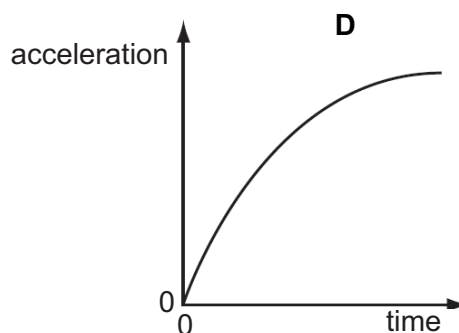
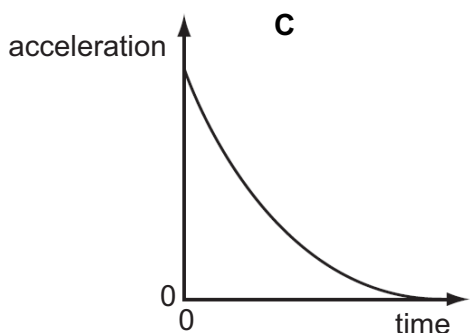
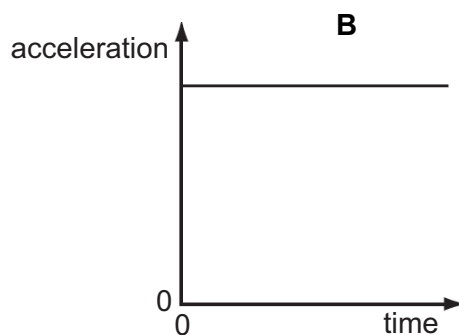
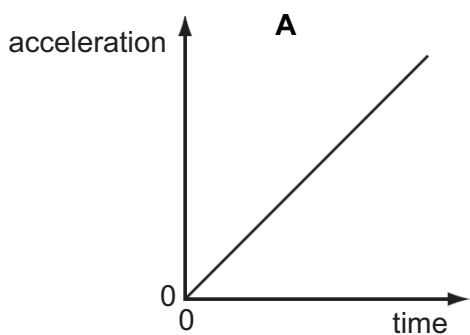
What is the effect, if any, of the acceleration on the object's speed and direction?

|   | speed    | direction |
|---|----------|-----------|
| A | changes  | changes   |
| B | changes  | constant  |
| C | constant | changes   |
| D | constant | constant  |

42 9702/01/M/J/08/Q8

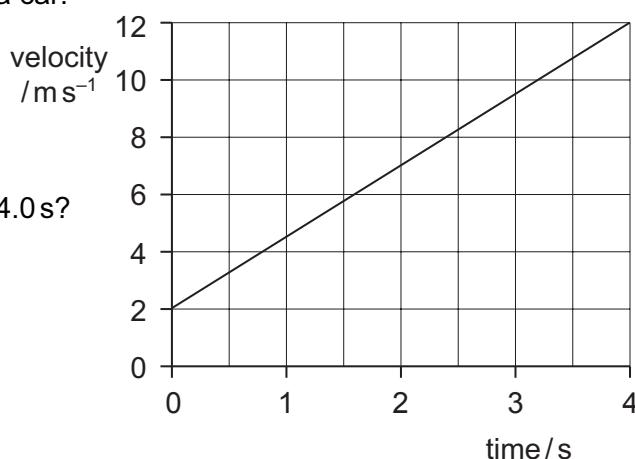
A football is dropped from the top of a tall building.

Which acceleration-time graph best represents the motion of the football through the air?



43 9702/01/O/N/08/Q6

The diagram shows a velocity-time graph for a car.



What is the distance travelled during the first 4.0 s?

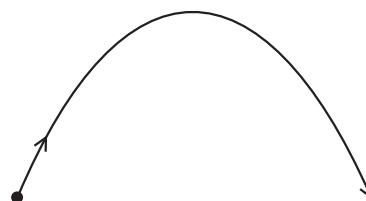
- A 2.5 m                      B 3.0 m  
C 20 m                      D 28 m

44 9702/01/O/N/08/Q7

A stone is thrown upwards and follows a curved path.

Air resistance is negligible.

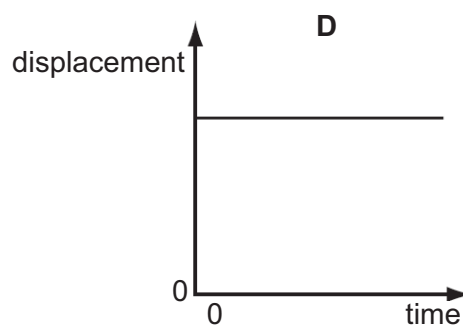
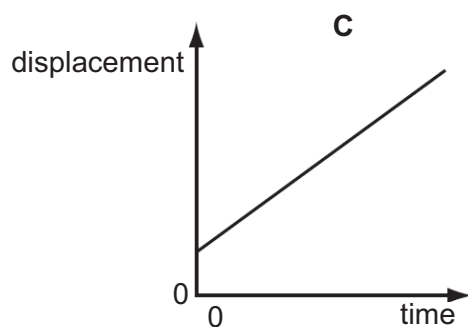
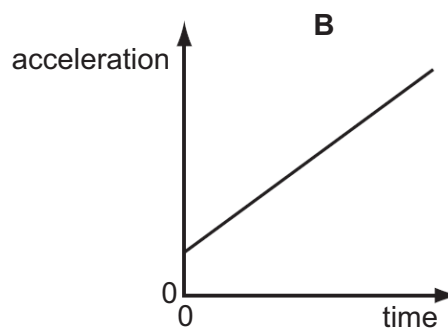
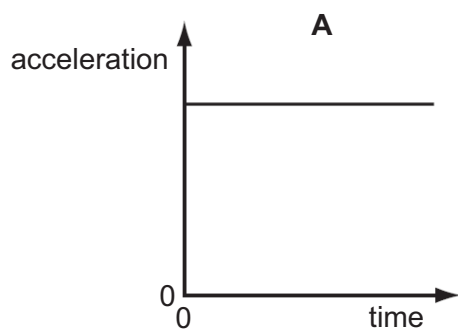
Why does the path have this shape?



- A The stone has a constant horizontal velocity and constant vertical acceleration.  
B The stone has a constant horizontal acceleration and constant vertical velocity.  
C The stone has a constant upward acceleration followed by a constant downward acceleration.  
D The stone has a constant upward velocity followed by a constant downward velocity.

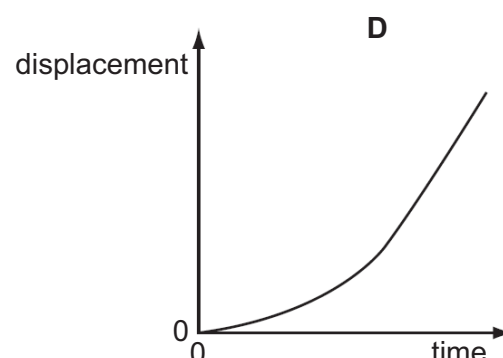
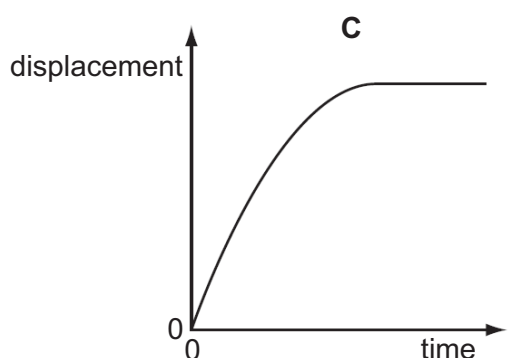
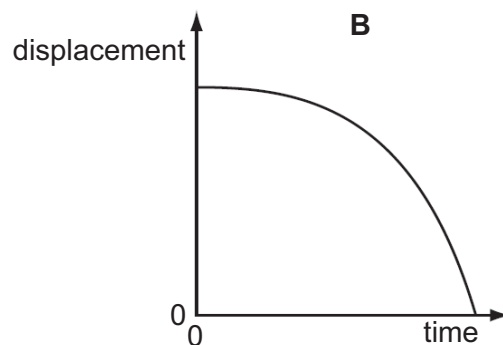
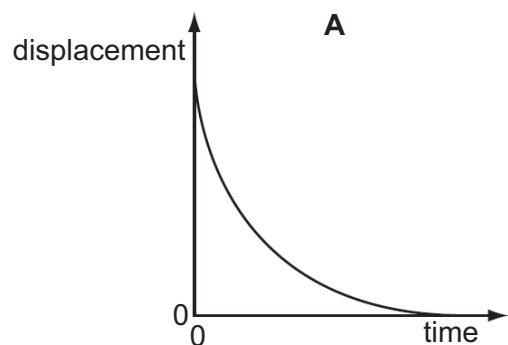
45 9702/11/O/N/14/Q6\*

Which graph represents the motion of a car that is travelling along a straight road with a speed that increases uniformly with time?



46 9702/01/M/J/09/Q5

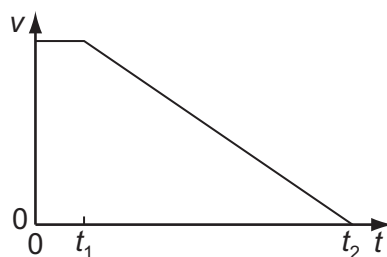
Which displacement-time graph best represents the motion of a falling sphere, the initial acceleration of which eventually reduces until it begins to travel at constant terminal velocity?



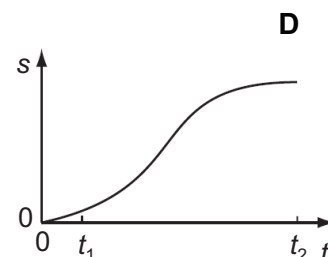
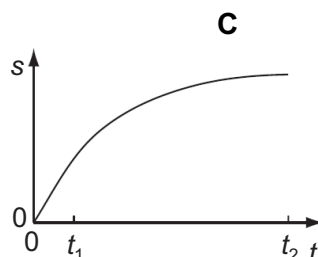
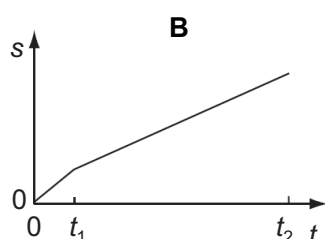
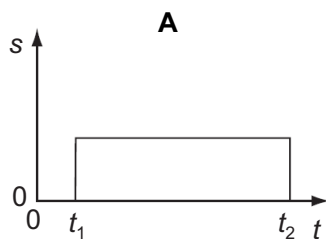
47 9702/01/M/J/09/Q6

When a car driver sees a hazard ahead, she applies the brakes as soon as she can and brings the car to rest.

The graph shows how the speed  $v$  of the car varies with time  $t$  after she sees the hazard.



Which graph represents the variation with time  $t$  of the distance  $s$  travelled by the car after she has seen the hazard?



**48 9702/12/O/N/09/Q4\***

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where it stops.

The maximum deceleration of the train is  $0.2 \text{ m s}^{-2}$ .

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

- A**  $20 \text{ m s}^{-1}$       **B**  $40 \text{ m s}^{-1}$       **C**  $200 \text{ m s}^{-1}$       **D**  $400 \text{ m s}^{-1}$

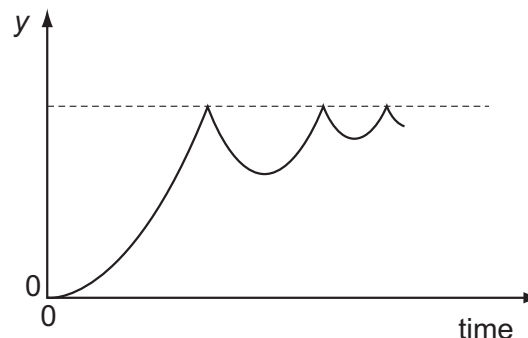
**49 9702/12/O/N/09/Q5**

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity  $y$  varies with time.

What is the quantity  $y$ ?

- A** acceleration      **B** displacement  
**C** kinetic energy      **D** velocity

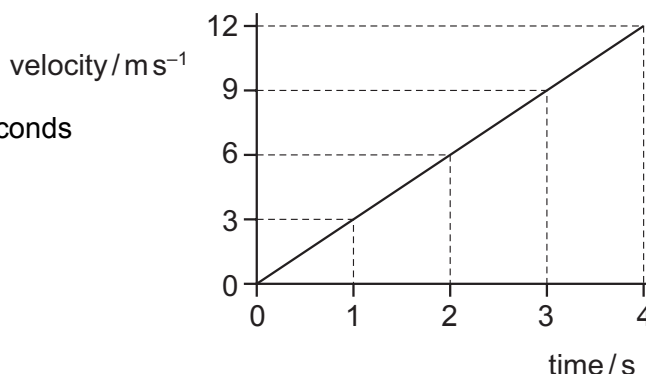


**50 9702/12/O/N/09/Q6**

The diagram shows a velocity-time graph.

What is the displacement during the last 2 seconds of the motion?

- A** 6 m      **B** 12 m  
**C** 18 m      **D** 24 m



**51 9702/12/M/J/10/Q4**

An ion is accelerated by a series of electrodes in a vacuum. A graph of the power supplied to the ion is plotted against time.

What is represented by the area under the graph between two times?

- A** the change in kinetic energy of the ion  
**B** the average force on the ion  
**C** the change in momentum of the ion  
**D** the change in velocity of the ion

**52 9702/11/O/N/10/Q9**

In order that a train can stop safely, it will always pass a signal showing a yellow light before it reaches a signal showing a red light. Drivers apply the brake at the yellow light and this results in a uniform deceleration to stop exactly at the red light.

The distance between the red and yellow lights is  $x$ .

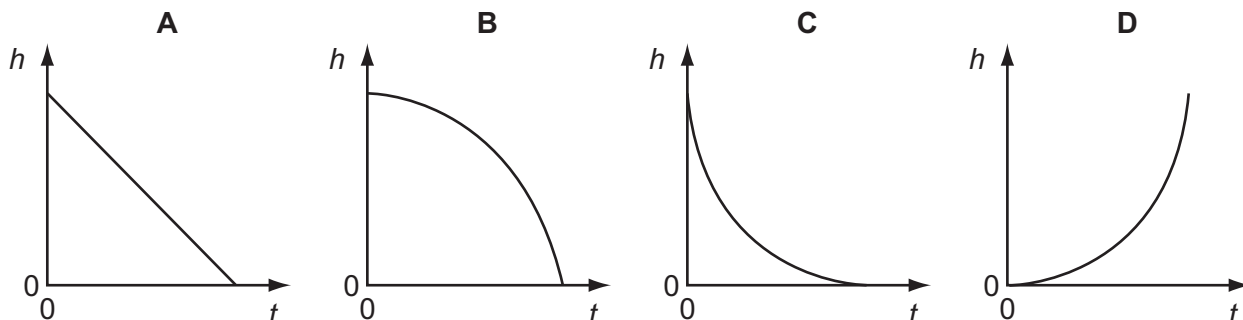
What must be the minimum distance between the lights if the train speed is increased by 20 %, without changing the deceleration of the trains?

- A**  $1.20x$       **B**  $1.25x$       **C**  $1.44x$       **D**  $1.56x$

53 9702/12/M/J/10/Q8

A small steel ball falls freely under gravity after being released from rest.

Which graph best represents the variation of the height  $h$  of the ball with time  $t$ ?



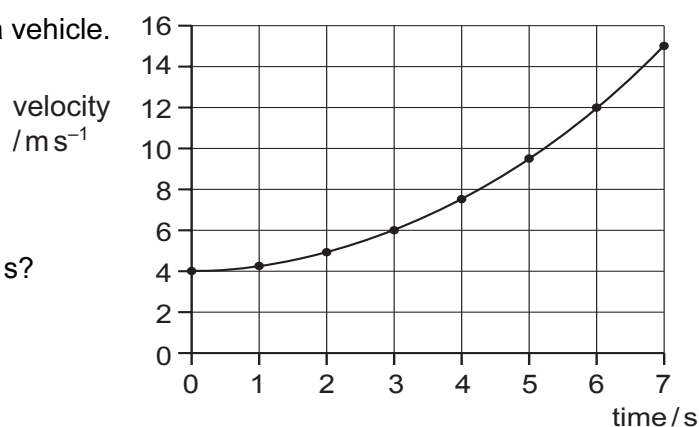
54 9702/12/M/J/10/Q9

The diagram shows a velocity-time graph for a vehicle.

The vehicle, moving at  $4.0 \text{ m s}^{-1}$ , begins to accelerate at time = 0.

What is the vehicle's acceleration at time = 3.0 s?

- A  $0.67 \text{ m s}^{-2}$       B  $1.0 \text{ m s}^{-2}$   
C  $1.3 \text{ m s}^{-2}$       D  $2.0 \text{ m s}^{-2}$



55 9702/11/O/N/10/Q7

A ball is thrown horizontally in still air from the top of a very tall building. The ball is affected by air resistance.

What happens to the horizontal and to the vertical components of the ball's velocity?

|   | horizontal component of velocity | vertical component of velocity |
|---|----------------------------------|--------------------------------|
| A | decreases to zero                | increases at a constant rate   |
| B | decreases to zero                | increases to a constant value  |
| C | remains constant                 | increases at a constant rate   |
| D | remains constant                 | increases to a constant value  |

56 9702/11/O/N/10/Q8

The velocity-time graph below is for a stone thrown vertically up into the air. Air resistance is negligible.

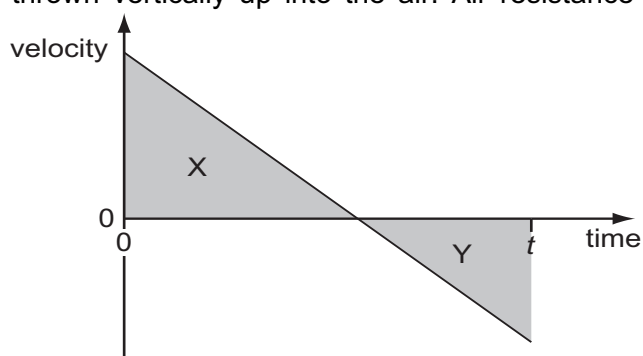
The stone is thrown up at time zero.

Area X represents a distance of 5 m.

Area Y represents a distance of 3 m.

What is the displacement of the stone from its initial position at time  $t$ ?

- A 2 m      C 5 m      B 3 m      D 8 m



57 9702/12/O/N/10/Q6

A football is dropped from the top of a three-storey building. It falls through air until it reaches the ground.

What remains constant throughout the fall?

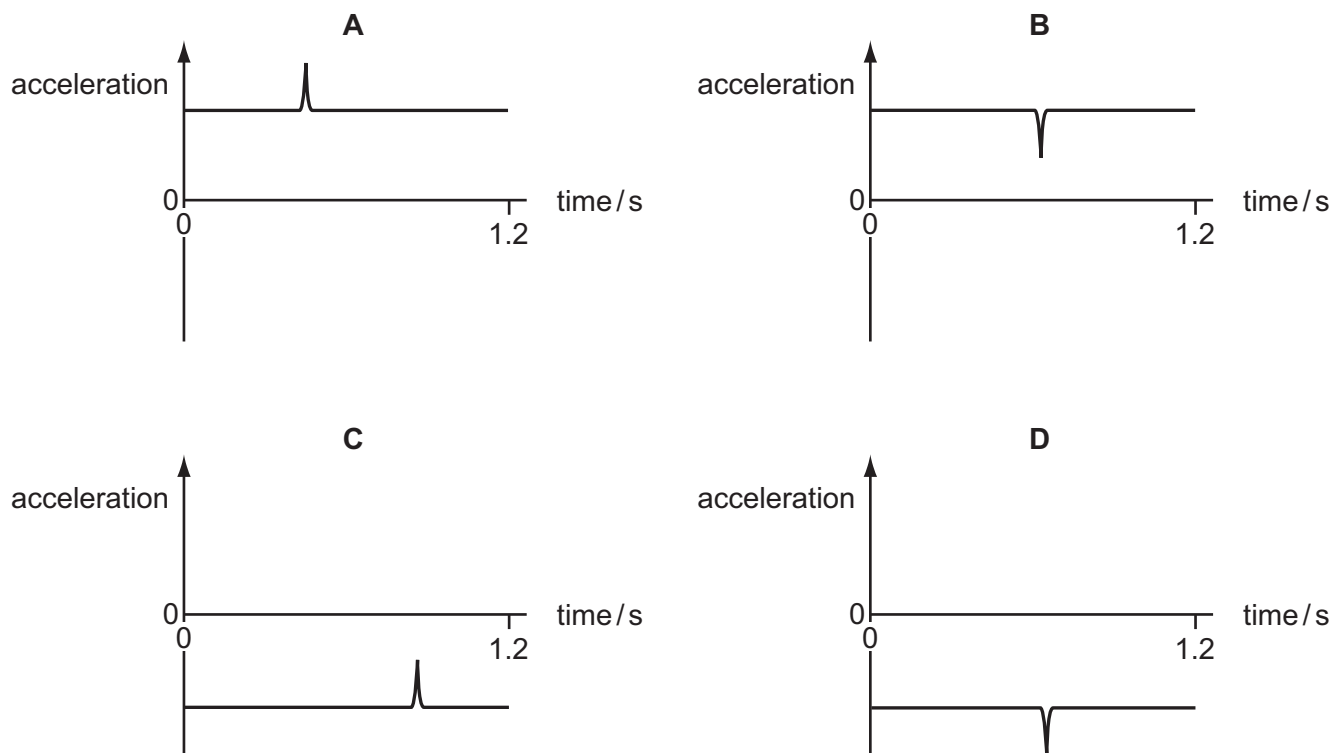
- A** acceleration of the football                      **C** velocity of the football  
**B** air resistance on the football                      **D** weight of the football

58 9702/12/O/N/10/Q7

A student throws a ball in the positive direction vertically upwards.

The ball makes an elastic collision with the ceiling, rebounds and accelerates back to the student's hand in a time of 1.2 s.

Which graph best represents the acceleration of the ball from the moment it leaves the hand to the instant just before it returns to the hand?



59 9702/12/O/N/10/Q8

A moving body undergoes uniform acceleration while travelling in a straight line between points X, Y and Z. The distances XY and YZ are both 40 m. The time to travel from X to Y is 12 s and from Y to Z is 6.0 s.

What is the acceleration of the body?

- A**  $0.37 \text{ ms}^{-2}$       **B**  $0.49 \text{ ms}^{-2}$       **C**  $0.56 \text{ ms}^{-2}$       **D**  $1.1 \text{ ms}^{-2}$

60 9702/13/O/N/13/Q9

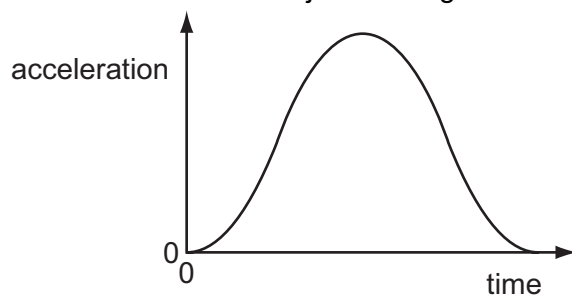
A person, travelling on a motorway a total distance of 200 km, travels the first 90 km at an average speed of  $80 \text{ km h}^{-1}$ .

Which average speed must be obtained for the rest of the journey if the person is to reach the destination in a total time of 2 hours 0 minutes?

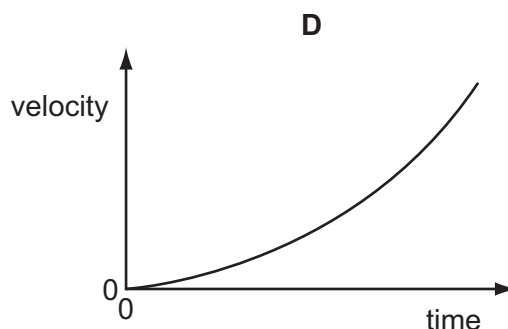
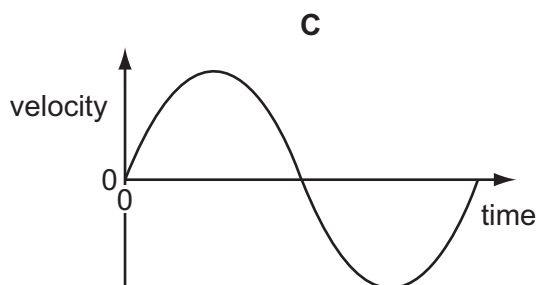
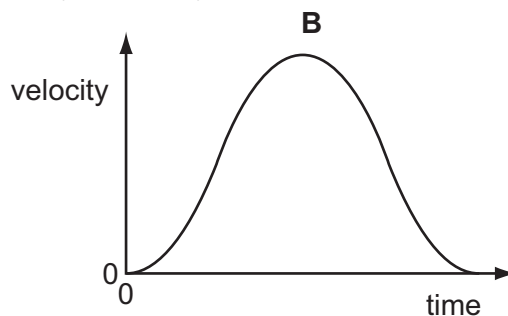
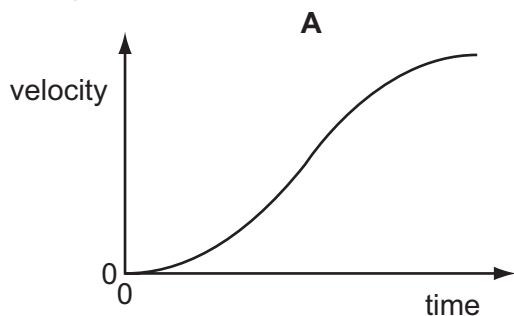
- A**  $110 \text{ km h}^{-1}$       **B**  $120 \text{ km h}^{-1}$       **C**  $122 \text{ km h}^{-1}$       **D**  $126 \text{ km h}^{-1}$

61 9702/11/M/J/14/Q8\*

The graph shows how the acceleration of an object moving in a straight line varies with time.



Which graph shows the variation with time of the velocity of the object?

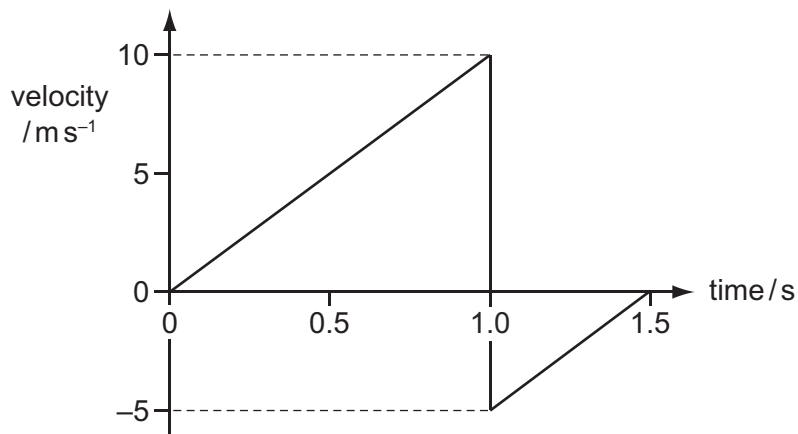


62 9702/11/M/J/11/Q7

A ball is released from rest at time zero. After 1.0 s it bounces inelastically from a horizontal surface and rebounds, reaching the top of its first bounce after 1.5 s.

What is the total displacement of the ball from its original position after 1.5 s?

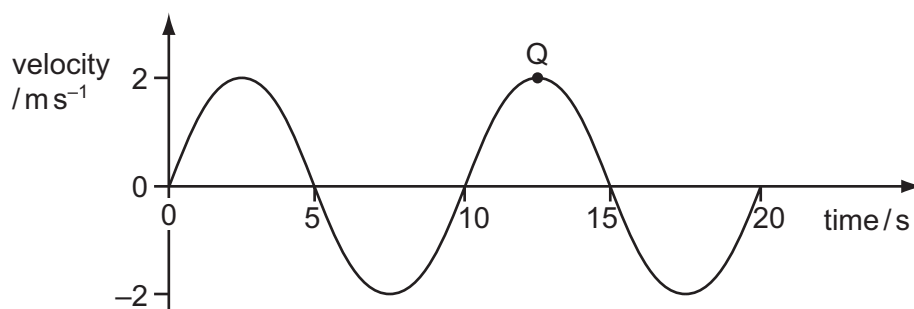
- A** 1.25 m      **B** 3.75 m  
**C** 5.00 m      **D** 6.25 m



63 9702/12/M/J/11/Q7

A particle moves in the manner shown by the velocity-time graph.

The displacement of the particle has been measured so that it is zero at  $t = 0$ . Point Q refers to a point in its motion.



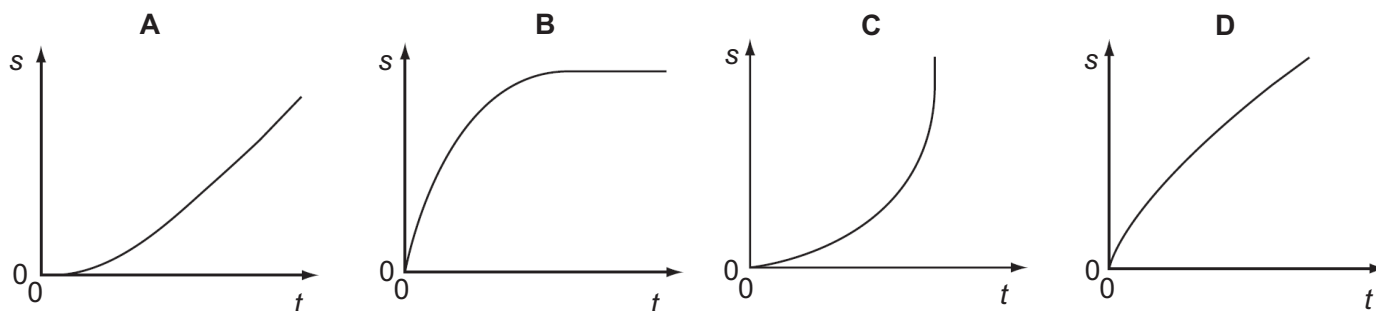
Which row of the table is correct?

|          | times for maximum displacement/s |      | acceleration at point Q / $\text{ms}^{-2}$ |
|----------|----------------------------------|------|--------------------------------------------|
| <b>A</b> | 2.5                              | 12.5 | 2                                          |
| <b>B</b> | 5                                | 15   | 2                                          |
| <b>C</b> | 2.5                              | 12.5 | 0                                          |
| <b>D</b> | 5                                | 15   | 0                                          |

64 9702/12/M/J/11/Q8\*

A tennis ball falls freely, in air, from the top of a tall building.

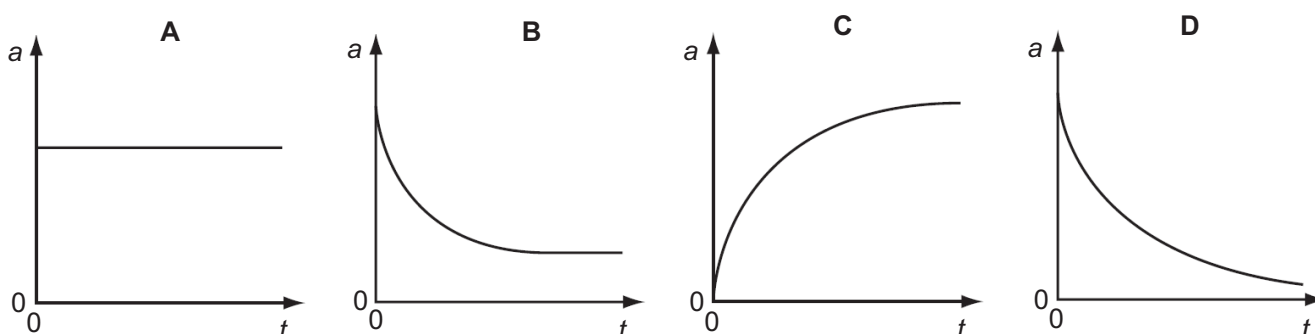
Which graph best represents the variation of distance  $s$  fallen with time  $t$ ?



65 9702/11/O/N/11/Q7

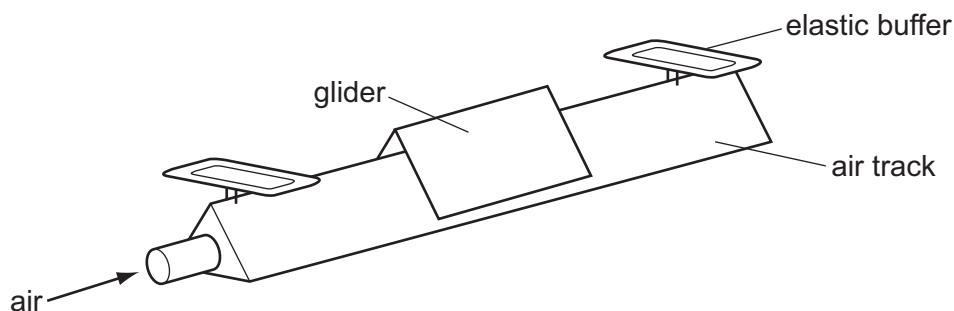
A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time  $t$  of the acceleration  $a$  of the ball as it falls, assuming that the effect of air resistance is **not** negligible?



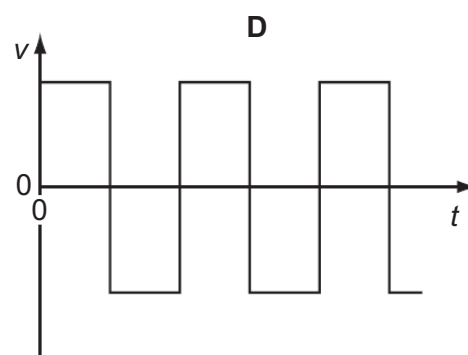
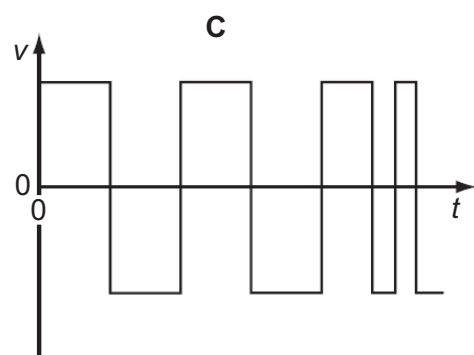
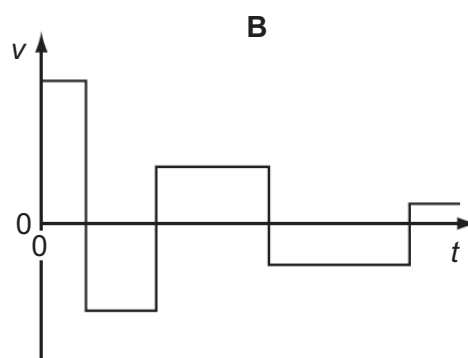
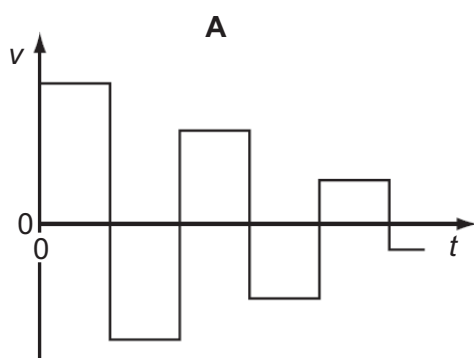
66 9702/12/M/J/11/Q9

A small glider moves along a friction-free horizontal air track as shown below.



At each end of the air track there is a perfectly elastic buffer.

Which graph represents the variation with time  $t$  of the velocity  $v$  of the glider as it moves between the two buffers?

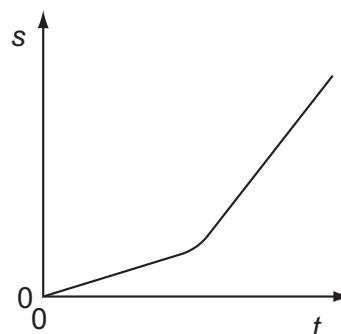


67 9702/12/O/N/11/Q8

The variation with time  $t$  of the distance  $s$  moved by a body is shown below.

What can be deduced from the graph about the motion of the body?

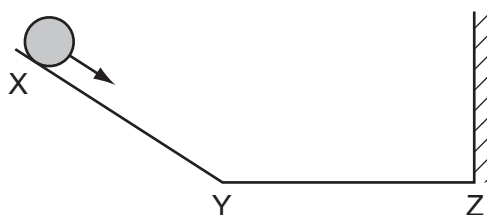
- A It accelerates continuously.
- B It starts from rest.
- C The distance is proportional to time.
- D The speed changes.



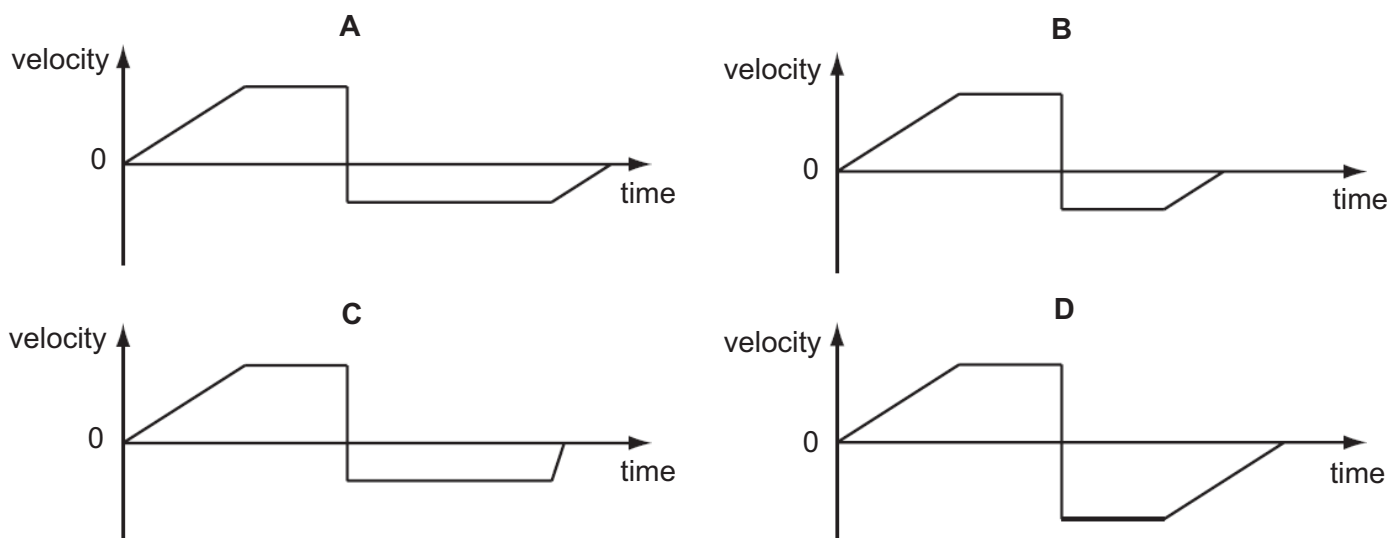
68 9702/13/O/N/14/Q8\*

A ball is released from rest on a smooth slope XY.

It moves down the slope, along a smooth horizontal surface YZ and rebounds inelastically at Z. Then it moves back to Y and comes to rest momentarily somewhere on XY.



Which velocity-time graph represents the motion of the ball?



69 9702/12/O/N/15/Q7

A boy throws a ball vertically upwards. It rises to a maximum height, where it is momentarily at rest, and then falls back to his hands.

Which row gives the acceleration of the ball at various stages in its motion? (Take vertically upwards as positive. Ignore air resistance.)

|   | rising                  | at maximum height       | falling                 |
|---|-------------------------|-------------------------|-------------------------|
| A | $-9.81 \text{ ms}^{-2}$ | 0                       | $+9.81 \text{ ms}^{-2}$ |
| B | $-9.81 \text{ ms}^{-2}$ | $-9.81 \text{ ms}^{-2}$ | $-9.81 \text{ ms}^{-2}$ |
| C | $+9.81 \text{ ms}^{-2}$ | $+9.81 \text{ ms}^{-2}$ | $+9.81 \text{ ms}^{-2}$ |
| D | $+9.81 \text{ ms}^{-2}$ | 0                       | $-9.81 \text{ ms}^{-2}$ |

70 9702/11/O/N/11/Q9\*

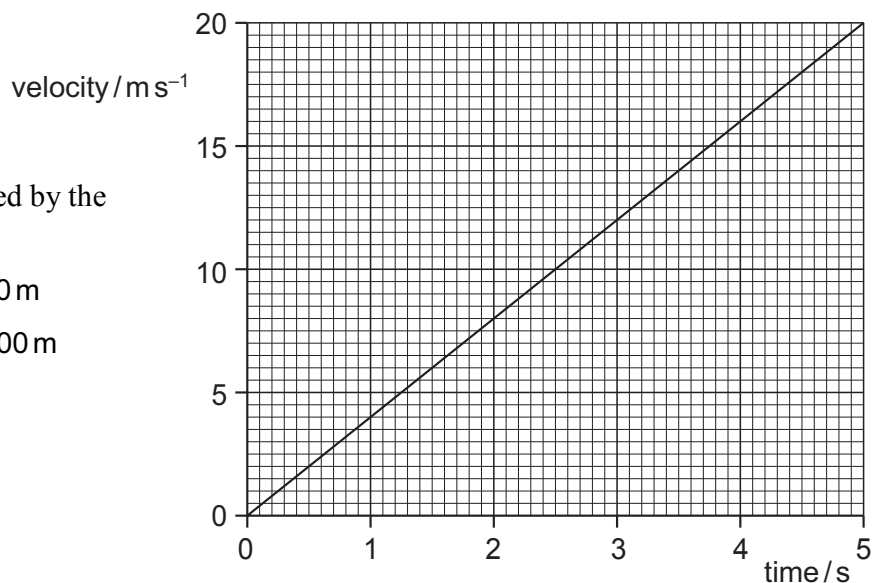
A body falling in a uniform gravitational field encounters air resistance. The air resistance increases until terminal velocity is reached.

Which factor does **not** affect its terminal velocity?

- A the density of the air                      C the mass of the body  
B the height from which the body falls      D the shape of the body

71 9702/12/O/N/11/Q6

The velocity of an object during the first five seconds of its motion is shown on the graph.



What is the distance travelled by the object in this time?

- A 4 m                      B 20 m  
C 50 m                    D 100 m

72 9702/12/O/N/11/Q7\*

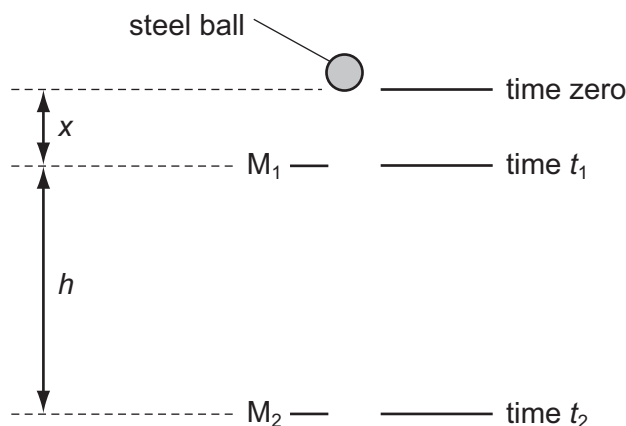
A stone of mass  $m$  is dropped from a tall building. There is significant air resistance. The acceleration of free fall is  $g$ .

When the stone reaches its terminal velocity, which information is correct?

|   | magnitude of the acceleration of the stone | magnitude of the force of gravity on the stone | magnitude of the force of air resistance on the stone |
|---|--------------------------------------------|------------------------------------------------|-------------------------------------------------------|
| A | $g$                                        | $mg$                                           | $mg$                                                  |
| B | zero                                       | $mg$                                           | $mg$                                                  |
| C | zero                                       | zero                                           | $mg$                                                  |
| D | zero                                       | zero                                           | zero                                                  |

73 9702/01/O/N/05/Q7

Two markers  $M_1$  and  $M_2$  are set up a vertical distance  $h$  apart.



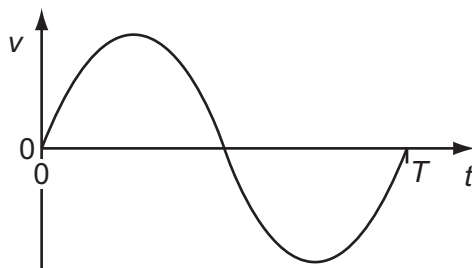
A steel ball is released at time zero from a point a distance  $x$  above  $M_1$ . The ball reaches  $M_1$  at time  $t_1$  and reaches  $M_2$  at time  $t_2$ . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

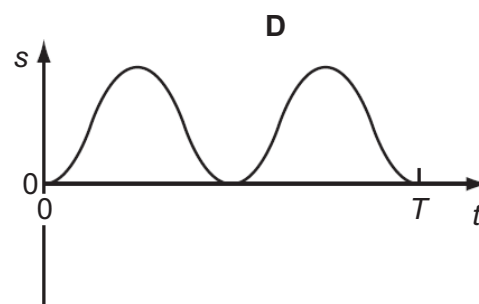
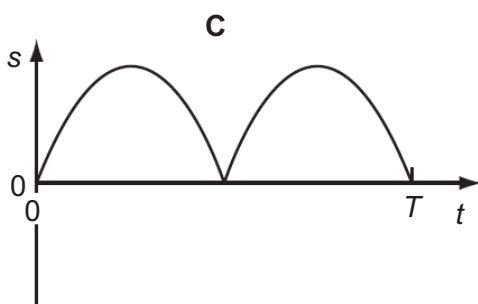
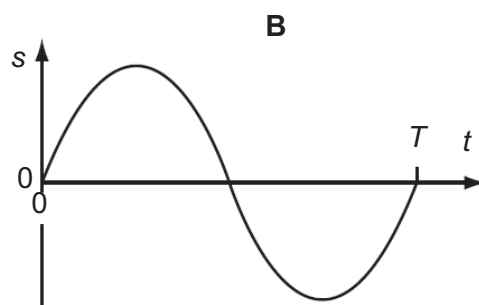
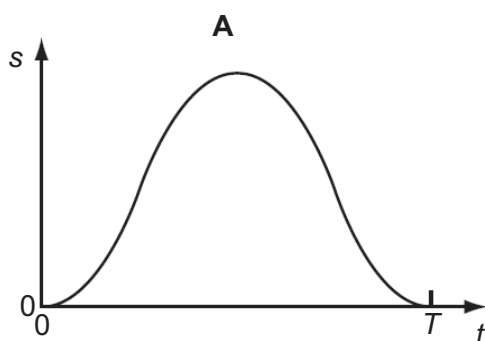
- A  $\frac{2h}{t^2}$                       B  $\frac{2h}{(t_1 + t_2)}$                       C  $\frac{2h}{(t_1^2 - t_2^2)}$                       D  $\frac{2h}{(t_1^2 - t_2^2)}$

74 9702/12/M/J/12/Q7

The graph shows how the velocity  $v$  of an object moving in a straight line varies over time  $t = 0$  to  $t = T$ .



Which graph represents the displacement  $s$  of the object in the time  $t = 0$  to  $t = T$ ?

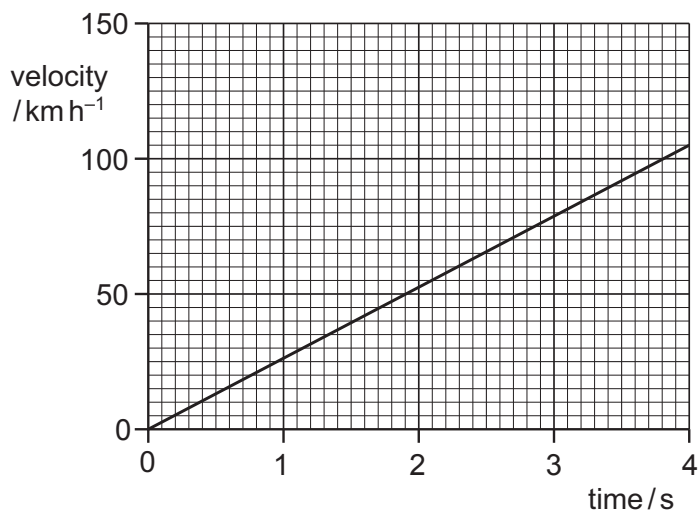


75 9702/13/M/J/15/Q8

The velocity of an electric car changes as shown.

What is the acceleration of the car?

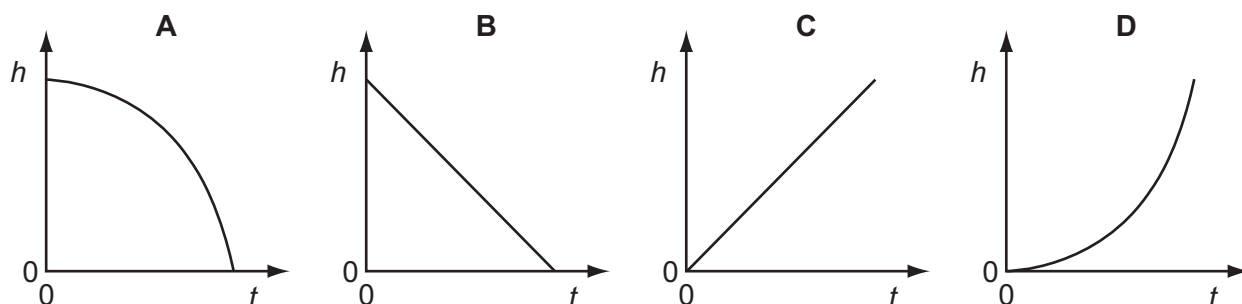
- A**  $210 \text{ ms}^{-2}$       **B**  $58 \text{ ms}^{-2}$   
**C**  $26 \text{ ms}^{-2}$       **D**  $7.3 \text{ ms}^{-2}$



76 9702/12/M/J/12/Q9

A brick is dislodged from a building and falls vertically under gravity.

Which graph best represents the variation of its height  $h$  above the ground with time  $t$  if air resistance is negligible?



77 9702/13/M/J/12/Q6

Which feature of a graph allows acceleration to be determined?

- A** the area under a displacement-time graph      **C** the slope of a displacement-time graph  
**B** the area under a velocity-time graph      **D** the slope of a velocity-time graph

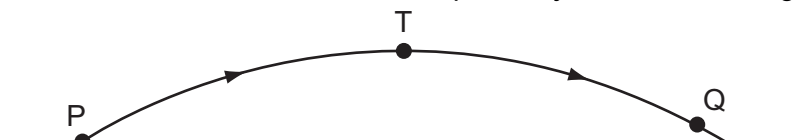
78 9702/13/M/J/12/Q7

In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

- A** air resistance      **C** measured distance longer than true distance  
**B** dimensions of the body are too large      **D** measured time longer than true time

79 9702/13/M/J/12/Q8

In the absence of air resistance, a stone is thrown from P and follows a parabolic path in which the highest point reached is T. The stone reaches point Q just before landing.



The vertical component of acceleration of the stone is

- A** zero at T.      **C** larger at Q than at T.  
**B** larger at T than at Q.      **D** the same at Q as at T.

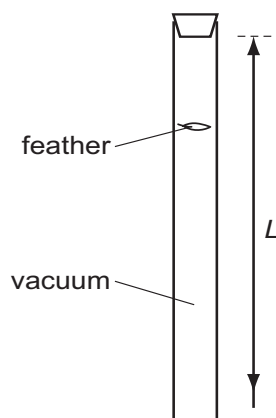
80 9702/13/M/J/12/Q9

The diagram shows a laboratory experiment in which a feather falls from rest in a long evacuated vertical tube of length  $L$ .

The feather takes time  $T$  to fall from the top to the bottom of the tube.

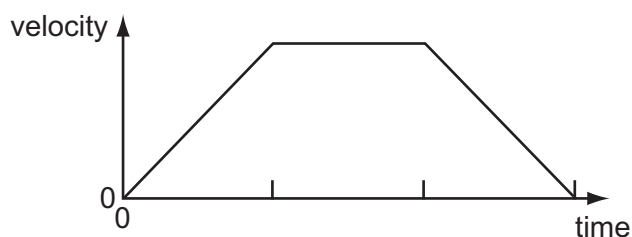
How far will the feather have fallen from the top of the tube in time  $0.50 T$ ?

- A**  $0.13 L$       **B**  $0.25 L$       **C**  $0.38 L$       **D**  $0.50 L$

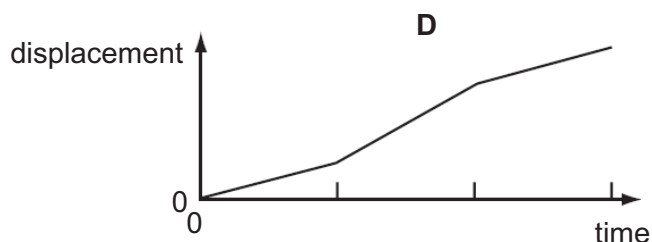
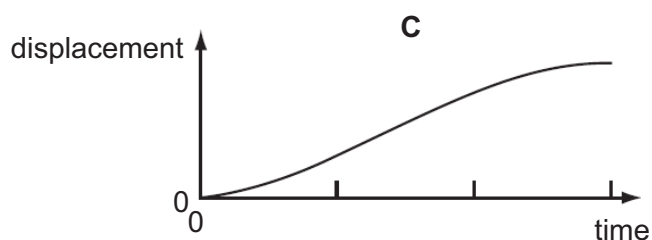
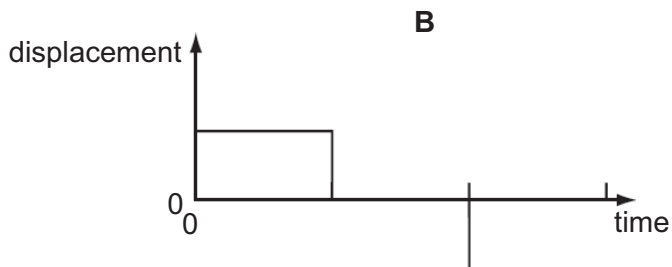
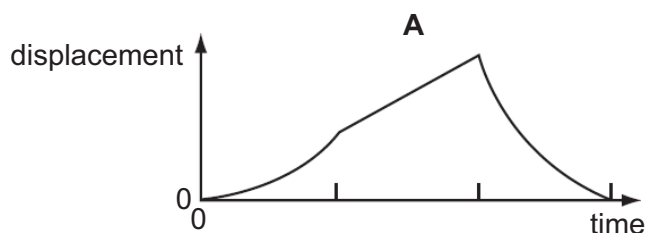


81 9702/12/O/N/12/Q9

The graph of velocity against time for an object moving in a straight line is shown.

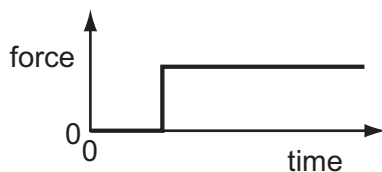


What is the corresponding graph of displacement against time?

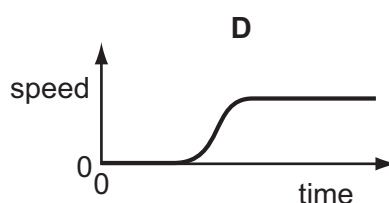
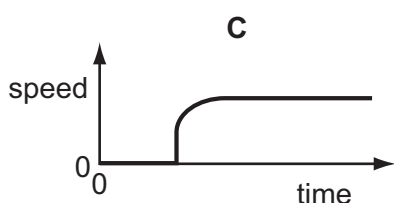
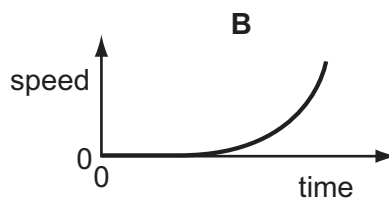
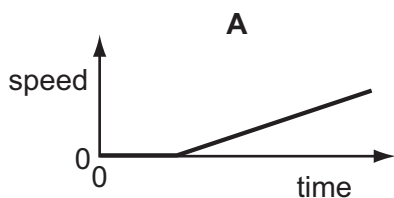


82 9702/11/M/J/13/Q9

A car is stationary at traffic lights. When the traffic lights go green, the driver presses down sharply on the accelerator. The resultant horizontal force acting on the car varies with time as shown.

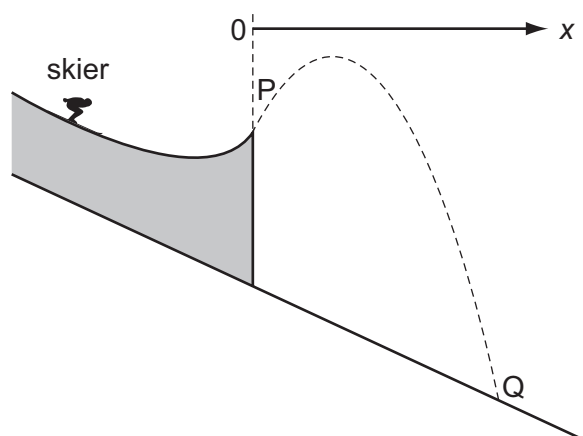


Which graph shows the variation with time of the speed of the car?

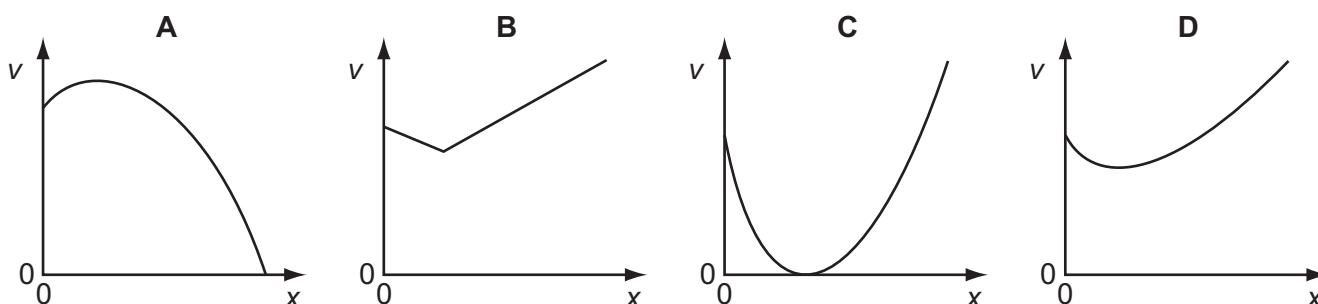


83 9702/12/O/N/12/Q10

The dotted line shows the path of a competitor in a ski-jumping competition.



Ignoring air resistance, which graph best represents the variation of his speed  $v$  with the horizontal distance  $x$  covered from the start of his jump at P before landing at Q?

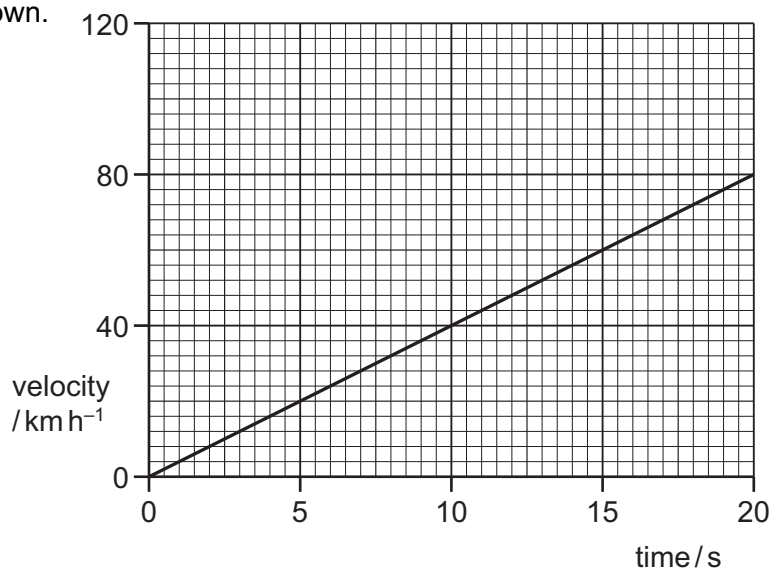


84 9702/12/O/N/12/Q11

The velocity of a car changes as shown.

What is the acceleration of the car?

- A  $1.1 \text{ ms}^{-2}$
- B  $4.0 \text{ ms}^{-2}$
- C  $224 \text{ ms}^{-2}$
- D  $800 \text{ ms}^{-2}$



85 9702/11/M/J/13/Q8

A goods train passes through a station at a steady speed of  $10 \text{ ms}^{-1}$ . An express train is at rest at the station. The express train leaves the station with a uniform acceleration of  $0.5 \text{ ms}^{-2}$  just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

How much time passes before the express train overtakes the goods train?

- A 6 s
- B 10 s
- C 20 s
- D 40 s

86 9702/13/O/N/12/Q8

A bicycle brakes so that it undergoes uniform deceleration from a speed of  $8 \text{ ms}^{-1}$  to  $6 \text{ ms}^{-1}$  over a distance of 7 m.

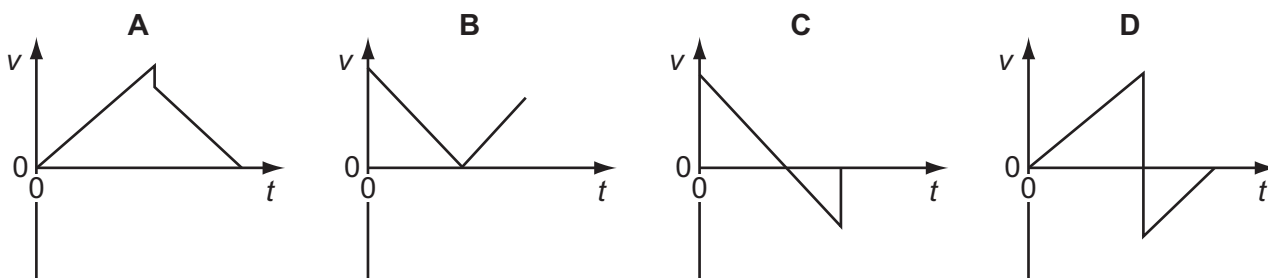
If the deceleration of the bicycle remains constant, what further distance will it travel before coming to rest?

- A 7 m                      B 9 m                      C 16 m                      D 21 m

87 9702/13/O/N/12/Q9

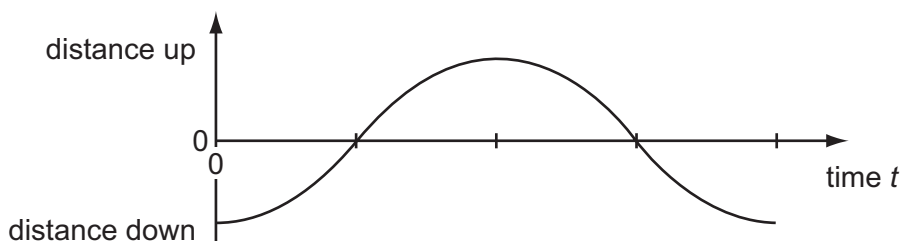
A ball is released from rest above a horizontal surface. It bounces once and is caught.

Which graph represents the variation with time  $t$  of the velocity  $v$  of the ball?

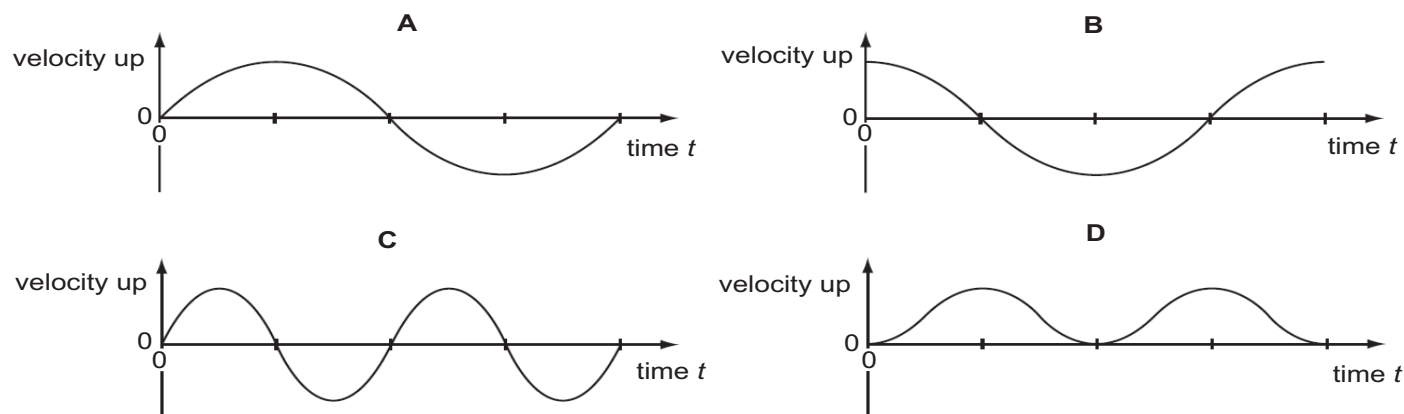


88 9702/13/O/N/12/Q10

A mass on the end of a spring bounces up and down as shown, after being released at time  $t = 0$ .



Which graph shows how the velocity varies with time?



89 9702/13/M/J/13/Q10

A stationary nucleus has nucleon number  $A$ .

The nucleus decays by emitting a proton with speed  $v$  to form a new nucleus with speed  $u$ . The new nucleus and the proton move away from one another in opposite directions.

Which equation gives  $v$  in terms of  $A$  and  $u$ ?

- A  $v = \left(\frac{A}{4} - 1\right)u$                       B  $v = (A - 1)u$                       C  $v = Au$                       D  $v = (A + 1)u$

90 9702/11/M/J/13/Q7

A body is released from rest and falls vertically in air of constant density.

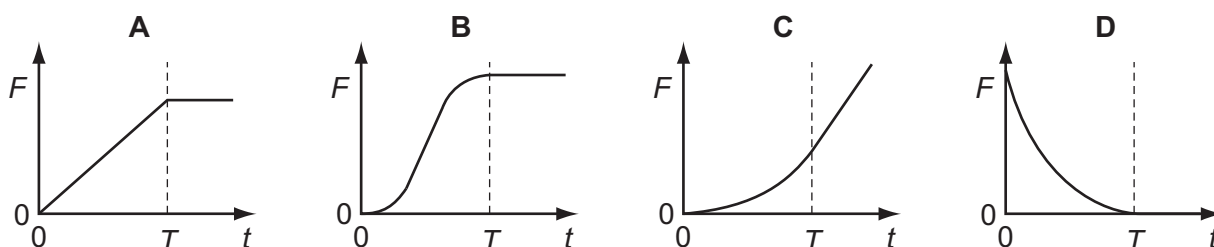
Which statement about the motion of the falling body is correct?

- A As it accelerates, its weight decreases so that its acceleration decreases until it travels with constant velocity.
- B It accelerates initially at  $9.8 \text{ m s}^{-2}$  but the drag force increases so its acceleration decreases.
- C Its velocity increases at a constant rate until its velocity becomes constant.
- D The drag force of the air increases continually and eventually the velocity decreases.

91 9702/12/M/J/13/Q6

A sky diver falls vertically from a stationary balloon. She leaves the balloon at time  $t = 0$ . At time  $t = T$ , she reaches terminal velocity. Beyond the time shown in the graphs, she opens her parachute.

Which graph shows the variation with time  $t$  of the force  $F$  due to air resistance?



92 9702/12/M/J/13/Q7

The diagram shows an arrangement to stop trains that are travelling too fast.



Trains coming from the left travel at a speed of  $50 \text{ ms}^{-1}$ . At marker 1, the driver must apply the brakes so that the train decelerates uniformly in order to pass marker 2 at no more than  $10 \text{ ms}^{-1}$ .

The train carries a detector that notes the times when the train passes each marker and will apply an emergency brake if the time between passing marker 1 and marker 2 is less than 20 s.

How far from marker 2 should marker 1 be placed?

- A 200 m
- B 400 m
- C 500 m
- D 600 m

93 9702/12/M/J/15/Q10

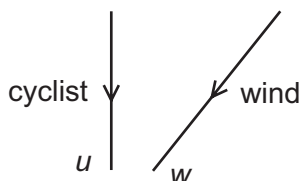
A firework rocket is fired vertically upwards. The fuel burns and produces a constant upwards force on the rocket. After 5 seconds there is no fuel left. Air resistance is negligible.

What is the acceleration before and after 5 seconds?

|   | before 5 seconds | after 5 seconds |
|---|------------------|-----------------|
| A | constant         | constant        |
| B | constant         | zero            |
| C | increasing       | constant        |
| D | increasing       | zero            |

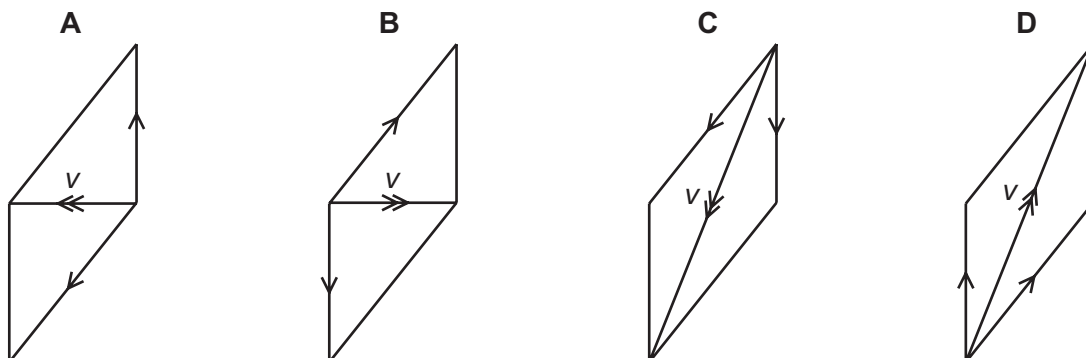
94 9702/13/M/J/13/Q4

A cyclist is travelling due south with velocity  $u$ . The wind is blowing from the north-east with velocity  $w$ .



The wind has a velocity  $v$  relative to the cyclist, where  $v = w - u$ .

Which vector diagram shows the magnitude and direction of velocity  $v$ ?

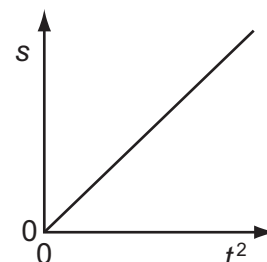


95 9702/13/M/J/13/Q8

At time  $t = 0$ , a body moves from rest with constant acceleration in a straight line. At time  $t$ , the body is distance  $s$  from its rest position. A graph is drawn of  $s$  against  $t^2$ , as shown.

Which statement describes the acceleration of the body?

- A It is equal to half the value of the gradient of the graph.
- B It is equal to the value of the gradient of the graph.
- C It is equal to twice the value of the gradient of the graph.
- D It is equal to the reciprocal of the gradient of the graph.

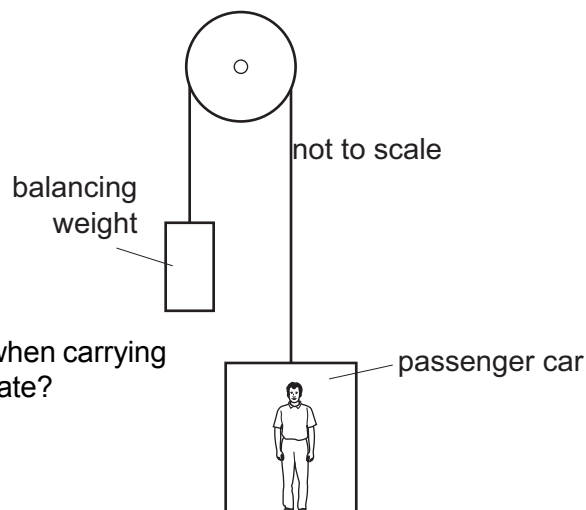


96 9702/13/M/J/13/Q9

A lift (elevator) consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.

Some masses are shown in the table.

|                  | mass / kg |
|------------------|-----------|
| passenger car    | 520       |
| balancing weight | 640       |
| passenger        | 80        |

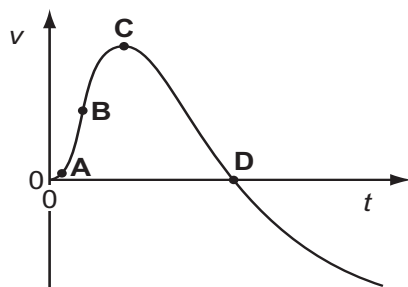


What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

- A  $0.032 \text{ ms}^{-2}$
- B  $0.32 \text{ ms}^{-2}$
- C  $0.61 \text{ ms}^{-2}$
- D  $0.65 \text{ ms}^{-2}$

97 9702/13/O/N/13/Q7

The graph shows how the velocity  $v$  of a firework rocket changes with time  $t$ .  
At which point on the graph does the rocket have the greatest acceleration?



98 9702/13/O/N/13/Q8

On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where the train stops.

The maximum deceleration of the train is  $0.20 \text{ m s}^{-2}$ .

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

- A  $14 \text{ m s}^{-1}$       B  $20 \text{ m s}^{-1}$       C  $40 \text{ m s}^{-1}$       D  $400 \text{ m s}^{-1}$

99 9702/11/M/J/14/Q6

A tennis ball is thrown horizontally in air from the top of a tall building.

If the effect of air resistance is **not** negligible, what happens to the horizontal and vertical components of the ball's velocity?

|   | horizontal component of velocity | vertical component of velocity |
|---|----------------------------------|--------------------------------|
| A | constant                         | constant                       |
| B | constant                         | increases at a constant rate   |
| C | decreases to zero                | increases at a constant rate   |
| D | decreases to zero                | increases to a maximum value   |

100 9702/11/M/J/14/Q7

An object is thrown with velocity  $5.2 \text{ m s}^{-1}$  vertically upwards on the Moon. The acceleration due to gravity on the Moon is  $1.62 \text{ m s}^{-2}$ .

What is the time taken for the object to return to its starting point?

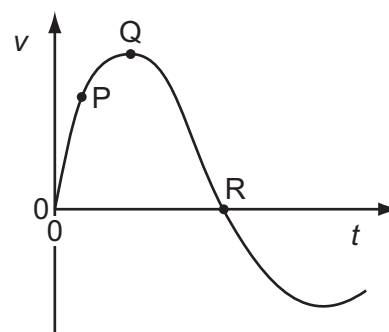
- A 2.5 s      B 3.2 s      C 4.5 s      D 6.4 s

101 9702/12/O/N/13/Q7

The graph shows how velocity  $v$  varies with time  $t$  for a bungee jumper.

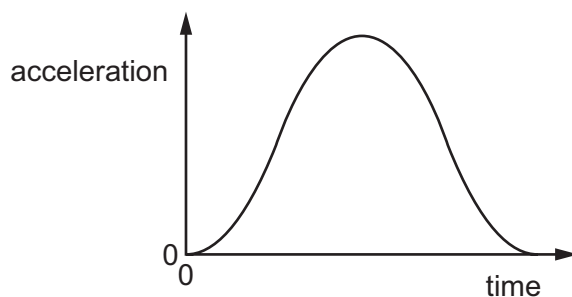
At which point is the bungee jumper momentarily at rest and at which point does she have zero acceleration?

|   | jumper at rest | jumper with zero acceleration |
|---|----------------|-------------------------------|
| A | Q              | P                             |
| B | Q              | R                             |
| C | R              | Q                             |
| D | R              | R                             |



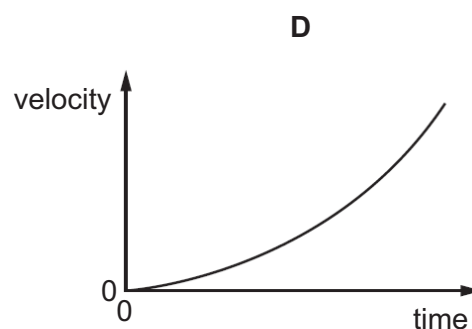
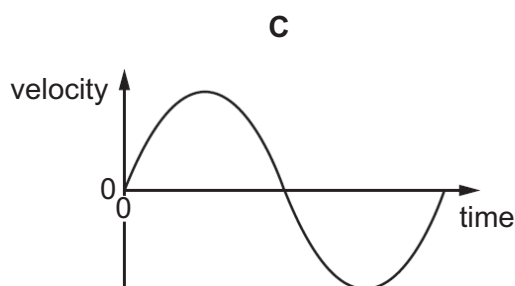
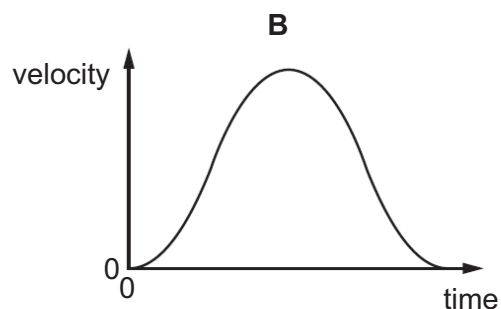
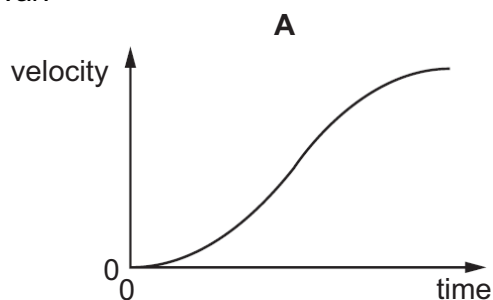
102 9702/11/M/J/14/Q8

The graph shows how the acceleration of an object moving in a straight line varies with time.



The object starts from rest.

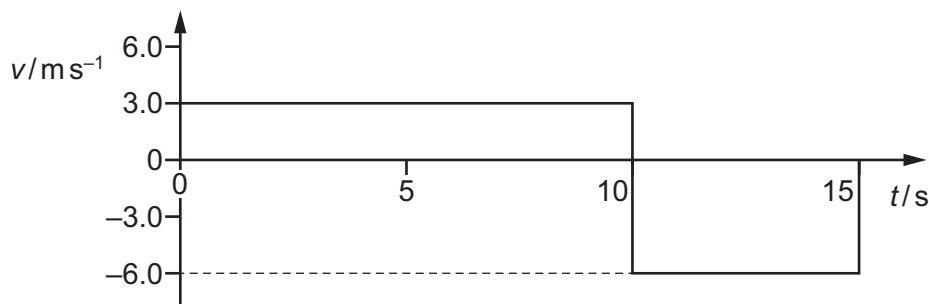
Which graph shows the variation with time of the velocity of the object over the same time interval?



103 9702/12/M/J/14/Q6

A radio-controlled toy car travels along a straight line for a time of 15 s.

The variation with time  $t$  of the velocity  $v$  of the car is shown below.

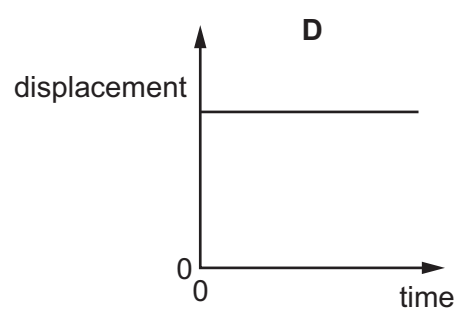
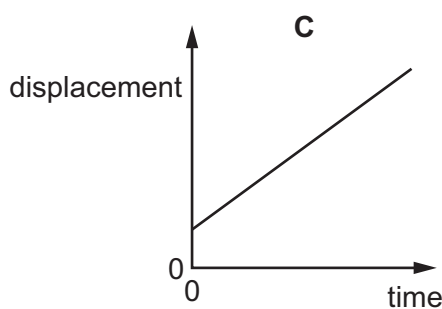
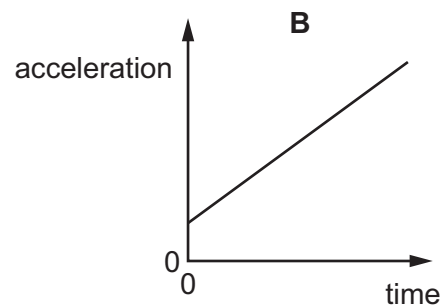
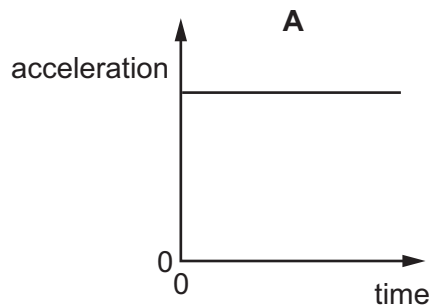


What is the average velocity of the toy car for the journey shown by the graph?

- A**  $-1.5 \text{ ms}^{-1}$       **B**  $0.0 \text{ ms}^{-1}$       **C**  $4.0 \text{ ms}^{-1}$       **D**  $4.5 \text{ ms}^{-1}$

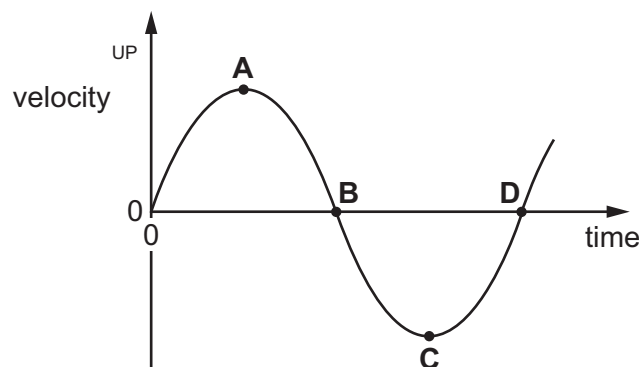
104 9702/12/O/N/14/Q6

Which graph represents the motion of a car that is travelling along a straight road with a speed that increases uniformly with time?



105 9702/13/O/N/14/Q7

The diagram shows a velocity-time graph for a mass moving up and down on the end of a spring. Which point represents the velocity of the mass when at the lowest point of its motion?



106 9702/13/M/J/15/Q9

A body falling in a uniform gravitational field encounters air resistance. The air resistance increases until terminal velocity is reached.

Which factor does **not** affect its terminal velocity?

- |                                               |                                |
|-----------------------------------------------|--------------------------------|
| <b>A</b> the density of the air               | <b>C</b> the mass of the body  |
| <b>B</b> the height from which the body falls | <b>D</b> the shape of the body |

107 9702/12/M/J/15/Q9

A sprinter runs a 100 m race in a straight line. He accelerates from the starting block at a constant acceleration of  $2.5 \text{ m s}^{-2}$  to reach his maximum speed of  $10 \text{ m s}^{-1}$ . He maintains this speed until he crosses the finish line.

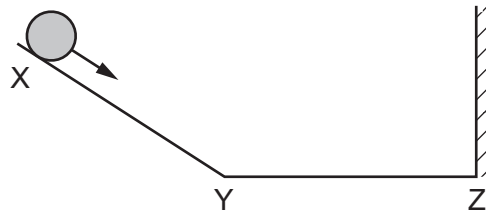
Which time does it take the sprinter to run the race?

- |              |               |               |               |
|--------------|---------------|---------------|---------------|
| <b>A</b> 4 s | <b>B</b> 10 s | <b>C</b> 12 s | <b>D</b> 20 s |
|--------------|---------------|---------------|---------------|

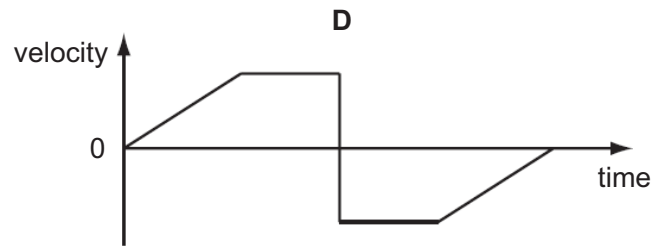
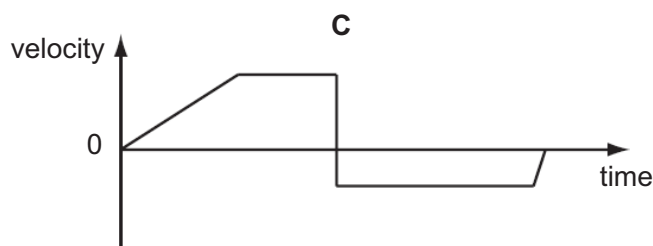
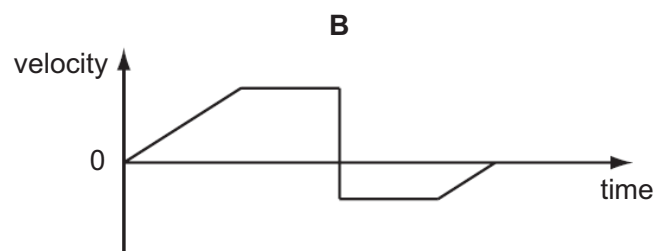
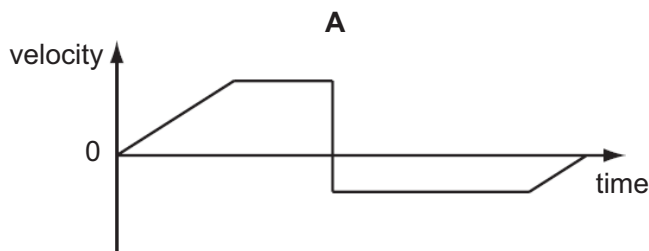
108 9702/13/O/N/14/Q8

A ball is released from rest on a smooth slope XY.

It moves down the slope, along a smooth horizontal surface YZ and rebounds inelastically at Z. Then it moves back to Y and comes to rest momentarily somewhere on XY.



Which velocity-time graph represents the motion of the ball?



109 9702/13/O/N/14/Q9

A man weighs 240 N on Mars where the acceleration of free fall  $g$  is  $4 \text{ ms}^{-2}$ . On the Moon,  $g$  is  $2 \text{ ms}^{-2}$ .

Which statement is correct?

- A** The man has a mass on Mars of 60 N.
- B** The man has a mass on the Moon of 120 kg.
- C** The man weighs 120 N on the Moon.
- D** The man weighs 240 N on the Moon.

110 9702/11/M/J/15/Q8

An insect jumps with an initial vertical velocity of  $1.0 \text{ ms}^{-1}$ , reaching a maximum height of  $3.5 \times 10^{-2} \text{ m}$ . Assume the deceleration is uniform.

What is the magnitude of the deceleration?

- A**  $3.6 \text{ ms}^{-2}$
- B**  $9.8 \text{ ms}^{-2}$
- C**  $14 \text{ ms}^{-2}$
- D**  $29 \text{ ms}^{-2}$

111 9702/11/M/J/15/Q9

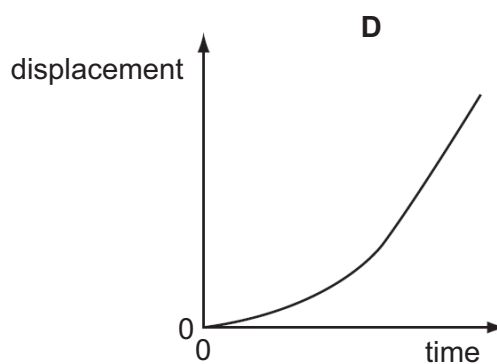
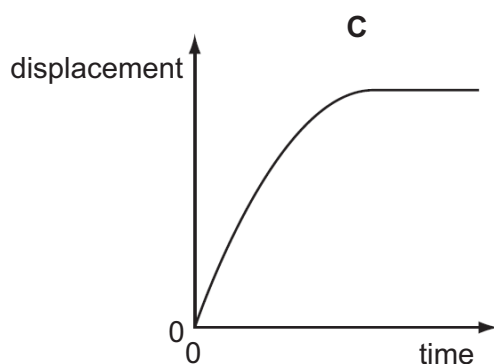
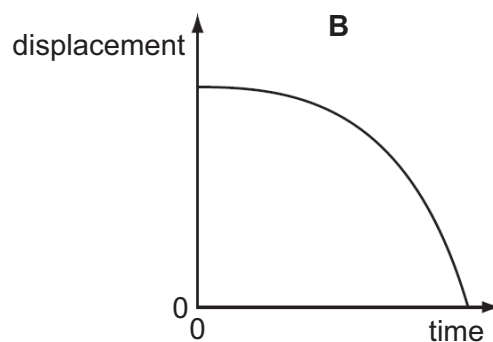
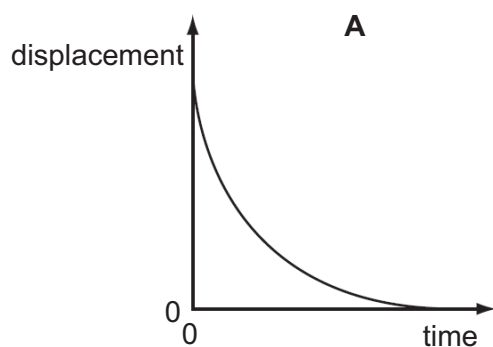
A body having uniform acceleration  $a$  increases its velocity from  $u$  to  $v$  in time  $t$ .

Which expression would **not** give a correct value for the body's displacement during time  $t$ ?

- A**  $ut + \frac{1}{2}at^2$
- B**  $vt - \frac{1}{2}at^2$
- C**  $\frac{(v+u)(v-u)}{2a}$
- D**  $\frac{(v-u)t}{2}$

**112 9702/11/M/J/15/Q7**

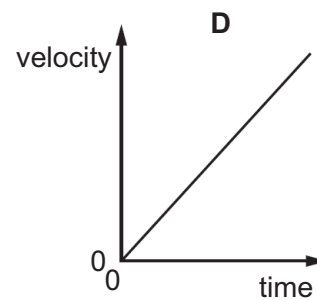
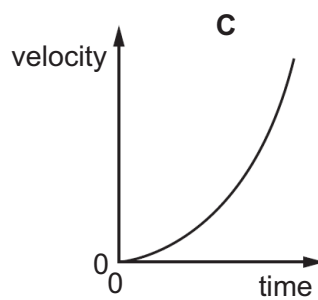
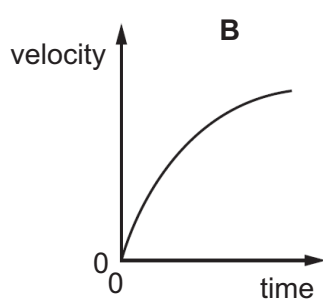
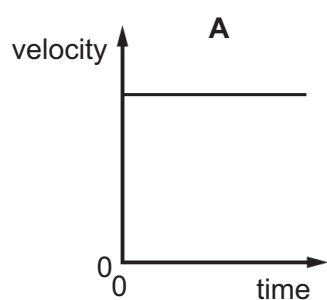
A sphere is released and falls. Its initial acceleration reduces until it eventually begins to travel at constant terminal velocity. Which displacement-time graph best represents the motion of the sphere?



**113 9702/12/M/J/15/Q8**

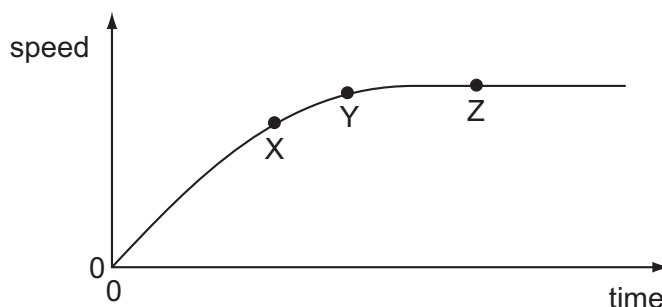
A stone is thrown horizontally from the top of a cliff. Air resistance is negligible.

Which graph shows the variation with time of the vertical component of the stone's velocity?



**114 9702/13/M/J/15/Q7**

A raindrop falls vertically from rest in air. The variation with time of the speed of the raindrop is shown in the graph.



Which statement about the raindrop is correct?

- A** At point X, the raindrop has an acceleration of  $9.81 \text{ m s}^{-2}$ .
- B** At point Z, the force on the raindrop due to air resistance has reached its maximum value and so the acceleration of the raindrop has also reached its maximum value.
- C** At point Z, the force due to air resistance is equal and opposite to the weight of the raindrop and so the speed of the raindrop is zero.
- D** The resultant force on the raindrop at point Y is less than the resultant force on the raindrop at point X.

**115 9702/11/M/J/17/Q4**

The values of displacement, velocity and acceleration of a vehicle can be deduced from graphs representing its motion. Often the areas under these graphs, or the gradients of the graphs, are used.

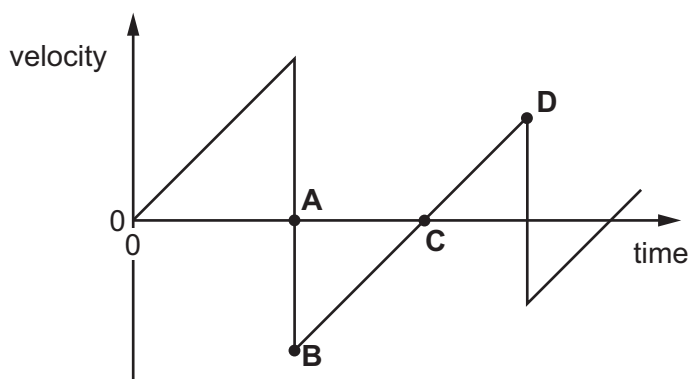
What would **not** give a value for a displacement, a velocity or an acceleration?

- A** area under a velocity-time graph
- B** gradient of a displacement-time graph
- C** gradient of a velocity-time graph
- D** gradient of an acceleration-time graph

**116 9702/11/M/J/17/Q5**

A ball is released from rest above a hard, horizontal surface. The graph shows how the velocity of the bouncing ball varies with time.

At which point on the graph does the ball reach its maximum height after the first bounce?



117 9702/11/M/J/17/Q6

A ball is kicked upwards at an angle of  $45^\circ$  to horizontal ground. After a short flight, the ball returns to the ground.

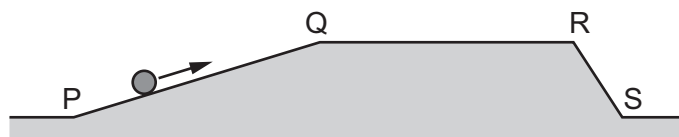
It may be assumed that air resistance is negligible.

What is **never** zero during the flight of the ball?

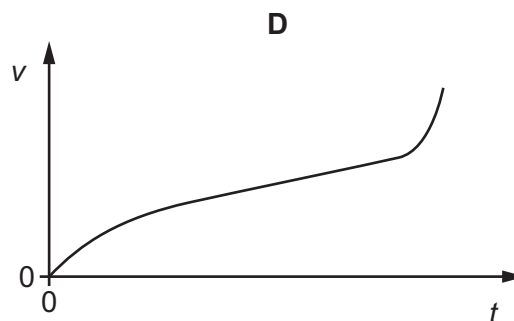
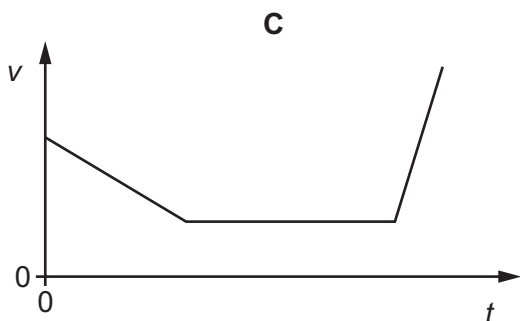
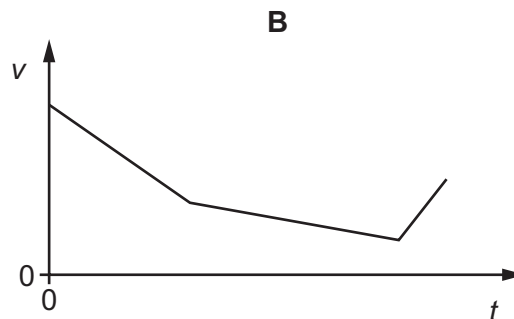
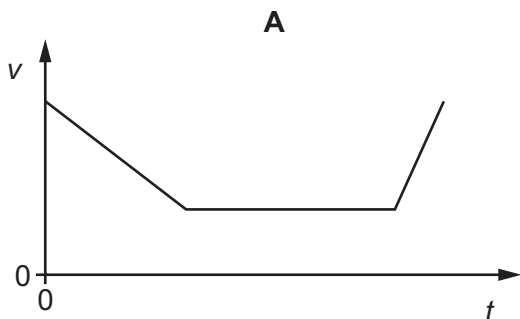
- A the horizontal component of the ball's acceleration
- B the horizontal component of the ball's velocity
- C the vertical component of the ball's momentum
- D the vertical component of the ball's velocity

118 9702/12/M/J/17/Q6

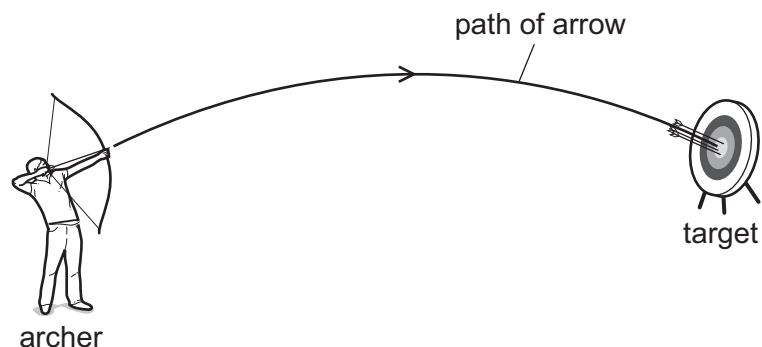
A ball is set in motion at P on a frictionless surface. It moves up slope PQ, along the horizontal surface QR and finally descends slope RS.



Which graph could represent the variation with time  $t$  of the ball's speed  $v$  as the ball moves from P to S?

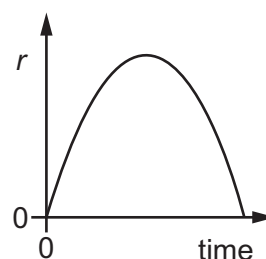
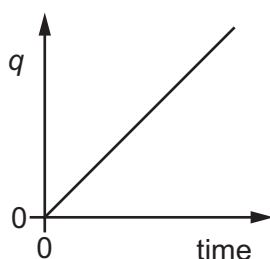
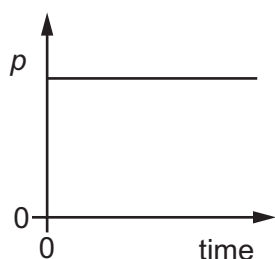


An archer shoots an arrow at a target. The diagram shows the path of the arrow.



Air resistance is assumed to be negligible.

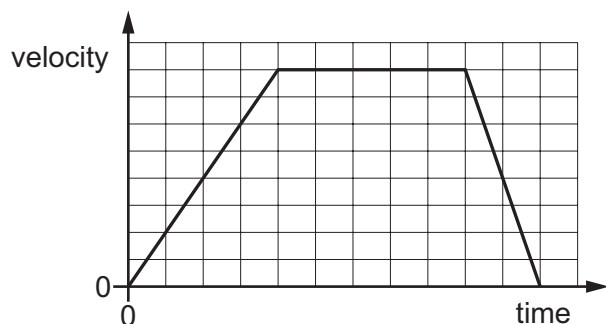
The graphs show how three different quantities  $p$ ,  $q$  and  $r$ , relating to the motion of the arrow, vary with time.



Which quantity is the horizontal component of displacement and which quantity is the vertical component of displacement of the arrow?

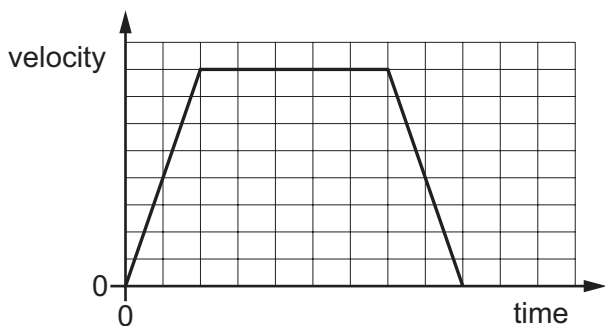
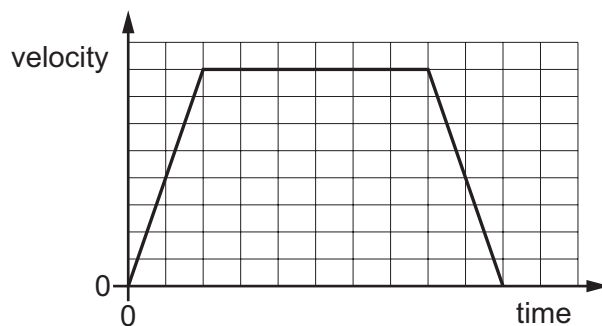
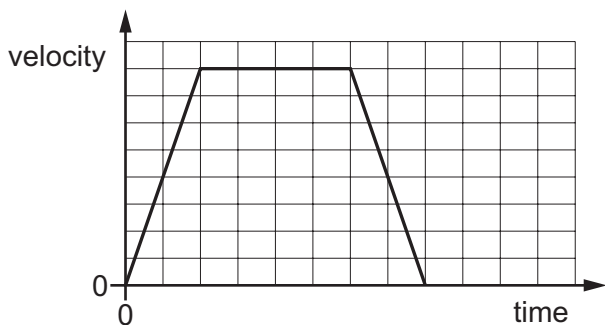
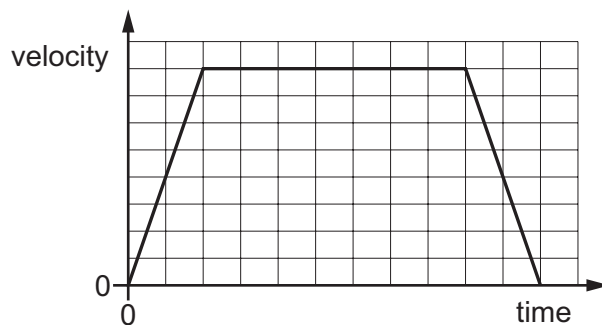
|          | horizontal<br>component of<br>displacement | vertical<br>component of<br>displacement |
|----------|--------------------------------------------|------------------------------------------|
| <b>A</b> | $p$                                        | $q$                                      |
| <b>B</b> | $q$                                        | $r$                                      |
| <b>C</b> | $r$                                        | $p$                                      |
| <b>D</b> | $r$                                        | $q$                                      |

The velocity-time graph for a train starting at one station and stopping at the next is shown.



Another train has double the acceleration but the same maximum speed and the same deceleration.

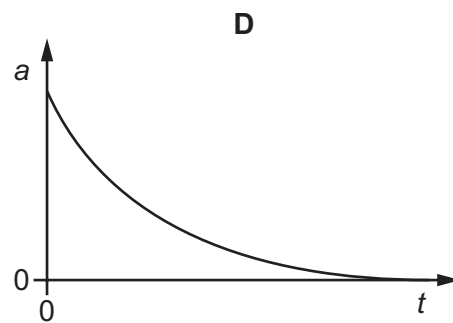
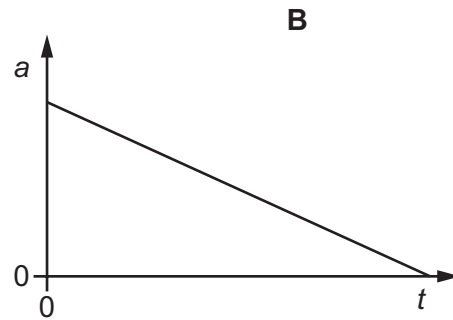
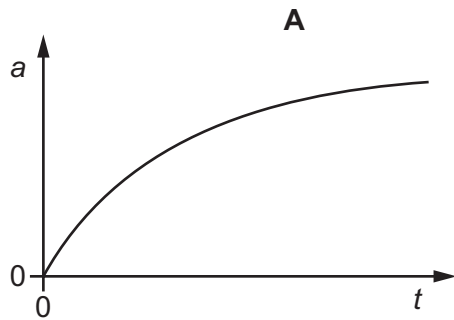
Which velocity-time graph, on the same scale, shows the motion of this train between the same stations?

**A****B****C****D**

121 9702/12/O/N/17/Q8

A stone is released from rest and falls a long distance in air.

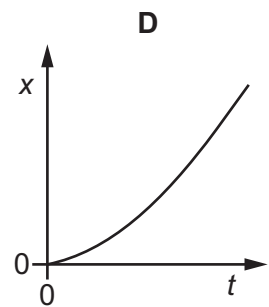
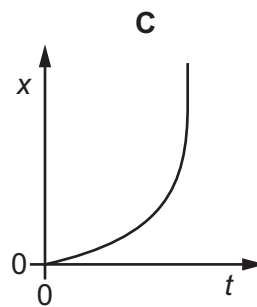
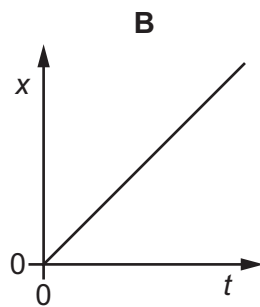
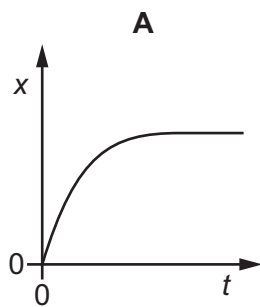
Which graph could show the variation with time  $t$  of the acceleration  $a$  of the stone?



122 9702/13/O/N/17/Q6

A football falls from the top of a tall building.

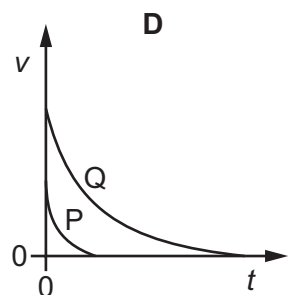
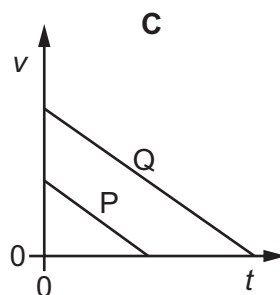
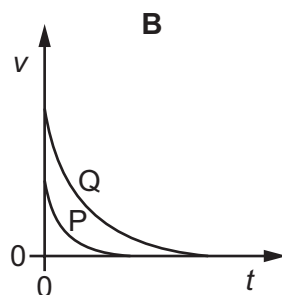
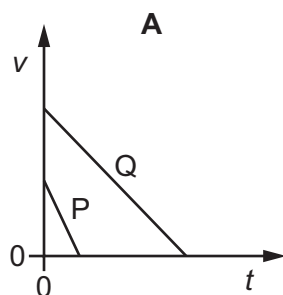
Which graph best represents the way in which the distance  $x$  fallen varies with time  $t$ ?



123 9702/13/O/N/17/Q7

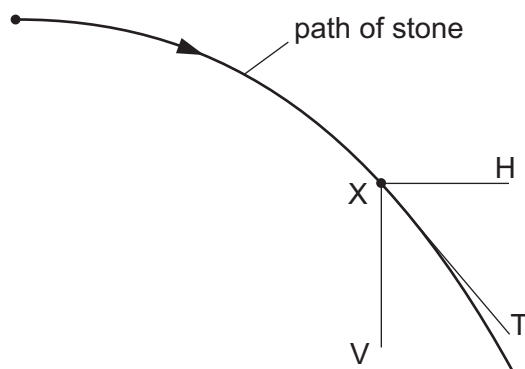
Two identical cars P and Q are travelling along a straight road. Car Q is travelling at twice the speed of car P. The brakes are applied to both cars, producing the same constant deceleration.

Which graph shows how the velocity  $v$  of each car varies with time  $t$ ?



**124 9702/13/O/N/17/Q11**

A stone is projected horizontally in a vacuum and moves along the path shown.



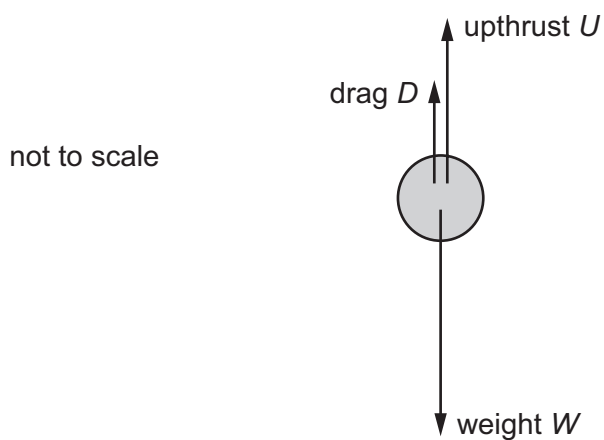
X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.

Along which directions do forces act on the stone at X?

- A** XV only      **B** XH only      **C** XV and XH      **D** XT only

**125 9702/13/O/N/17/Q13**

A solid sphere falls at constant (terminal) velocity in a liquid. The three forces acting on the sphere are shown in the diagram.



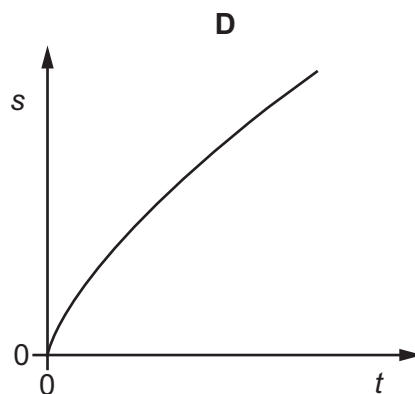
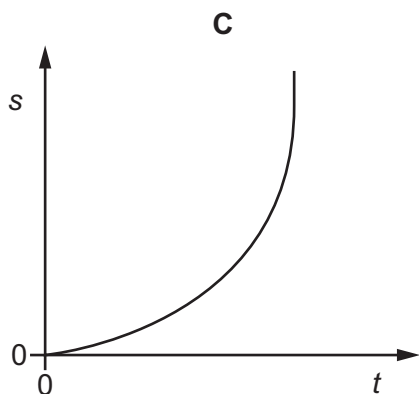
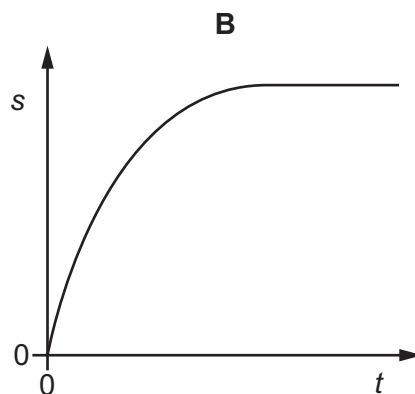
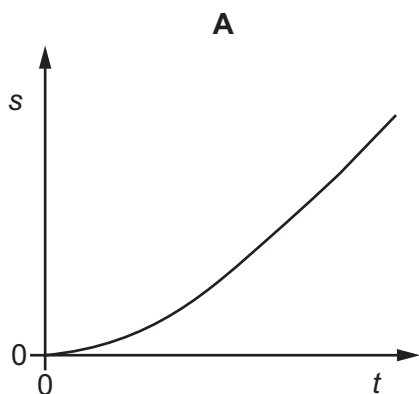
How are the three forces related?

- A**  $W + D = U$   
**B**  $W > U + D$   
**C**  $W - U = D$   
**D**  $W < D + U$

126 9702/11/M/J/18/Q6

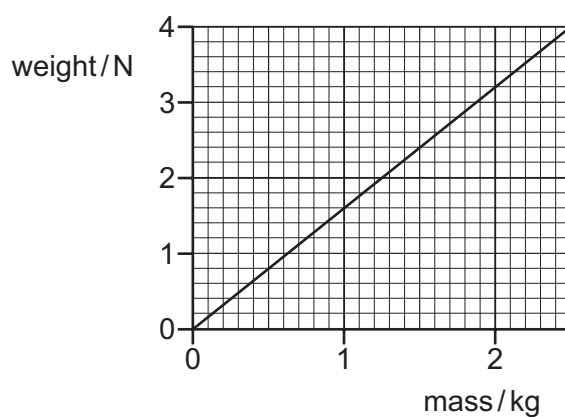
A tennis ball falls freely, in air, from the top of a tall building.

Which graph best represents the variation with time  $t$  of the distance  $s$  fallen?



127 9702/11/M/J/18/Q7

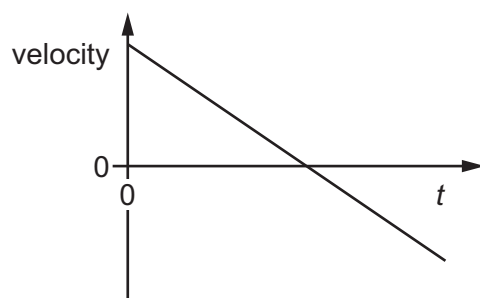
The graph shows the variation with mass of the weight of objects on a particular planet.



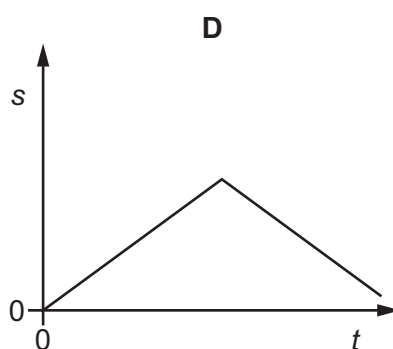
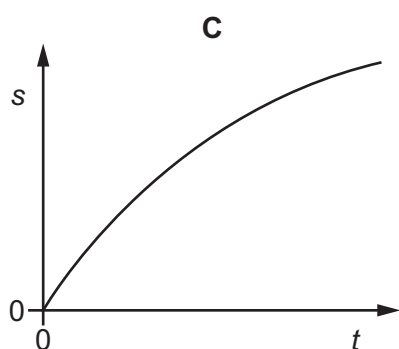
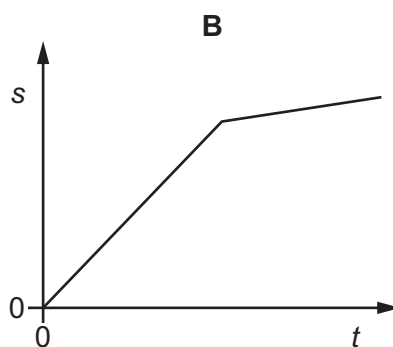
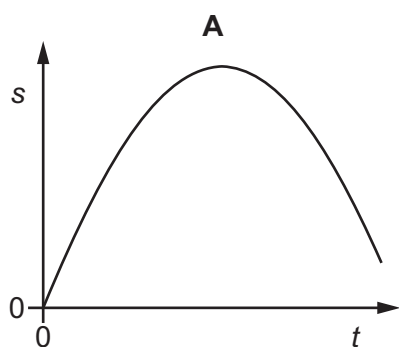
What is the value of the acceleration of free fall on the planet?

- A**  $0.63\text{ ms}^{-2}$     **B**  $1.6\text{ ms}^{-2}$     **C**  $3.2\text{ ms}^{-2}$     **D**  $9.8\text{ ms}^{-2}$

The velocity of an object changes with time  $t$  as shown.

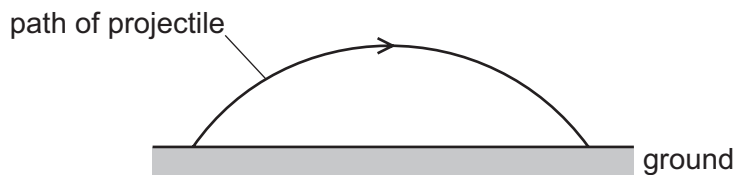


Which graph best shows the variation with time  $t$  of the displacement  $s$  of the object?



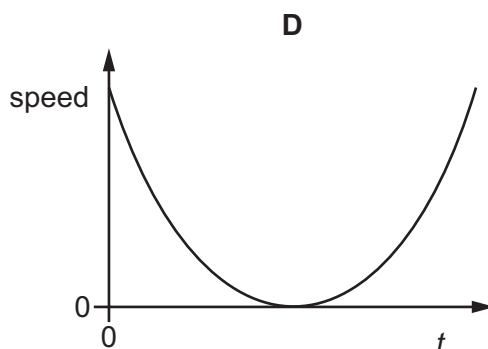
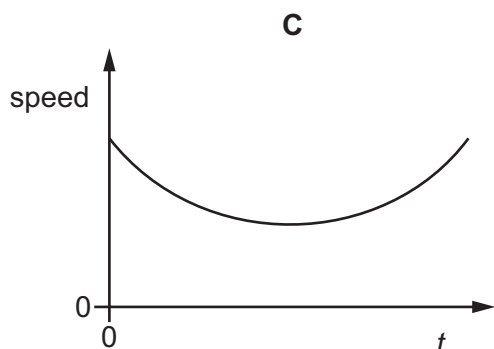
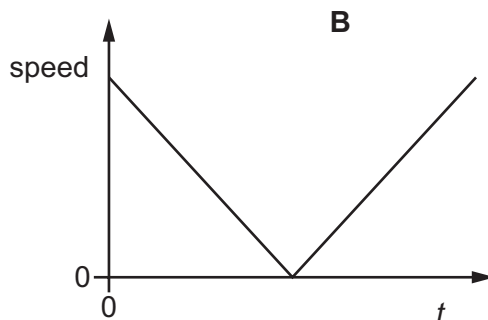
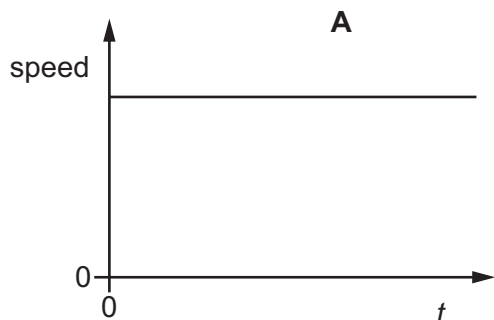
129 9702/12/M/J/18/Q6

A projectile is launched at an angle to the horizontal at time  $t = 0$ . It travels over horizontal ground, as shown.



Assume that air resistance is negligible.

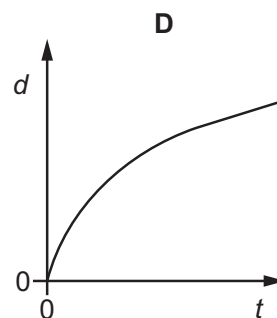
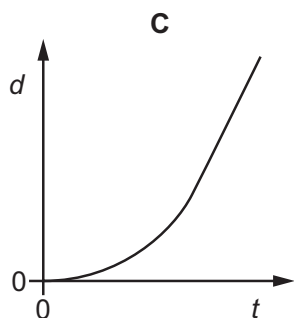
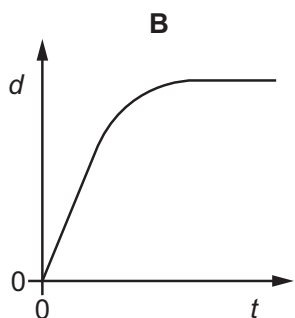
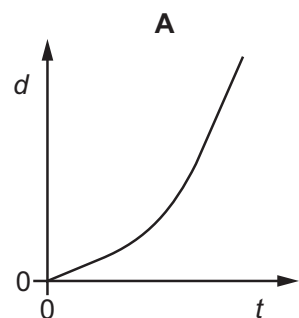
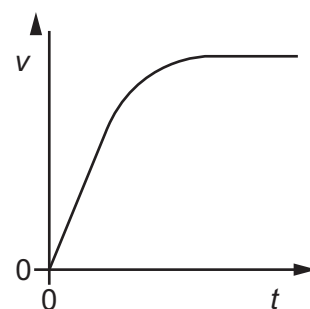
Which graph best shows the variation with  $t$  of the speed of the projectile from when it is launched to when it lands on the ground?



130 9702/12/M/J/18/Q7

A sky-diver falls vertically from a helicopter and reaches constant (terminal) velocity. The graph shows the variation with time  $t$  of the speed  $v$  of the sky-diver.

Which graph shows the variation with time  $t$  of the distance  $d$  fallen by the sky-diver?



**131 9702/13/M/J/18/Q6**

A rock on the surface of Mars is projected vertically upwards with an initial speed of  $9.4 \text{ ms}^{-1}$ . The rock rises to a height of 12 m above the surface.

Assume there is no atmosphere on Mars.

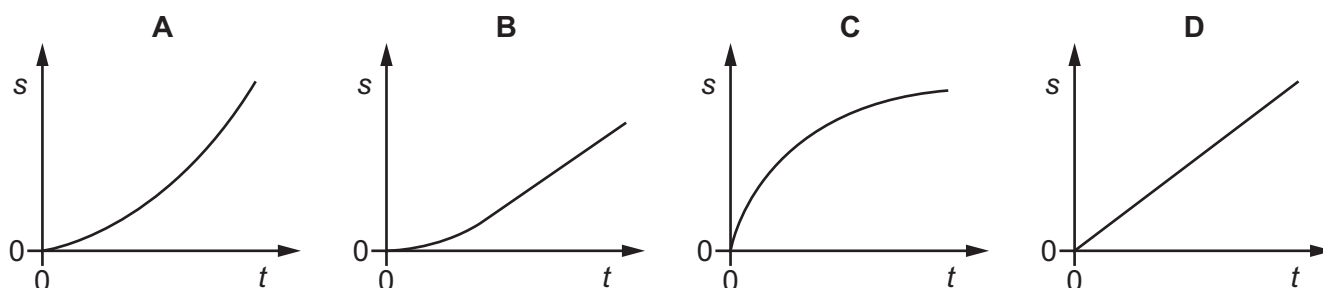
What is the acceleration of free fall near the surface of Mars?

- A**  $0.39 \text{ ms}^{-2}$       **B**  $3.7 \text{ ms}^{-2}$       **C**  $7.4 \text{ ms}^{-2}$       **D**  $9.8 \text{ ms}^{-2}$

**132 9702/13/M/J/18/Q8**

A sky-diver falls from a stationary balloon at time  $t = 0$ . As the sky-diver falls, her speed and the air resistance increase until the force of the air resistance is equal to her weight.

Which graph best shows the variation with time  $t$  of the displacement  $s$  for the motion of the sky-diver?



**133 9702/12/F/M/18/Q7**

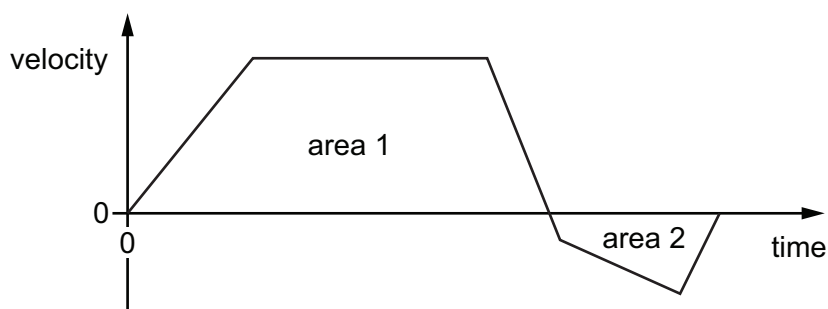
A stone of mass  $m$  is dropped from a tall building. There is significant air resistance. The acceleration of free fall is  $g$ .

When the stone is falling at a constant (terminal) velocity, which information is correct?

|          | magnitude of the acceleration of the stone | magnitude of the force of gravity on the stone | magnitude of the force of air resistance on the stone |
|----------|--------------------------------------------|------------------------------------------------|-------------------------------------------------------|
| <b>A</b> | $g$                                        | zero                                           | $mg$                                                  |
| <b>B</b> | zero                                       | $mg$                                           | $mg$                                                  |
| <b>C</b> | zero                                       | zero                                           | $mg$                                                  |
| <b>D</b> | zero                                       | $mg$                                           | zero                                                  |

134 9702/12/F/M/18/Q8

The velocity-time graph for an object is shown.



How can the total displacement of the object be determined?

- A area 1 – area 2    B  $\frac{(\text{area 1} + \text{area 2})}{2}$     C area 1 + area 2    D area 2 – area 1

135 9702/12/F/M/18/Q9

A girl throws a ball vertically upwards. It takes a time of 3.20 s to return to her hand.

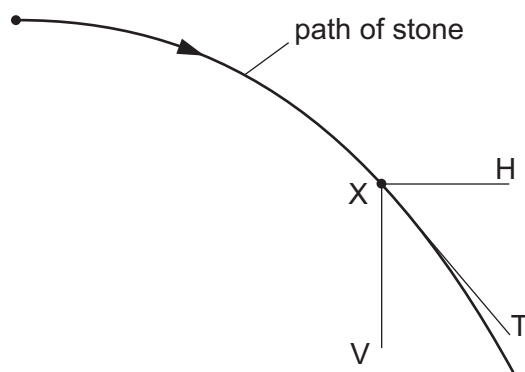
Assume air resistance is negligible.

What is the initial speed with which the ball is thrown?

- A  $3.07 \text{ ms}^{-1}$     B  $7.85 \text{ ms}^{-1}$     C  $15.7 \text{ ms}^{-1}$     D  $31.4 \text{ ms}^{-1}$

136 9702/13/O/N/17/Q11

A stone is projected horizontally in a vacuum and moves along the path shown.



X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.

Along which directions do forces act on the stone at X?

- A XV and XH    B XV only    C XH only    D XT only

137 9702/11/O/N/18/Q6

A tennis ball is thrown horizontally in air from the top of a tall building.

The effect of air resistance is **not** negligible.

What happens to the horizontal and to the vertical components of the ball's velocity?

|          | horizontal component of velocity | vertical component of velocity |
|----------|----------------------------------|--------------------------------|
| <b>A</b> | constant                         | constant                       |
| <b>B</b> | constant                         | increases at a constant rate   |
| <b>C</b> | decreases to zero                | increases at a constant rate   |
| <b>D</b> | decreases to zero                | increases to a maximum value   |

138 9702/12/O/N/18/Q6

A sprinter runs a 100 m race. The sprinter has a constant acceleration from rest of  $2.5 \text{ ms}^{-2}$  until reaching a speed of  $10 \text{ ms}^{-1}$ . The speed then remains constant until the end of the race.

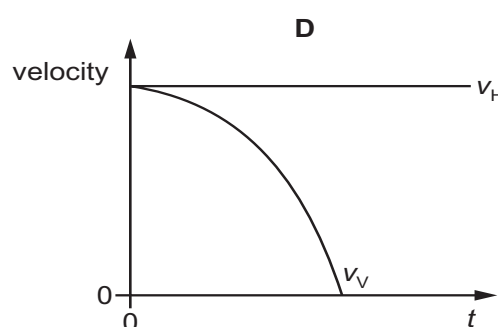
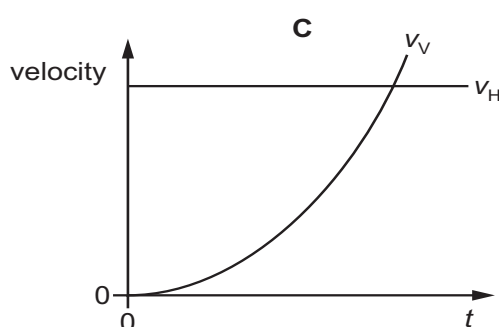
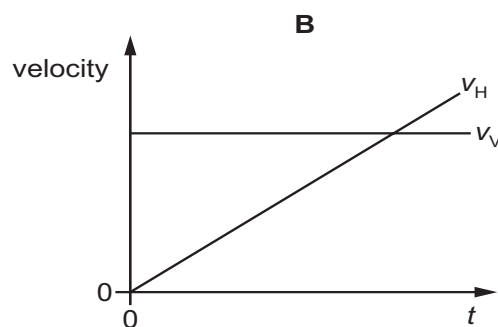
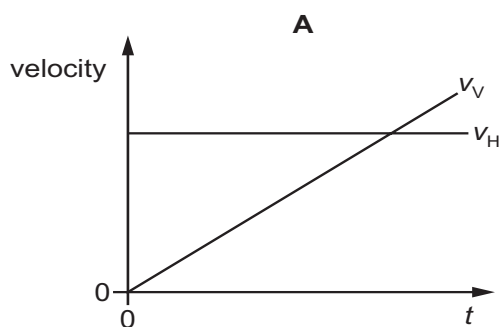
Which time does it take the sprinter to run the race?

- A** 8.9 s      **B** 10 s      **C** 12 s      **D** 14 s

139 9702/13/O/N/18/Q6

A stone is projected horizontally at time  $t = 0$  and falls. Air resistance is negligible. The stone has a horizontal component of velocity  $v_H$  and a vertical component of velocity  $v_V$ .

Which graph shows how  $v_H$  and  $v_V$  vary with time  $t$ ?



140 9702/13/O/N/18/Q8

A mass is placed on a frictionless slope inclined at  $30^\circ$  to the horizontal. The mass is then released.

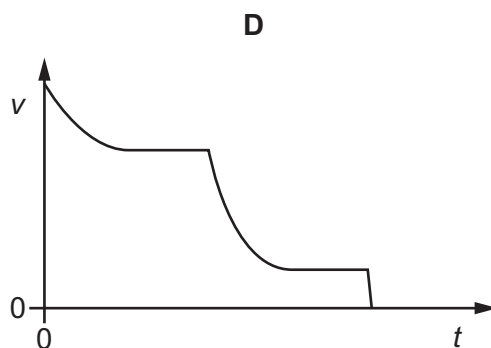
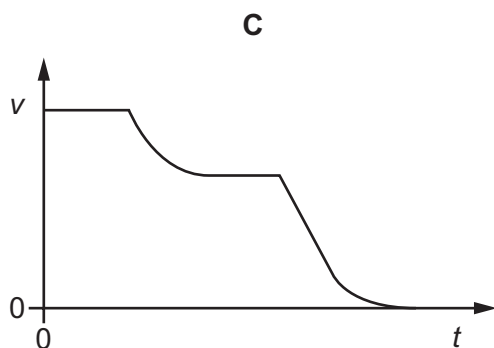
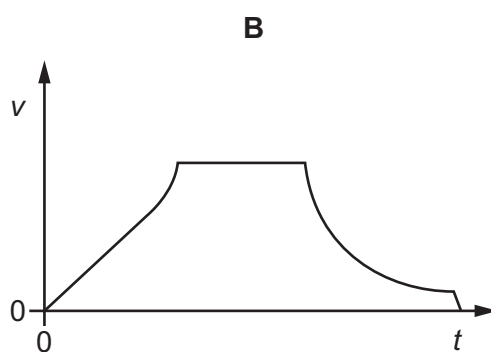
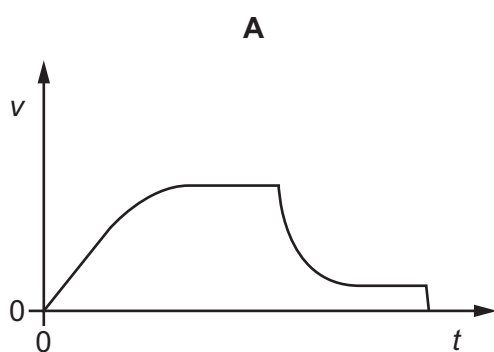
What is its acceleration down the slope?

- A  $4.9 \text{ ms}^{-2}$       B  $5.7 \text{ ms}^{-2}$       C  $8.5 \text{ ms}^{-2}$       D  $9.8 \text{ ms}^{-2}$

141 9702/13/O/N/18/Q9

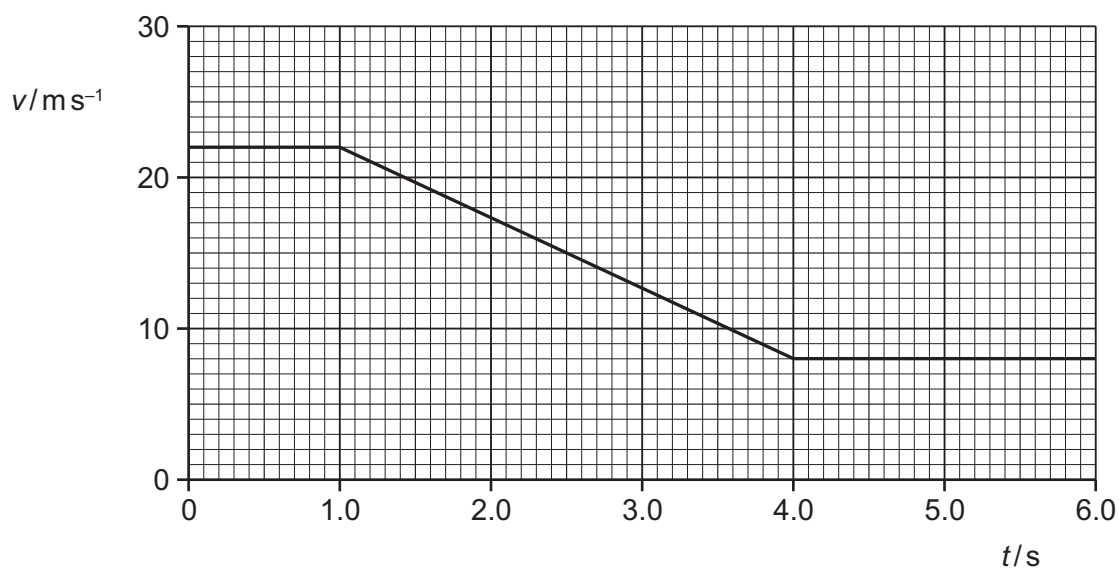
A parachutist falls vertically from rest at time  $t = 0$  from a hot-air balloon. She falls for some distance before opening her parachute.

Which graph best shows the variation with time  $t$  of the speed  $v$  of the parachutist?



142 9702/11/M/J/19/Q6

A car travels along a straight horizontal road. The graph shows the variation of the velocity  $v$  of the car with time  $t$  for 6.0 s of its journey.



The brakes of the car are applied from  $t = 1.0$  s to  $t = 4.0$  s.

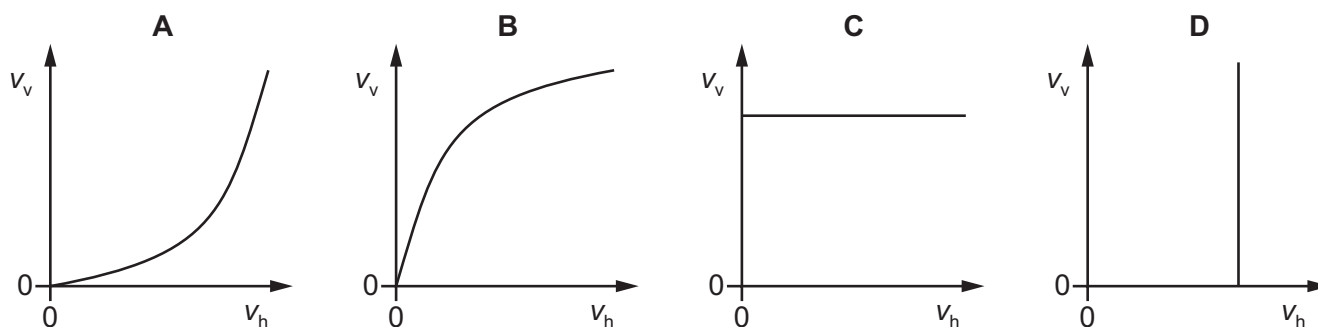
How far does the car travel while the brakes are applied?

- A 21 m                      B 45 m                      C 67 m                      D 83 m

143 9702/11/M/J/19/Q7

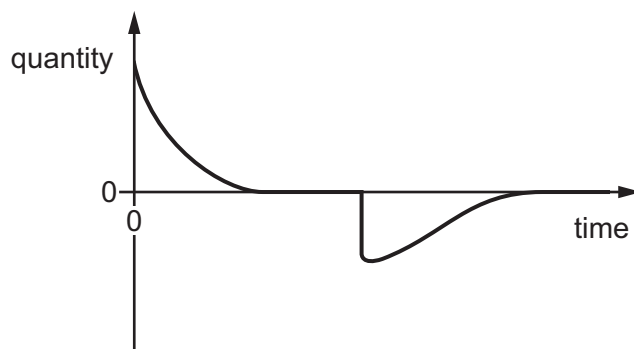
A stone is thrown horizontally from the top of a cliff and falls into the sea some time later. Air resistance is negligible.

Which graph shows how the vertical component  $v_v$  of velocity of this stone varies with its horizontal component  $v_h$  of velocity as it moves through the air?



144 9702/12/M/J/19/Q8

The graph shows how a physical quantity varies with time.

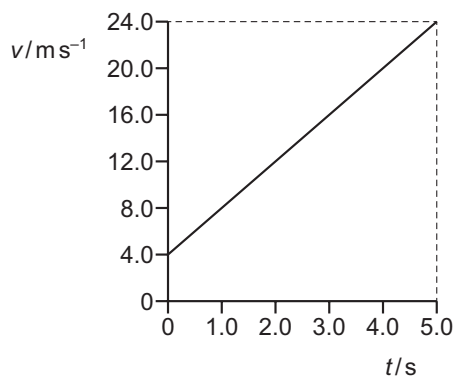


Which event could best be represented by the graph?

- A** the acceleration of a firework rising to a maximum height and falling to the ground
- B** the acceleration of a skydiver leaving an aircraft, falling, opening a parachute and falling to the ground
- C** the speed of a javelin as it leaves an athlete's hand, falls and sinks into the ground
- D** the speed of a high jump athlete leaving the ground, jumping over a bar and descending to the ground

145 9702/13/M/J/19/Q7

The graph shows the variation of velocity  $v$  with time  $t$  for an object.



The object passes a fixed point at time  $t = 0$ .

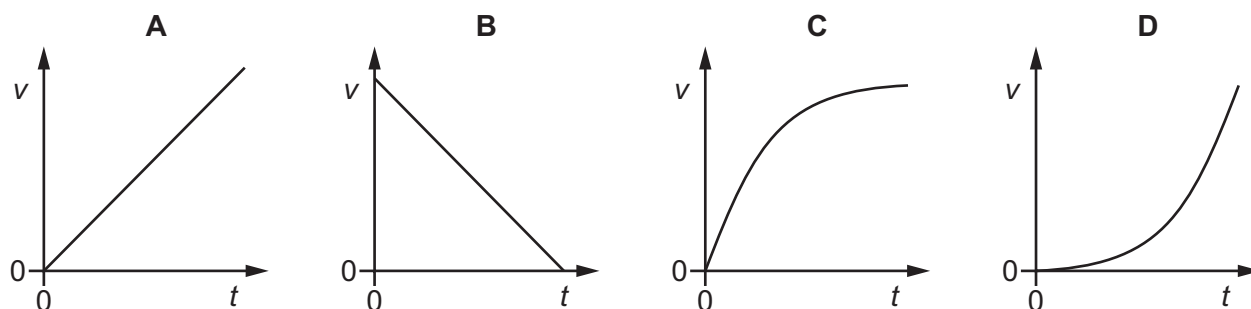
What is the displacement of the object from the fixed point at time  $t = 5.0$ s and what is the acceleration of the object?

|          | displacement<br>/m | acceleration<br>/ms <sup>-2</sup> |
|----------|--------------------|-----------------------------------|
| <b>A</b> | 60                 | 4.0                               |
| <b>B</b> | 70                 | 4.0                               |
| <b>C</b> | 60                 | 4.8                               |
| <b>D</b> | 70                 | 4.8                               |

**146 9702/13/M/J/19/Q8**

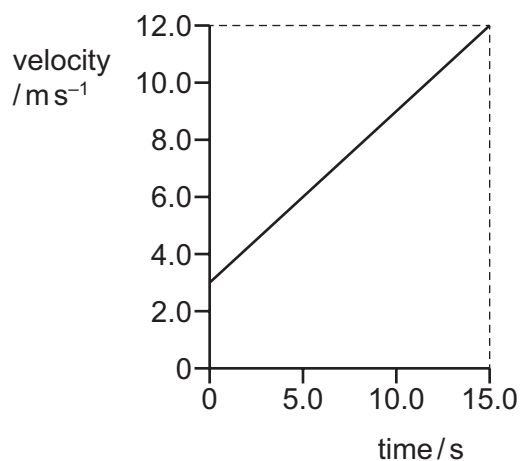
A skydiver jumps from an aeroplane and falls vertically through the air.

Which graph shows the variation with time  $t$  of the skydiver's vertical velocity  $v$ ?



**147 9702/11/O/N/19/Q6**

The velocity-time graph for an object of mass 2.5 kg is shown.



What is the resultant force acting on the object?

- A** 0.60 N      **B** 0.80 N      **C** 1.5 N      **D** 2.0 N

**148 9702/12/O/N/19/Q6**

A ball is thrown vertically upwards from ground level and reaches a maximum height of 12.7 m before falling back to ground level.

Assume air resistance is negligible.

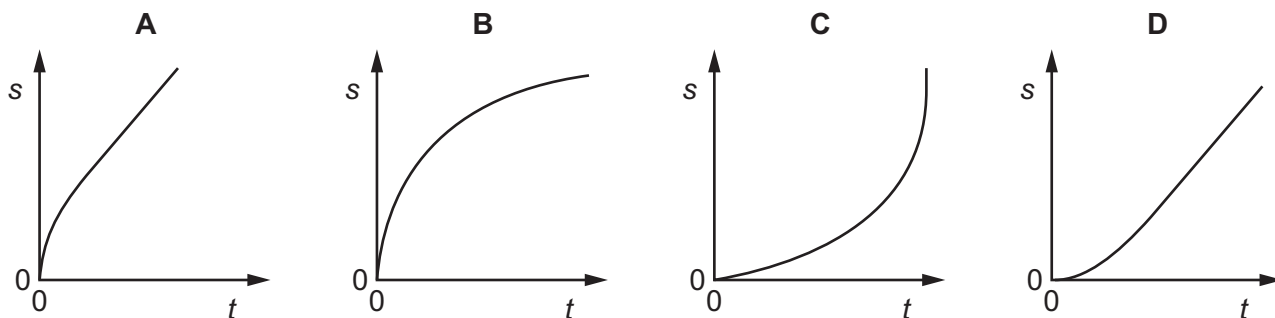
What is the total time for which the ball is in the air?

- A** 1.61 s      **B** 3.22 s      **C** 3.88 s      **D** 5.18 s

**149 9702/12/O/N/19/Q8**

A tennis ball is released from rest at time  $t = 0$  and falls through air for a long time.

Which graph of its displacement  $s$  against time  $t$  best represents the motion of the ball?



**150 9702/13/O/N/19/Q6**

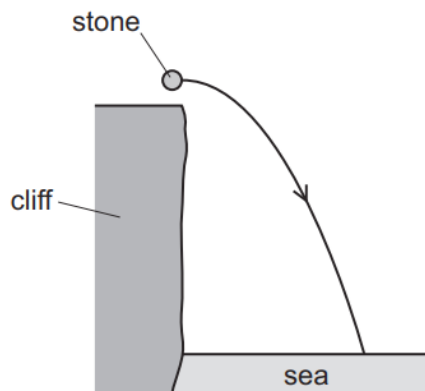
A lead sphere is released from rest at point X, a long way above the surface of a planet. The sphere falls in a vacuum. After a time of 4.0 s, it has fallen through a vertical distance of 3.0 m. Assume the acceleration of free fall is constant.

How far will the sphere have fallen from point X at a time of 20 s after its release?

- A** 15 m      **B** 75 m      **C** 80 m      **D** 2000 m

**151. 9702/12/F/M/20/Q6**

A stone is thrown horizontally from the top of a cliff and falls into the sea below. Air resistance is negligible. The path of the stone is shown.



In which direction does the resultant force on the stone act during its fall?

- A** horizontally to the right
- B** parallel to its velocity
- C** perpendicular to its velocity
- D** vertically downwards

**152. 9702/12/F/M/20/Q7**

A car moves with uniform acceleration along a straight road. Oil leaks from the car at the rate of one drop every two seconds. The diagram shows the distances between three successive oil drops on the road.



What is the acceleration of the car?

- A**  $0.75 \text{ ms}^{-2}$       **B**  $1.5 \text{ ms}^{-2}$       **C**  $3.0 \text{ ms}^{-2}$       **D**  $6.0 \text{ ms}^{-2}$

**153. 9702/12/F/M/20/Q8**

A person of mass 60 kg stands on accurate bathroom scales, placed on the floor of an elevator (lift) which operates in a tall building.

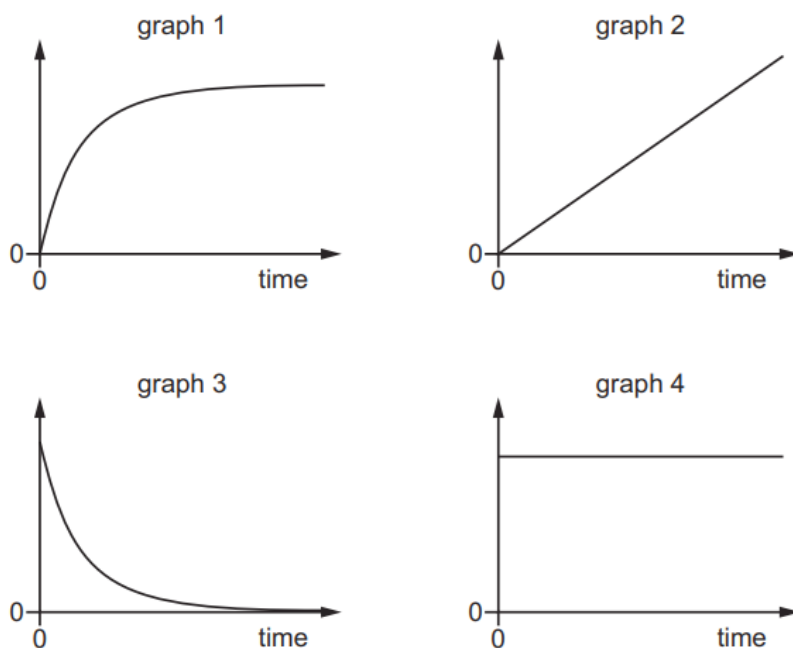
At a certain instant the bathroom scales read 58 kg.

Which row could give the person's direction of movement and type of motion?

|          | direction | motion         |
|----------|-----------|----------------|
| <b>A</b> | downwards | constant speed |
| <b>B</b> | downwards | slowing down   |
| <b>C</b> | upwards   | constant speed |
| <b>D</b> | upwards   | slowing down   |

**154. 9702/12/F/M/20/Q9**

The diagram shows graphs of various quantities plotted against time for an object dropped from a stationary balloon high in the atmosphere.

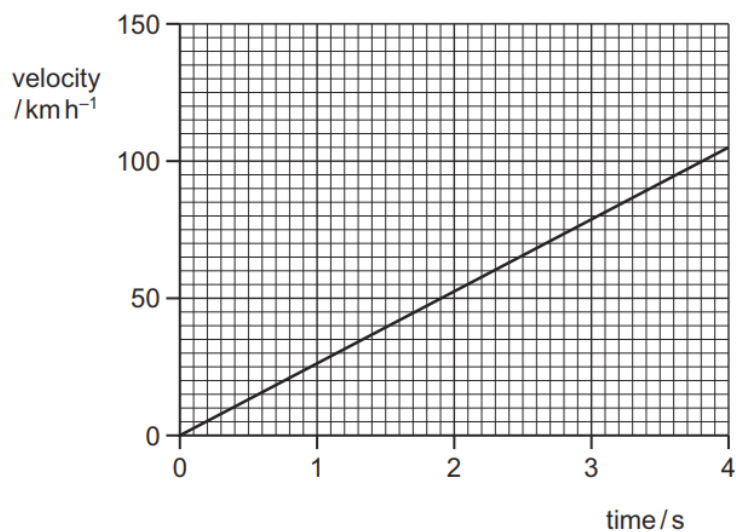


Which statement could be correct?

- A** Graph 1 is acceleration against time and graph 3 is resultant force against time.
- B** Graph 1 is acceleration against time and graph 4 is resultant force against time.
- C** Graph 3 is acceleration against time and graph 1 is velocity against time.
- D** Graph 3 is acceleration against time and graph 2 is velocity against time.

**155. 9702/11/M/J/20/Q6**

The velocity of an electric car changes as shown.

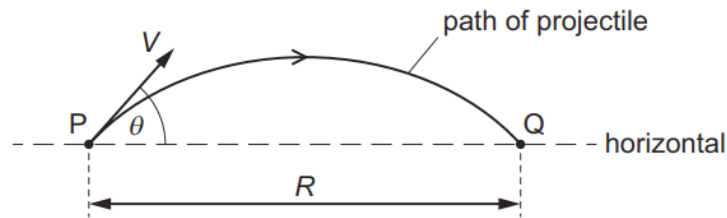


What is the acceleration of the car?

- A**  $210 \text{ ms}^{-2}$
- B**  $58 \text{ ms}^{-2}$
- C**  $26 \text{ ms}^{-2}$
- D**  $7.3 \text{ ms}^{-2}$

**156. 9702/11/M/J/20/Q7**

A projectile is fired from point P with velocity  $V$  at an angle  $\theta$  to the horizontal. It lands at point Q, a horizontal distance  $R$  from P, after time  $T$ .



The acceleration of free fall is  $g$ . Air resistance is negligible.

Which equation is correct?

- A**  $R = VT \cos \theta$
- B**  $R = VT \sin \theta$
- C**  $R = VT \cos \theta - \frac{1}{2} g T^2$
- D**  $R = VT \sin \theta - \frac{1}{2} g T^2$

**157. 9702/11/M/J/20/Q9**

The resultant force  $F$  on a raindrop of mass  $m$  falling with velocity  $v$  is given by the equation

$$F = mg - kv^2$$

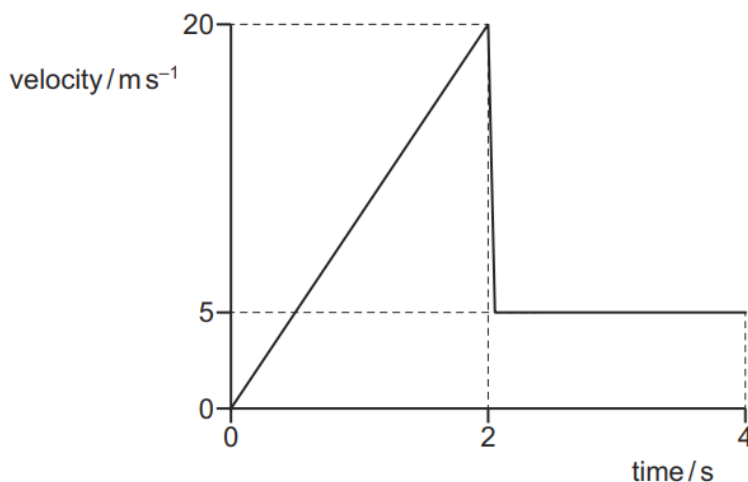
where  $k$  is a constant and  $g$  is the acceleration of free fall.

What is the velocity of the raindrop when it reaches a constant (terminal) velocity?

- A**  $\sqrt{\frac{k}{mg}}$
- B**  $\frac{k}{mg}$
- C**  $\sqrt{\frac{mg}{k}}$
- D**  $\frac{mg}{k}$

**158. 9702/12/M/J/20/Q5**

A stone is dropped from a height of 20 m above water. The graph shows the variation with time of the velocity of the stone.

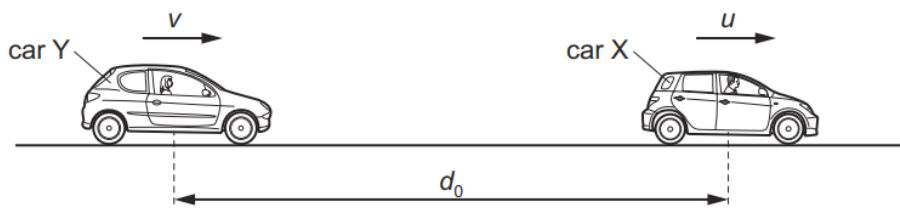


Which statement describes the approximate position of the stone four seconds after it is dropped?

- A** It is at a distance of 10 m above the surface of the water.
- B** It is at a distance of 10 m below the surface of the water.
- C** It is at a distance of 20 m below the surface of the water.
- D** It is at a distance of 30 m below the surface of the water.

**159. 9702/12/M/J/20/Q6**

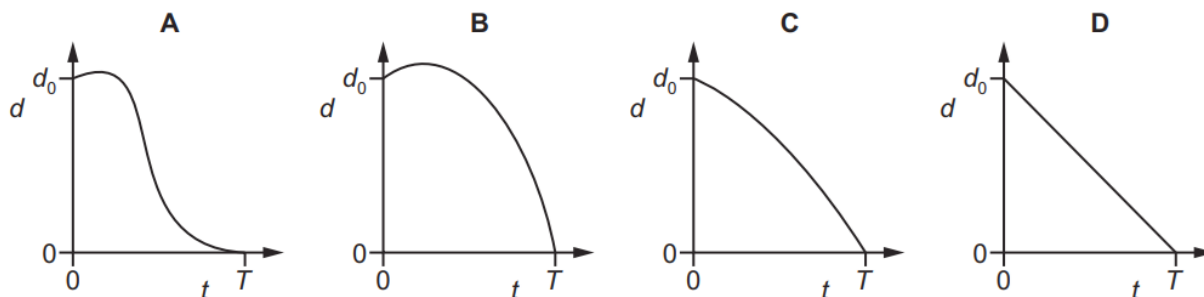
A car X is travelling at a constant speed  $u$  along a straight road. At time  $t = 0$  a second car Y is a distance  $d_0$  behind car X and travelling at a speed  $v$  in the same direction. Speed  $v$  is less than speed  $u$ .



At time  $t = 0$  car Y begins to accelerate with a constant acceleration.

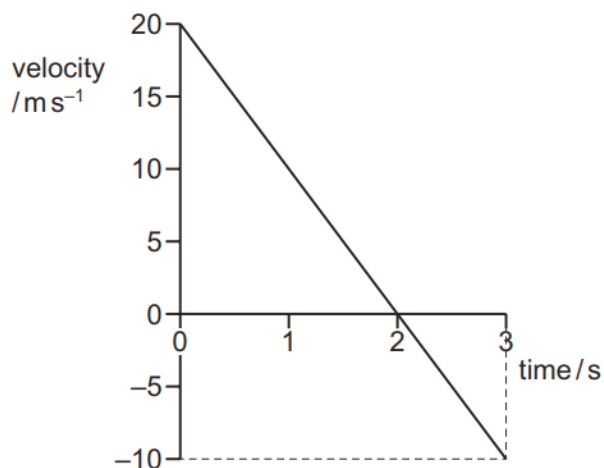
Car Y overtakes car X at time  $t = T$ .

Which graph could best show the variation with time  $t$  of the distance  $d$  between the cars?



**160. 9702/13/M/J/20/Q6**

The graph shows how the velocity of a ball varies with time from the moment it is hit vertically upwards from the ground.



What is the displacement of the ball from the ground after a time of 3.0 s?

- A** 15 m                      **B** 25 m                      **C** 30 m                      **D** 45 m

**161. 9702/13/M/J/20/Q7**

Two students each throw a ball horizontally from the top of a tower. The two balls are released at the same time.

The first student throws her ball with a speed of  $20 \text{ m s}^{-1}$  and the second student throws his ball with a speed of  $10 \text{ m s}^{-1}$ .

Assume air resistance is negligible and that the balls land on horizontal ground.

Which row describes the horizontal distances travelled and the landing times of the two balls on the ground?

|          | horizontal distances | landing times |
|----------|----------------------|---------------|
| <b>A</b> | same                 | same          |
| <b>B</b> | same                 | different     |
| <b>C</b> | different            | same          |
| <b>D</b> | different            | different     |

162. 9702/13/M/J/20/Q9

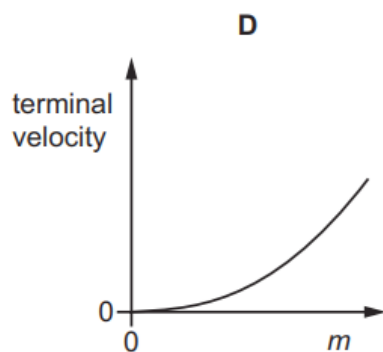
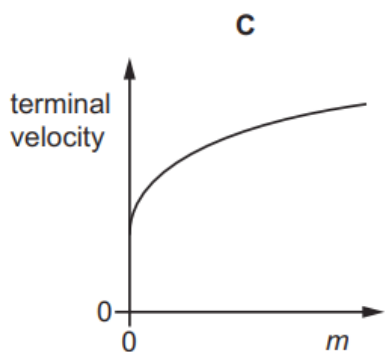
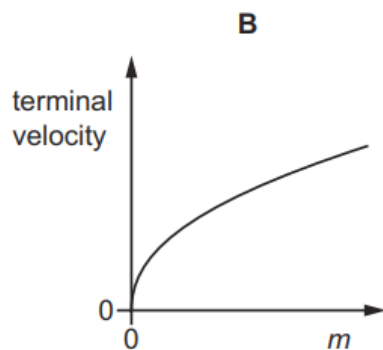
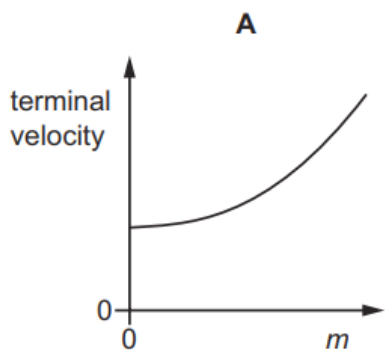
The resultant force  $F$  on a raindrop of mass  $m$  falling vertically with velocity  $v$  is given by the equation

$$F = mg - kv^2$$

where  $k$  is a constant and  $g$  is the acceleration of free fall.

The falling raindrop eventually reaches a constant (terminal) velocity.

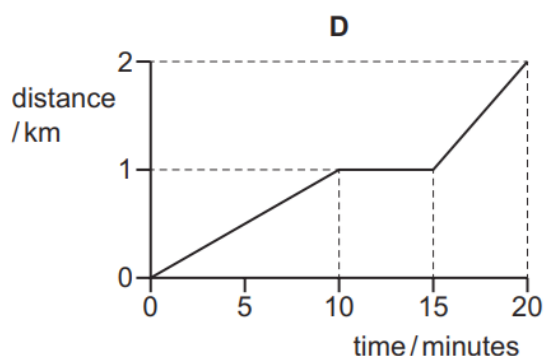
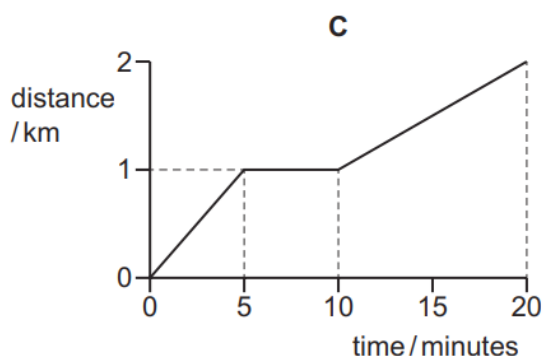
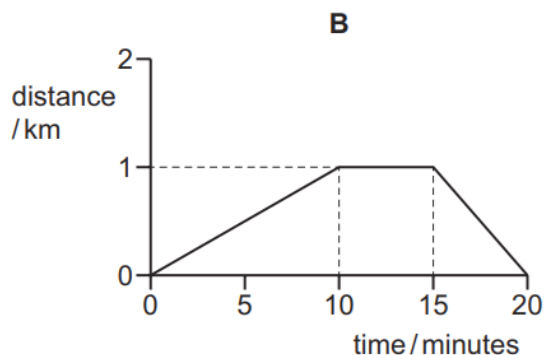
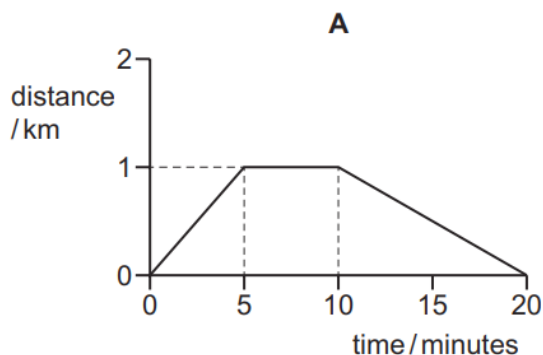
Which graph shows the variation of the terminal velocity of the raindrop with mass  $m$ ?



**163. 9702/11/O/N/20/Q6**

A student cycles uphill from home to a shop, taking 10 minutes. The student then spends 5 minutes in the shop, before cycling home downhill at twice the initial speed.

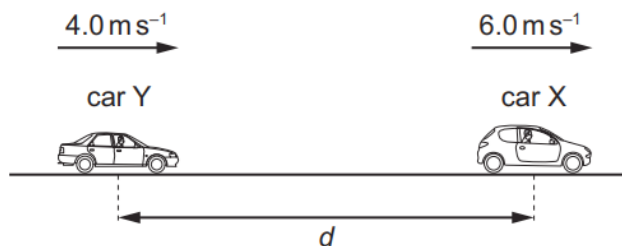
Which graph could show the variation with time of the distance travelled by the cyclist?



**164. 9702/11/O/N/20/Q7**

Two cars X and Y are travelling along the same straight road. Car X is travelling at a constant speed of  $6.0 \text{ m s}^{-1}$ . Car Y has a constant acceleration of  $0.50 \text{ m s}^{-2}$ .

At the instant shown, car X is a distance  $d$  ahead of car Y. Car Y is travelling at a speed of  $4.0 \text{ m s}^{-1}$ .



Car Y is level with car X after a time of 20 seconds.

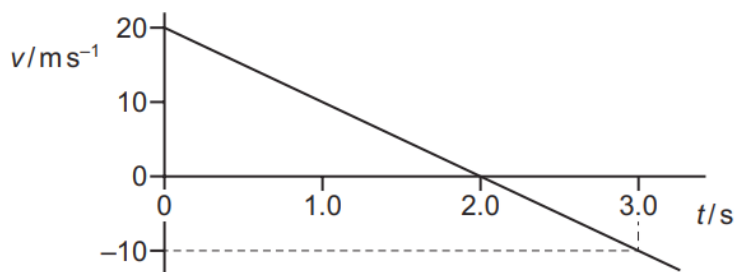
What is the distance  $d$ ?

- A** 40 m      **B** 60 m      **C** 180 m      **D** 300 m

**165. 9702/12/O/N/20/Q6**

A stone is thrown vertically upwards from a point X at time  $t = 0$ .

The variation with time  $t$  of the velocity  $v$  of the stone is shown.

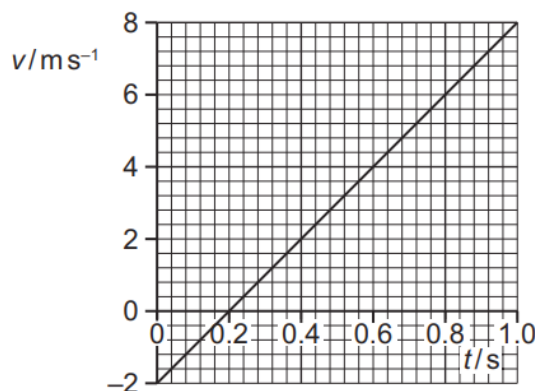


What is the displacement of the stone from point X at time  $t = 3.0$  s?

- A** 15 m above X
- B** 15 m below X
- C** 25 m above X
- D** 25 m below X

**166. 9702/13/O/N/20/Q6**

An object moves in a straight line. The graph shows the variation with time  $t$  of the velocity  $v$  of the object.



At time  $t = 0$  the object is at point X.

What is the displacement of the object from point X at time  $t = 0.80$  s?

- A** 1.6 m      **B** 1.8 m      **C** 2.0 m      **D** 3.2 m

**167. 9702/13/O/N/20/Q7**

An object accelerates uniformly from rest to speed  $v$ . It then moves at constant speed  $v$  for a time of 8.0 s before decelerating uniformly to rest. The total time taken is 12.0 s, and the total distance travelled is 60 m.

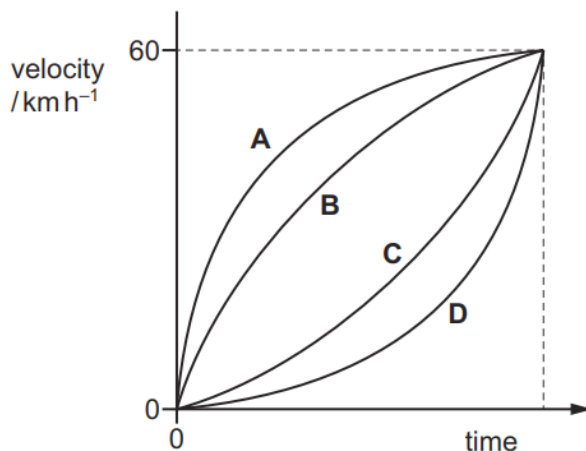
What is the speed  $v$ ?

- A**  $3.0 \text{ m s}^{-1}$       **B**  $5.0 \text{ m s}^{-1}$       **C**  $6.0 \text{ m s}^{-1}$       **D**  $15 \text{ m s}^{-1}$

**168. 9702/12/F/M/21/Q6**

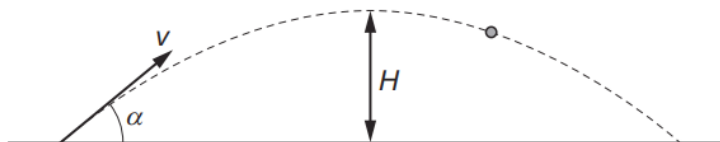
Four cars, **A**, **B**, **C** and **D**, move from rest in a straight line. The cars take the same time to accelerate to a velocity of  $60 \text{ km h}^{-1}$ . Their velocity–time graphs are shown.

Which car reaches a velocity of  $60 \text{ km h}^{-1}$  in the shortest distance?



**169. 9702/12/F/M/21/Q7**

A cannon fires a cannonball with an initial speed  $v$  at an angle  $\alpha$  to the horizontal.



Which equation is correct for the maximum height  $H$  reached?

**A**  $H = \frac{v \sin \alpha}{2a}$     **B**  $H = \frac{g \sin \alpha}{2v}$     **C**  $H = \frac{(v \sin \alpha)^2}{2a}$     **D**  $H = \frac{g^2 \sin \alpha}{2v}$

**170. 9702/12/F/M/21/Q9**

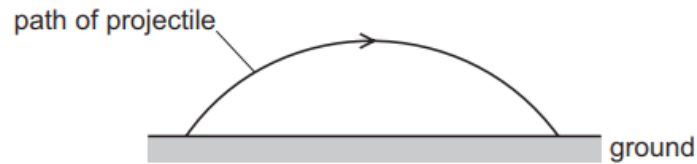
A skydiver, who is falling vertically through the air, opens his parachute.

Which row describes the velocity of the skydiver immediately after he opens his parachute?

|          | direction of velocity | magnitude of velocity |
|----------|-----------------------|-----------------------|
| <b>A</b> | downwards             | decreases             |
| <b>B</b> | downwards             | increases             |
| <b>C</b> | upwards               | decreases             |
| <b>D</b> | upwards               | increases             |

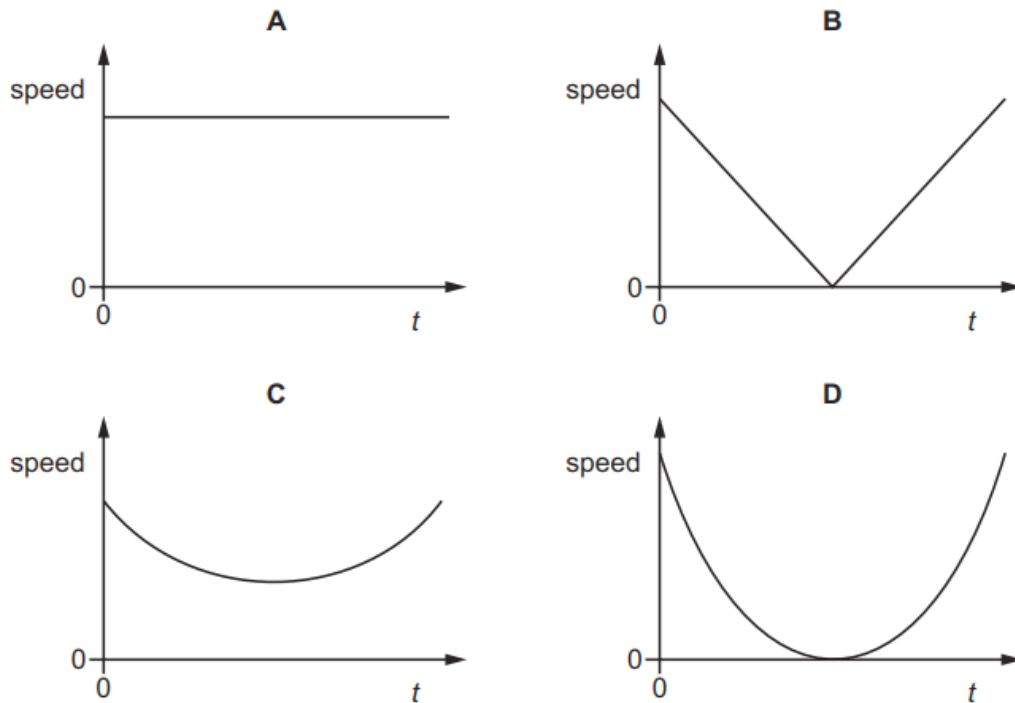
**171. 9702/11/M/J/21/Q6**

A projectile is launched at an angle to the horizontal at time  $t = 0$ . It travels over horizontal ground, as shown.



Air resistance is negligible.

Which graph best shows the variation with  $t$  of the speed of the projectile from when it is launched to when it lands on the ground?



**172. 9702/11/M/J/21/Q7**

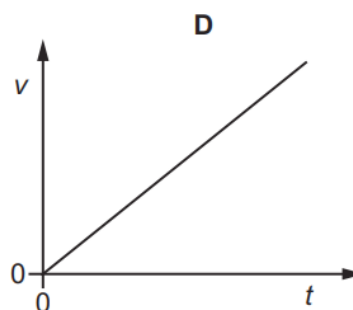
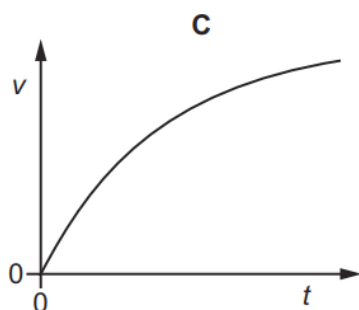
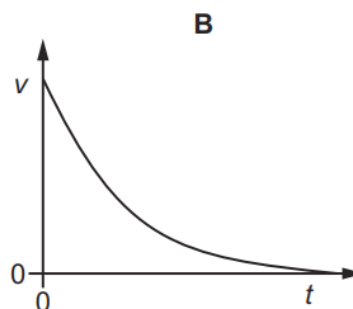
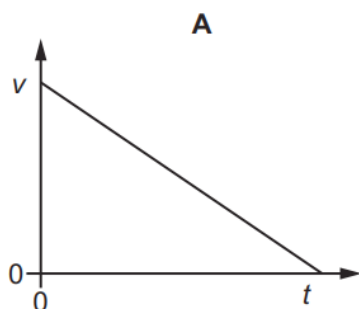
A train, initially at rest at a station, has a uniform acceleration of  $0.20 \text{ ms}^{-2}$  until it reaches a speed of  $20 \text{ ms}^{-1}$ . It travels for a time at this constant speed and then has a uniform deceleration of  $0.40 \text{ ms}^{-2}$  until it comes to rest at the next station. The distance between the two stations is 3000 m.

What is the time taken by the train to travel between the two stations?

- A** 75 s                      **B** 150 s                      **C** 230 s                      **D** 300 s

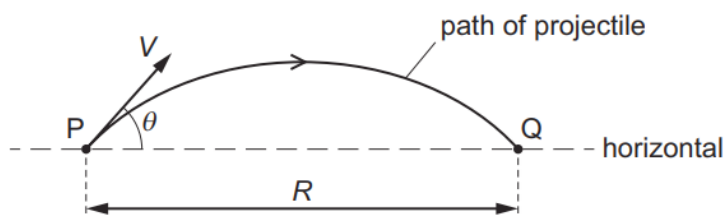
173. 9702/12/M/J/21/Q6

Which graph shows the variation with time  $t$  of the velocity  $v$  of an object falling vertically downwards in a vacuum?



174. 9702/12/M/J/21/Q7

A projectile is fired from point P with velocity  $V$  at an angle  $\theta$  to the horizontal. It lands at point Q, a horizontal distance  $R$  from P. Air resistance is negligible.



The acceleration of free fall is  $g$ .

Which equation for  $R$  is correct?

**A**  $R = \frac{V^2 \sin \theta \cos \theta}{g}$

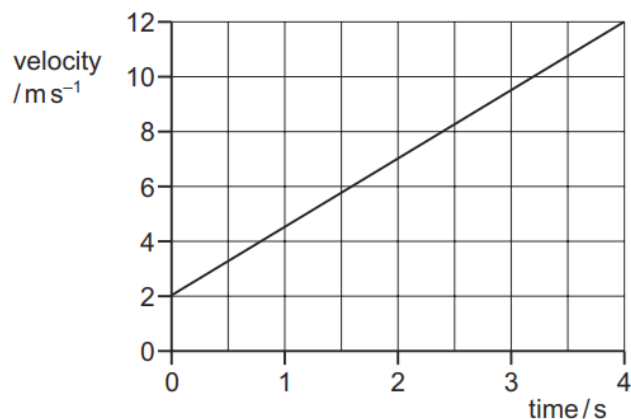
**B**  $R = \frac{2V^2 \sin \theta \cos \theta}{g}$

**C**  $R = \frac{V^2 \sin \theta \cos \theta}{2g}$

**D**  $R = \frac{V^2 g \sin \theta \cos \theta}{2}$

**175. 9702/13/M/J/21/Q6**

The diagram shows a velocity–time graph for a car.



What is the distance travelled during the first 4.0 s?

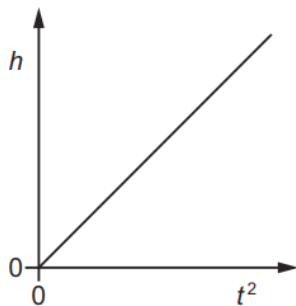
- A** 2.5 m      **B** 3.0 m      **C** 20 m      **D** 28 m

**176. 9702/13/M/J/21/Q7**

A steel ball is dropped from rest from a height  $h$  above the ground. The ball hits the ground after a time  $t$ .

This is repeated for a number of different heights.

The graph shows the variation of  $h$  with  $t^2$  for the ball.



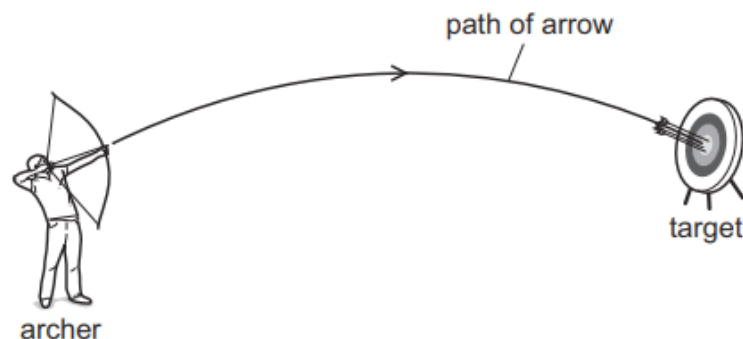
The gradient of the graph is  $G$ .

Which expression gives the acceleration of the ball?

- A**  $\frac{G}{2}$       **B**  $G$       **C**  $2G$       **D**  $G^2$

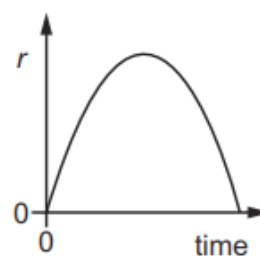
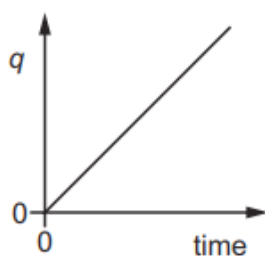
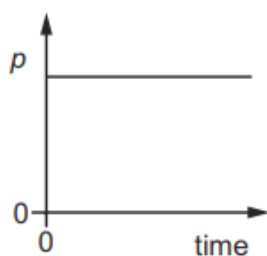
177. 9702/11/O/N/21/Q6

An archer shoots an arrow at a target. The diagram shows the path of the arrow.



Air resistance is negligible.

The graphs show how three different quantities  $p$ ,  $q$  and  $r$  vary with time.



Which quantity could be the horizontal component of displacement and which quantity could be the vertical component of displacement of the arrow?

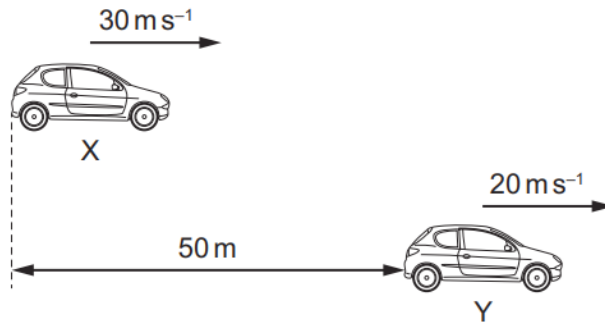
|          | horizontal<br>component of<br>displacement | vertical<br>component of<br>displacement |
|----------|--------------------------------------------|------------------------------------------|
| <b>A</b> | $p$                                        | $q$                                      |
| <b>B</b> | $q$                                        | $r$                                      |
| <b>C</b> | $r$                                        | $p$                                      |
| <b>D</b> | $r$                                        | $q$                                      |

**178. 9702/11/O/N/21/Q7**

Two cars X and Y are positioned as shown at time  $t = 0$ .

They are travelling in the same direction.

X is 50 m behind Y and has a constant velocity of  $30 \text{ m s}^{-1}$ . Y has a constant velocity of  $20 \text{ m s}^{-1}$ .

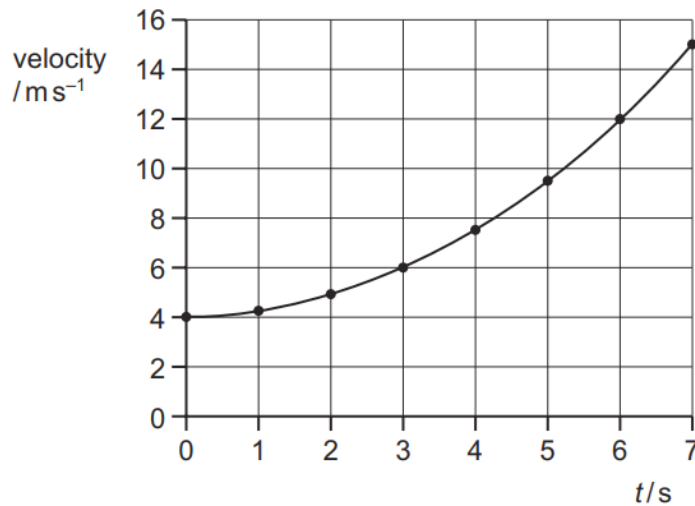


What is the value of  $t$  when X is level with Y?

- A** 1.0 s      **B** 1.7 s      **C** 2.5 s      **D** 5.0 s

**179. 9702/12/O/N/21/Q6**

The graph shows the variation with time  $t$  of the velocity of a vehicle moving in a straight line.



The vehicle, moving at  $4.0 \text{ m s}^{-1}$ , begins to accelerate at time  $t = 0$ .

What is the vehicle's acceleration at time  $t = 3.0 \text{ s}$ ?

- A**  $0.67 \text{ m s}^{-2}$       **B**  $1.0 \text{ m s}^{-2}$       **C**  $1.3 \text{ m s}^{-2}$       **D**  $2.0 \text{ m s}^{-2}$

**180. 9702/12/O/N/21/Q7**

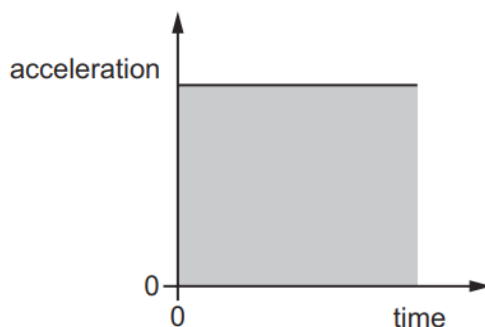
A stone is projected vertically upwards from the ground at an initial speed of  $15\text{ ms}^{-1}$ . Air resistance is negligible.

What is the maximum height reached by the stone?

- A** 0.76 m      **B** 11 m      **C** 23 m      **D** 110 m

**181. 9702/13/O/N/21/Q6**

The graph shows the variation with time of the acceleration of a car.



What **must** the shaded area under the graph represent?

- A** the average velocity of the car  
**B** the change in velocity of the car  
**C** the final velocity of the car  
**D** the initial velocity of the car

**182. 9702/13/O/N/21/Q7**

A stone is thrown horizontally off a cliff and then lands in the sea. Air resistance is negligible.

Which statement about the stone's motion is **not** correct?

- A** The final displacement of the stone depends upon its initial horizontal velocity.  
**B** The stone travels with a constant horizontal component of velocity until it hits the water.  
**C** The stone travels with an increasing vertical component of velocity.  
**D** The time taken for the stone to hit the surface of the water depends on its initial horizontal velocity.

**183. 9702/12/F/M/22/Q5**

How can the acceleration of an object be determined?

- A** from the area under a displacement–time graph  
**B** from the area under a velocity–time graph  
**C** from the gradient of a displacement–time graph  
**D** from the gradient of a velocity–time graph

**184. 9702/12/F/M/22/Q6**

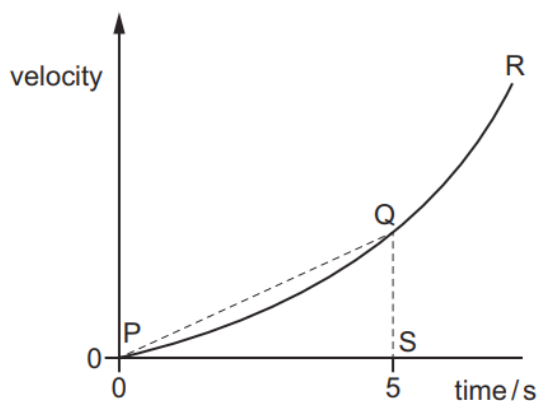
A sprinter takes a time of 11.0 s to run a 100 m race. She first accelerates uniformly from rest, reaching a speed of  $10 \text{ m s}^{-1}$ . She then runs at a constant speed of  $10 \text{ m s}^{-1}$  until the finish line.

What is the uniform acceleration of the sprinter for the first part of the race?

- A**  $0.5 \text{ m s}^{-2}$       **B**  $0.91 \text{ m s}^{-2}$       **C**  $1.7 \text{ m s}^{-2}$       **D**  $5.0 \text{ m s}^{-2}$

**185. 9702/11/M/J/22/Q5**

The curved line PQR is the velocity–time graph for a car starting from rest.



What is the average acceleration of the car over the first 5 s?

- A** the area below the curve PQ  
**B** the area of the triangle PQS  
**C** the gradient of the straight line PQ  
**D** the gradient of the tangent at Q

**186. 9702/11/M/J/22/Q6**

A ball is thrown horizontally with a speed of  $10.0 \text{ m s}^{-1}$  above horizontal ground. The ball hits the ground after a time of 3.0 s.

Air resistance is negligible.

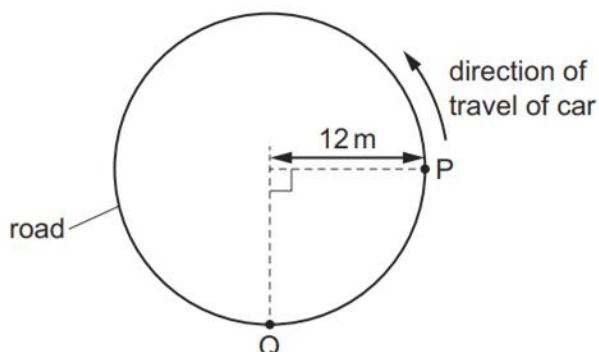
What is the speed of the ball just before it hits the ground?

- A**  $10 \text{ m s}^{-1}$       **B**  $29 \text{ m s}^{-1}$       **C**  $31 \text{ m s}^{-1}$       **D**  $39 \text{ m s}^{-1}$

**187. 9702/12/M/J/22/Q5**

A car travels anticlockwise along a horizontal circular road of radius 12 m, as shown.

The car takes a time of 4.0 s to move from position P to position Q.



What is the magnitude of the average velocity of the car for the journey from P to Q?

- A**  $4.2 \text{ m s}^{-1}$       **B**  $4.7 \text{ m s}^{-1}$       **C**  $6.0 \text{ m s}^{-1}$       **D**  $14 \text{ m s}^{-1}$

**188. 9702/12/M/J/22/Q6**

The water surface in a deep well is 78.0 m below the top of the well. A person at the top of the well drops a heavy stone down the well.

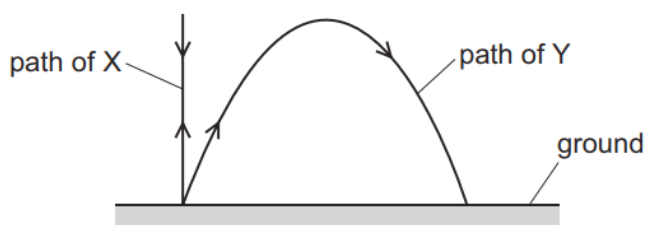
Air resistance is negligible. The speed of sound in the air is  $330 \text{ m s}^{-1}$ .

What is the time interval between the person dropping the stone and hearing it hitting the water?

- A** 3.75 s      **B** 3.99 s      **C** 4.19 s      **D** 4.22 s

**189. 9702/13/M/J/22/Q6**

Two projectiles, X and Y, are fired into the air from the same place on level ground and reach the same maximum height, as shown.



Projectile X is fired vertically upwards and projectile Y is fired at an angle to the horizontal.

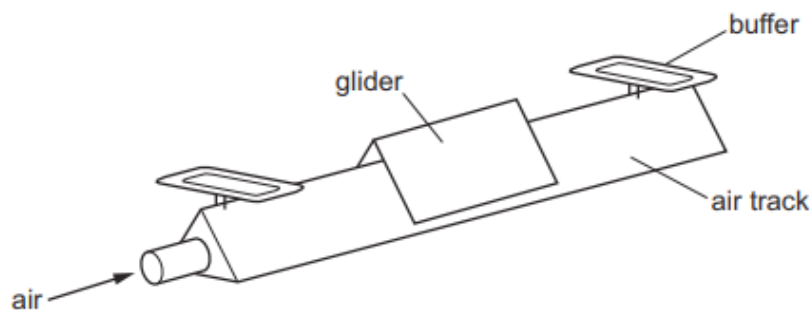
Air resistance is negligible.

Which statement is correct?

- A** X and Y are at rest at their maximum heights.  
**B** X and Y are fired with the same speed.  
**C** X and Y take the same time to return to the ground.  
**D** X and Y travel the same distance.

190. 9702/13/M/J/22/Q7

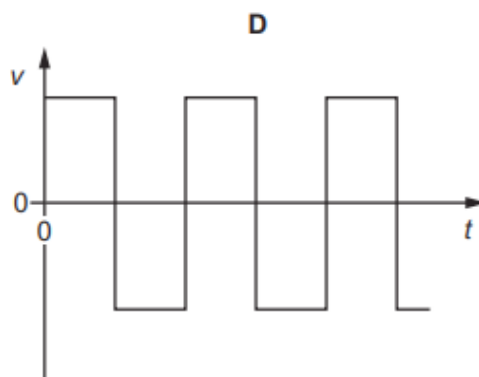
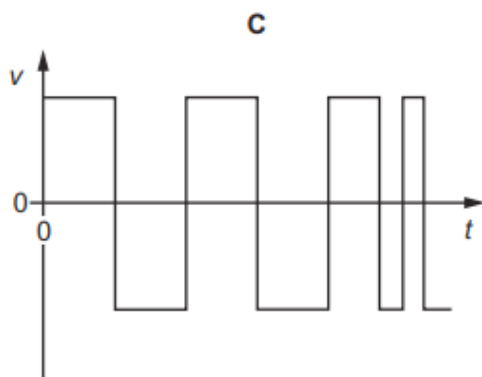
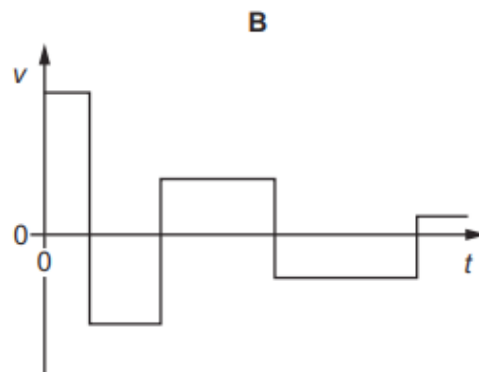
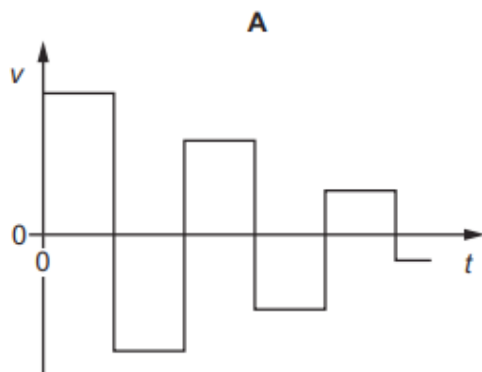
A small glider moves along a horizontal air track as shown.



At each end of the air track, the glider has a perfectly elastic collision with a fixed buffer.

The glider moves at a constant speed between collisions.

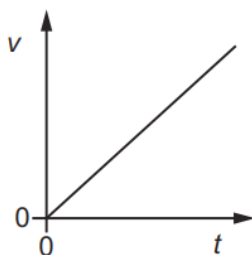
Which graph represents the variation with time  $t$  of the velocity  $v$  of the glider as it moves between the two buffers?



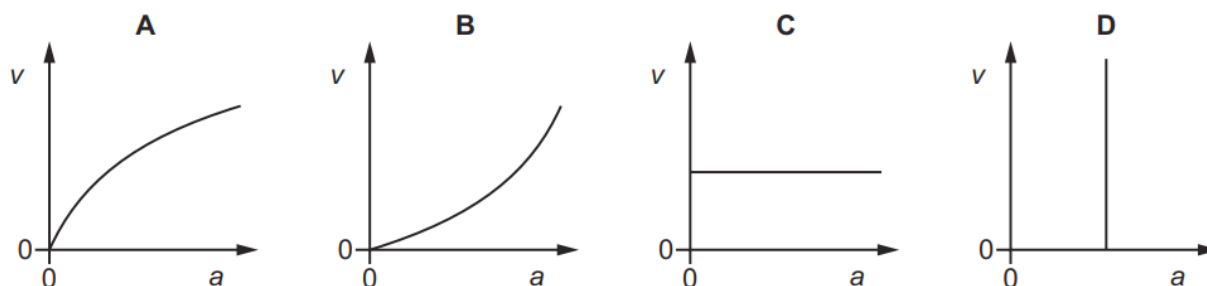
191. 9702/11/O/N/22/Q5

A particle accelerates from rest.

The graph shows the variation of the velocity  $v$  of the particle with time  $t$ .

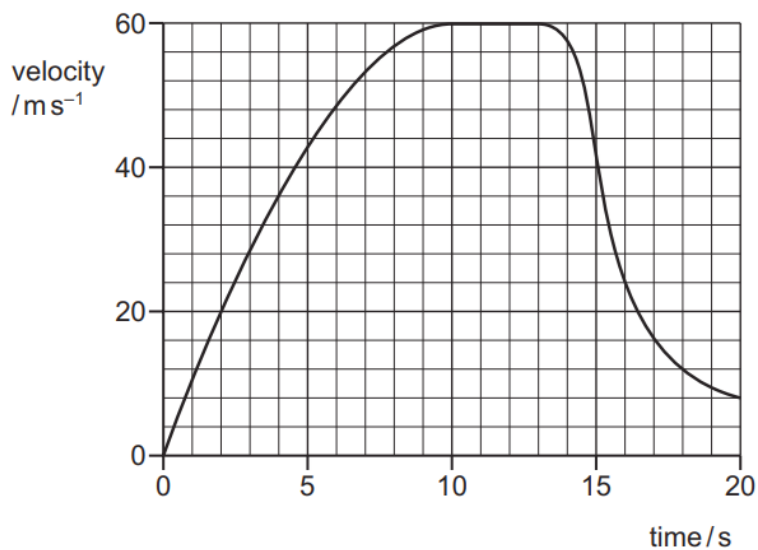


Which graph shows the variation of the velocity  $v$  with the acceleration  $a$  of the particle?



192. 9702/11/O/N/22/Q6

The graph shows the vertical velocity of a parachutist during the first 20 s of her jump.



Approximately how far does she fall **before** opening the parachute?

- A** 390 m      **B** 570 m      **C** 710 m      **D** 770 m

**193. 9702/12/O/N/22/Q5**

A toy car travels on a circular track at a constant speed of  $0.50 \text{ m s}^{-1}$ . It passes a point on the track at time  $t = 0$  and takes a time of  $40 \text{ s}$  to travel once around the track.

The magnitude of the average velocity of the car between  $t = 0$  and  $t = 20 \text{ s}$  is  $v_{20}$ .

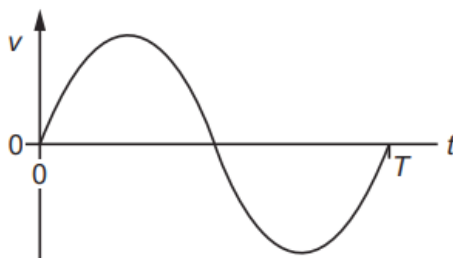
The magnitude of the average velocity of the car between  $t = 0$  and  $t = 40 \text{ s}$  is  $v_{40}$ .

What are  $v_{20}$  and  $v_{40}$ ?

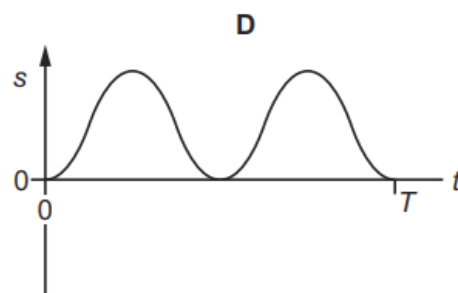
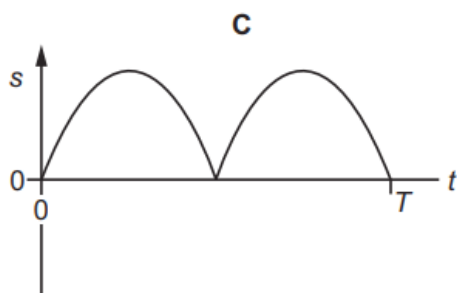
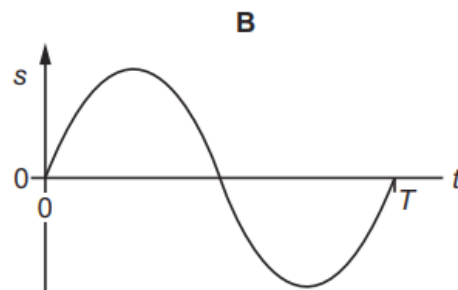
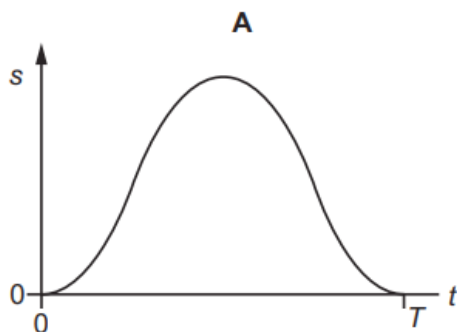
|          | $v_{20} / \text{m s}^{-1}$ | $v_{40} / \text{m s}^{-1}$ |
|----------|----------------------------|----------------------------|
| <b>A</b> | 0.32                       | 0                          |
| <b>B</b> | 0.32                       | 0.32                       |
| <b>C</b> | 0.50                       | 0                          |
| <b>D</b> | 0.50                       | 0.50                       |

**194. 9702/12/O/N/22/Q6**

The graph shows how the velocity  $v$  of an object moving in a straight line varies with time  $t$  from  $t = 0$  to  $t = T$ .



Which graph could represent the displacement  $s$  of the object from time  $t = 0$  to  $t = T$ ?



**195. 9702/12/O/N/22/Q7**

A goods train passes through a station at a steady speed of  $10 \text{ ms}^{-1}$  at time  $t = 0$ . An express train is at rest at the station. The express train leaves the station with a uniform acceleration of  $0.5 \text{ ms}^{-2}$  just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

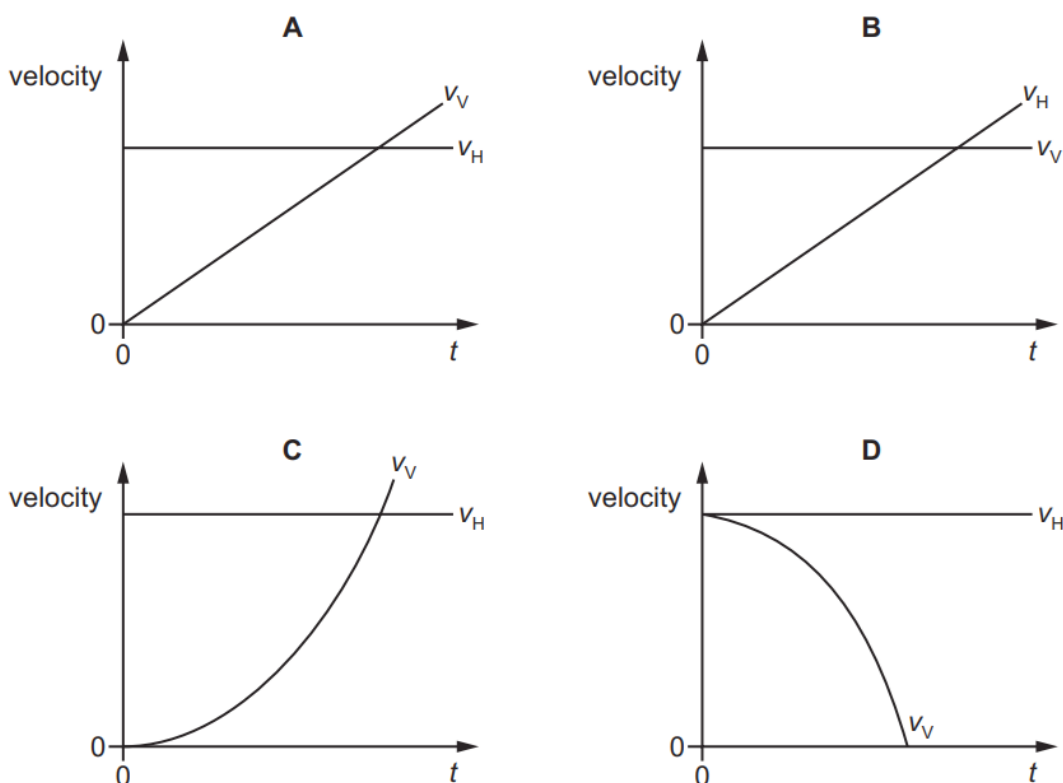
At which time  $t$  does the express train overtake the goods train?

- A** 6 s                      **B** 10 s                      **C** 20 s                      **D** 40 s

**196. 9702/13/O/N/22/Q5**

A stone is projected horizontally at time  $t = 0$  and falls. Air resistance is negligible. The stone has a horizontal component of velocity  $v_H$  and a vertical component of velocity  $v_V$ .

Which graph shows how  $v_H$  and  $v_V$  vary with time  $t$ ?



**197. 9702/13/O/N/22/Q6**

On the Earth, an object takes time  $T_E$  to fall from rest through a vertical distance  $h$ .

On the Moon, the same object takes time  $T_M$  to fall from rest through the same vertical distance  $h$ .

The ratio  $\frac{\text{acceleration of free fall on the Earth}}{\text{acceleration of free fall on the Moon}}$  is equal to 6.

Air resistance is negligible for the object on the Earth and on the Moon.

What is the ratio  $\frac{T_E}{T_M}$ ?

- A**  $\frac{1}{6}$                       **B**  $\frac{1}{\sqrt{6}}$                       **C**  $\sqrt{6}$                       **D** 6

**198. 9702/13/O/N/22/Q7**

A cyclist in still air pedals as hard as she can. She reaches a maximum speed. However, after a certain time her maximum speed increases.

What could be a possible cause for this?

- A** She cycles into a wind.
- B** She cycles over rougher ground.
- C** She sits more upright on the bicycle.
- D** She starts to travel downhill.

**199. 9702/12/F/M/23/Q5**

A ball is kicked upwards at an angle of  $45^\circ$  to horizontal ground. After a short flight, the ball returns to the ground.

It may be assumed that air resistance is negligible.

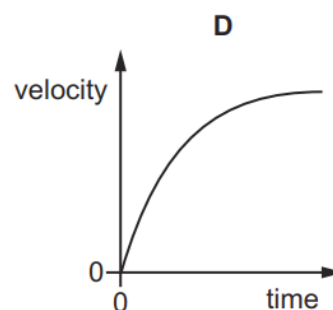
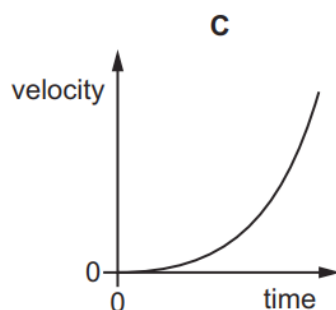
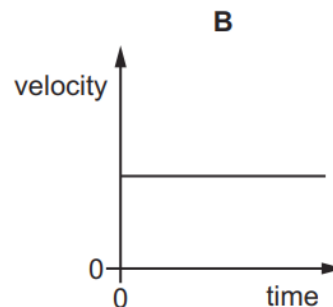
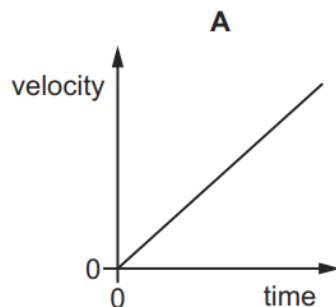
What is **never** zero during the flight of the ball?

- A** the horizontal component of the ball's acceleration
- B** the horizontal component of the ball's velocity
- C** the vertical component of the ball's momentum
- D** the vertical component of the ball's velocity

**200. 9702/12/F/M/23/Q9**

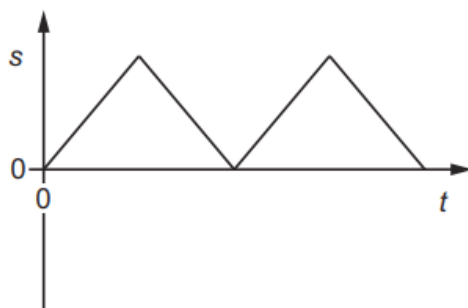
The velocity–time graphs of four different objects are shown.

Which graph represents an object falling from rest through a long distance in the Earth's atmosphere?

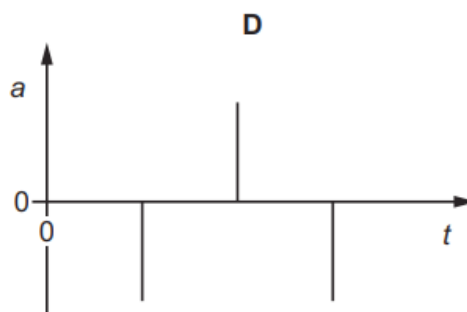
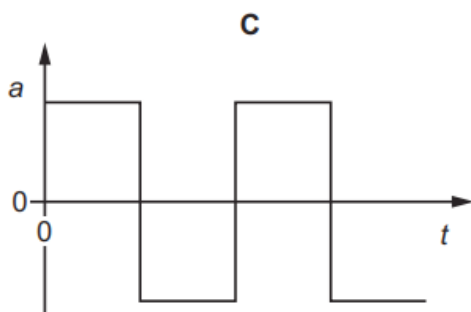
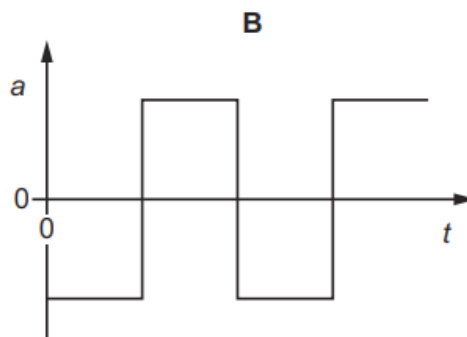
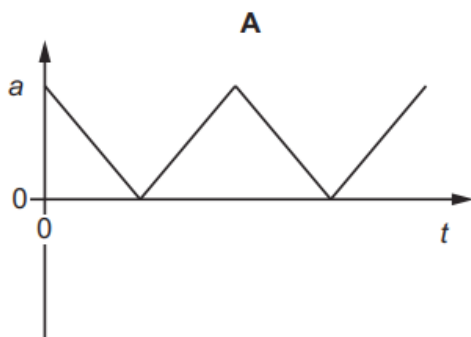


**201. 9702/12/F/M/23/Q6**

The graph shows the variation with time  $t$  of the displacement  $s$  of an object.



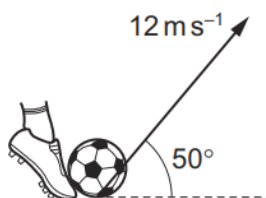
Which graph represents the variation with time  $t$  of the acceleration  $a$  of the object?



**202. 9702/11/M/J/23/Q6**

A ball is kicked so that it has an initial velocity of  $12 \text{ m s}^{-1}$  at an angle of  $50^\circ$  to horizontal ground.

Assume that air resistance is negligible.



What is the maximum height above the ground that is reached by the ball?

**A** 0.47 m

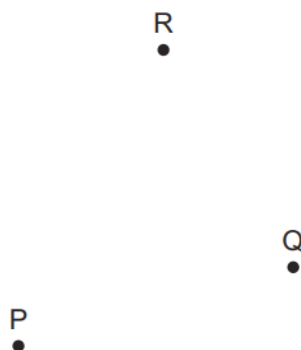
**B** 3.0 m

**C** 4.3 m

**D** 7.3 m

**203. 9702/11/M/J/23/Q5**

One object moves directly from P to R.



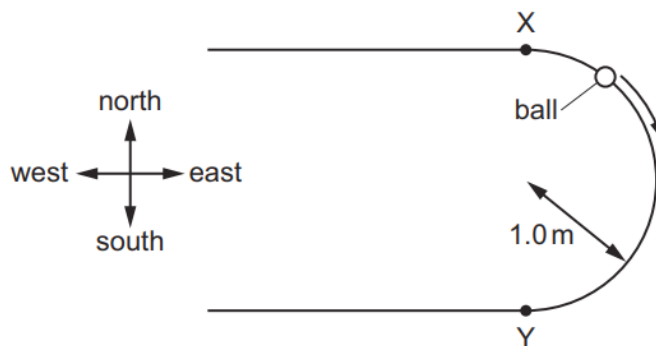
In a shorter time, a second object moves from P to Q to R.

Which statement about the two objects is correct for the journey from P to R?

- A** They have the same average speed.
- B** They have the same average velocity.
- C** They have the same displacement.
- D** They travel the same distance.

**204. 9702/12/M/J/23/Q5**

A ball travels from point X to point Y around a semicircular track of radius 1.0 m, as shown.



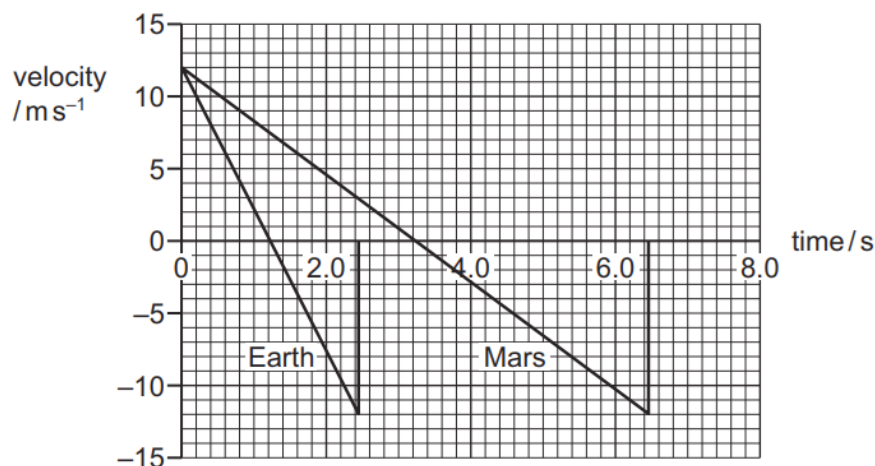
What is the displacement of the ball from X to Y?

- A** 2.0 m
- B** 2.0 m due south
- C** 3.1 m
- D** 3.1 m due south

**205. 9702/12/M/J/23/Q6**

A rock is launched vertically upwards from the surface of the Earth and an identical rock is launched vertically upwards from the surface of Mars. Each rock is launched with an initial velocity of  $12 \text{ m s}^{-1}$ .

Each rock then reaches its maximum height above the surface before returning back down to the surface. The velocity–time graph for each rock is shown. In both cases, air resistance is negligible.



What is the difference in the maximum heights of the two rocks?

- A** 12 m      **B** 15 m      **C** 19 m      **D** 24 m

**206. 9702/13/M/J/23/Q5**

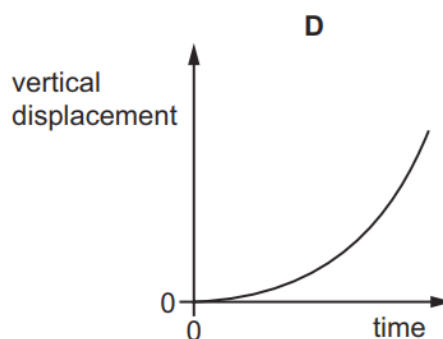
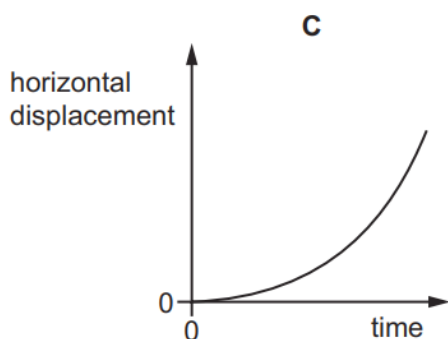
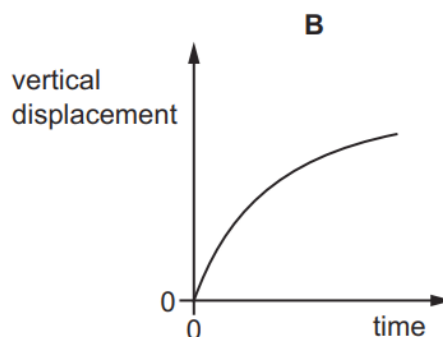
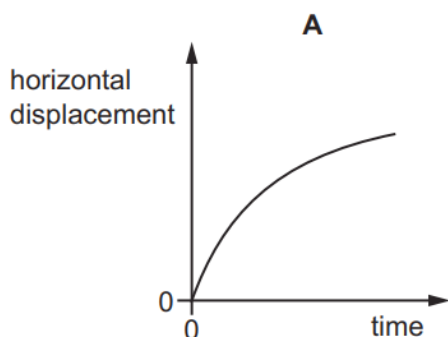
Which equation, representing uniformly accelerated motion in a straight line, can be determined using **only** the definition of acceleration?

- A**  $s = ut + \frac{1}{2}at^2$   
**B**  $s = \frac{1}{2}(u + v)t$   
**C**  $v = u + at$   
**D**  $v^2 = u^2 + 2as$

**207. 9702/13/M/J/23/Q6**

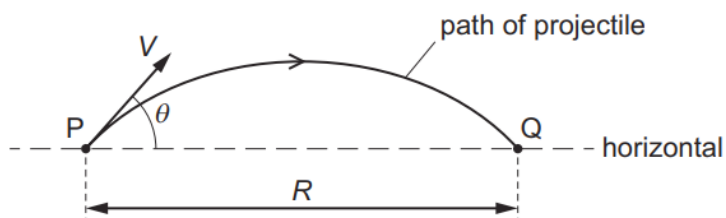
An object moves from rest with uniform velocity horizontally and uniform acceleration vertically.

Which graph showing the variation with time of the displacement of the object from its initial position is correct?



**208. 9702/11/O/N/23/Q6**

A projectile is fired from point P with velocity  $V$  at an angle  $\theta$  to the horizontal. It lands at point Q, a horizontal distance  $R$  from P, after time  $T$ .



The acceleration of free fall is  $g$ . Air resistance is negligible.

Which equation is correct?

**A**  $R = VT \cos \theta$

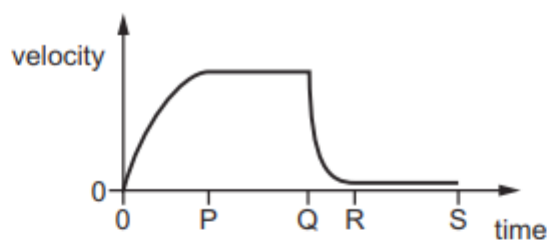
**B**  $R = VT \sin \theta$

**C**  $R = VT \cos \theta - \frac{1}{2} g T^2$

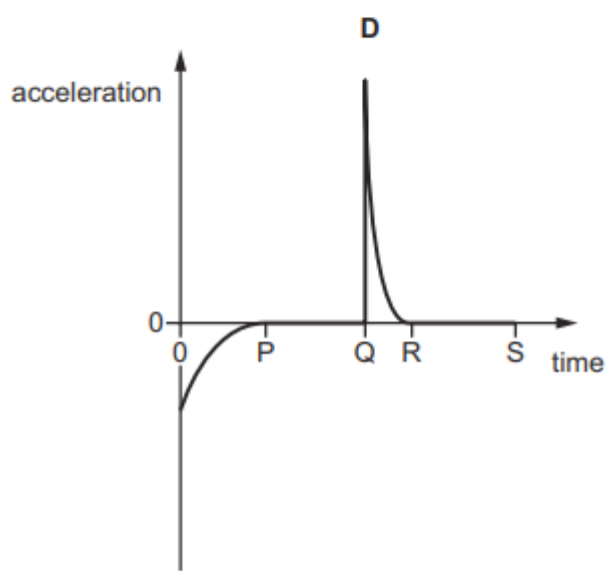
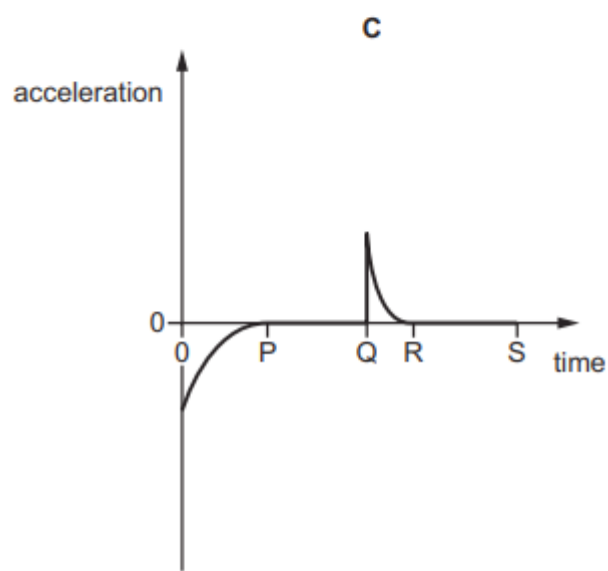
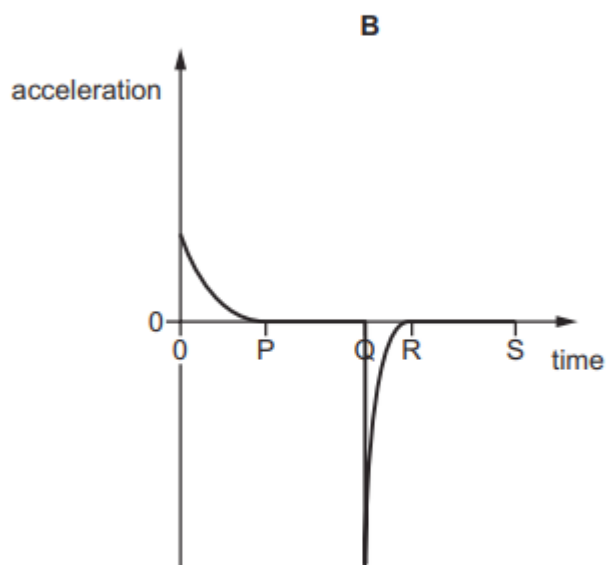
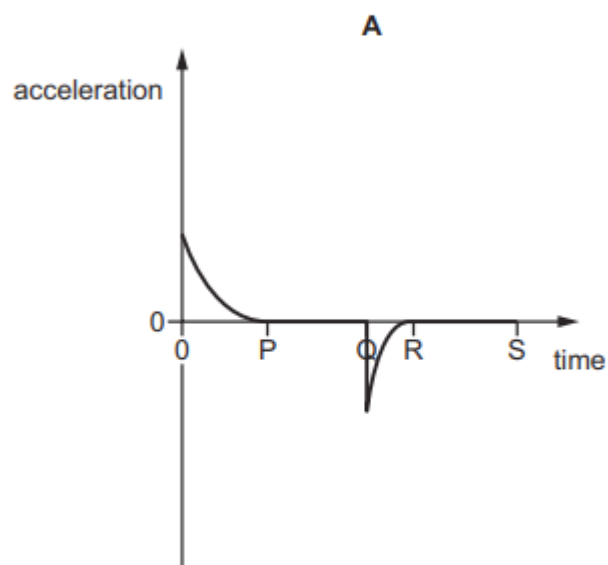
**D**  $R = VT \sin \theta - \frac{1}{2} g T^2$

209. 9702/11/O/N/23/Q5

A parachutist falls from a stationary balloon at time  $t = 0$ . The velocity–time graph for the parachutist from time  $t = 0$  until the time when he is just above the ground is shown.



Which graph best shows the variation with time of the acceleration of the parachutist?



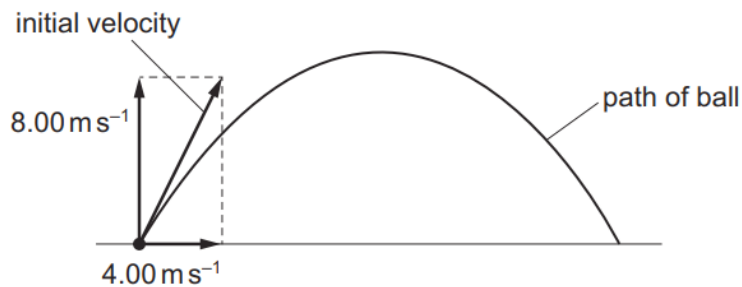
**210. 9702/12/O/N/23/Q5**

What is the definition of acceleration?

- A** the rate of change of displacement
- B** the rate of change of kinetic energy
- C** the rate of change of momentum
- D** the rate of change of velocity

**211. 9702/12/O/N/23/Q6**

An astronaut on the Moon, where there is no air resistance, throws a ball. The ball's initial velocity has a vertical component of  $8.00 \text{ m s}^{-1}$  and a horizontal component of  $4.00 \text{ m s}^{-1}$ , as shown.



The acceleration of free fall on the Moon is  $1.62 \text{ m s}^{-2}$ .

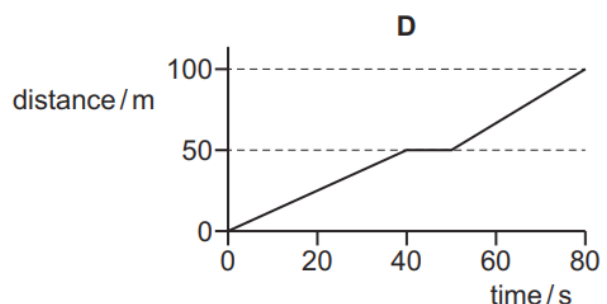
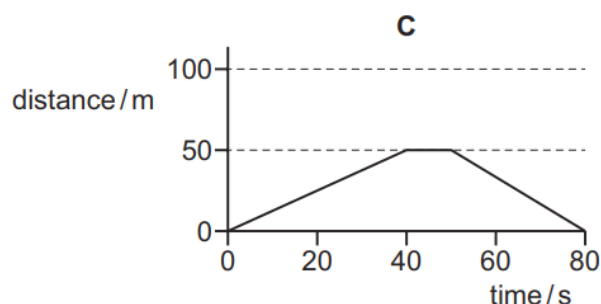
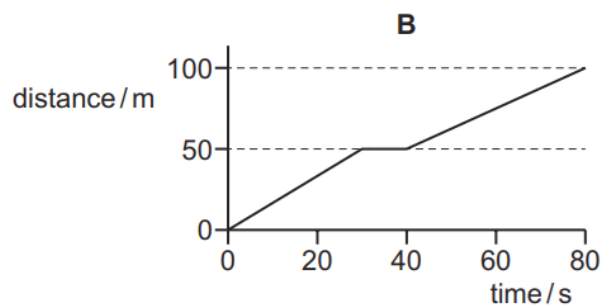
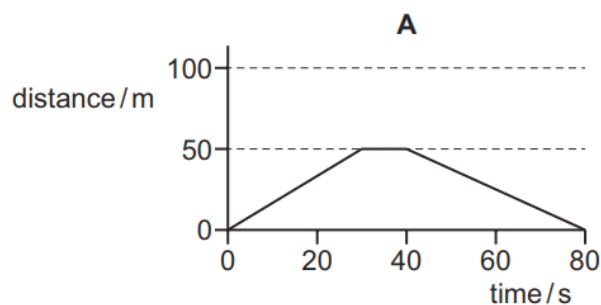
What is the speed of the ball  $9.00 \text{ s}$  after being thrown?

- A**  $6.58 \text{ m s}^{-1}$
- B**  $7.70 \text{ m s}^{-1}$
- C**  $10.6 \text{ m s}^{-1}$
- D**  $14.6 \text{ m s}^{-1}$

**212. 9702/13/O/N/23/Q5**

A student walks at a constant speed for a distance of  $50 \text{ m}$  in a time of  $40 \text{ s}$ . The student rests for a time of  $10 \text{ s}$  and then walks back to the starting point at a constant speed in a time of  $30 \text{ s}$ .

What is the distance–time graph for the motion of the student?



**213. 9702/13/O/N/23/Q6**

The time taken for an object to fall from rest through a certain distance on Mars is  $T_M$ . The time taken for the same object to fall from rest through the same distance on Earth is  $T_E$ . The acceleration of free fall on Mars is  $3.71 \text{ m s}^{-2}$ .

Assume that air resistance is negligible on both Earth and Mars.

What is the ratio  $\frac{T_M}{T_E}$ ?

**A** 0.378

**B** 0.615

**C** 1.63

**D** 2.64